

Unified Combinatorial Formulae and Inversion Theorems

**Stirling, Bell, Eulerian, Bernoulli, and Sheffer–Riordan Calculus;
Faà di Bruno, Lagrange–Good Inversion, and Asymptotic Coefficients**

A merged, deduplicated, and proof-complete synthesis

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Abstract

This monograph merges five overlapping draft articles into one coherent coefficient calculus. Repeated definitions, tables, and proofs have been consolidated; notation and boundary conventions have been normalized; and unique material from every draft has been retained. The supplied Wikipedia and MathWorld reference folders have been used as a topic audit, while authoritative online sources and primary papers have been used to verify and extend the special-function, multivariate-inversion, and asymptotic sections.

The principal combinatorial objects are the signed and unsigned Stirling numbers of the first kind, Stirling numbers of the second kind, Lah numbers, Bell numbers, Touchard and Bell polynomials, Eulerian numbers of types A and B , second-order Eulerian numbers, Catalan and Narayana numbers, and associahedral face numbers. The structural layer develops finite differences, factorial bases, partition-lattice Möbius inversion, moments and cumulants, polynomial sequences of binomial type, Sheffer sequences, and exponential Riordan arrays.

The analytic and compositional layer proves the scalar and Fréchet forms of Faà di Bruno's formula, higher product and inverse-function derivative formulas, ordinary and exponential composition laws, Lagrange–Bürmann reversion, Good's multivariate determinant formula, and the Fourier–Bessel solution of Kepler's equation. Bernoulli, Euler, Genocchi, second-kind Bernoulli, and complementary Bell sequences are developed from their generating functions. Euler–Maclaurin summation is proved with an explicit periodic-Bernoulli remainder. Pochhammer symbols, polygamma differences, the Barnes G -function, Darboux's principle, and generic square-root singularities connect the coefficient calculus to gamma products and asymptotic enumeration.

The treatment is deliberately proof-oriented. Generating functions are derived rather than merely recorded; formal and analytic hypotheses are separated; Jacobian factors and indexing conventions are made explicit; and schematic claims from the drafts are replaced by exact recurrences, controlled remainders, or reproducible coefficient algorithms. A final concordance records how the five drafts and the two supplied reference directories were absorbed.

How to read this work

The chapters are arranged by mathematical dependence, not by the order of the five drafts. Their common core has been written once, while genuinely distinct results have been placed where their prerequisites make them most transparent. Three master ideas organize the book:

1. a coefficient is often easiest to compute after changing to the basis in which the relevant operator is diagonal;
2. an exponential generating function remembers the decomposition of a labelled object into components; and
3. a residue or Jacobian converts implicit substitution into direct coefficient extraction.

Stirling and Eulerian arrays implement changes of basis. Bell polynomials and partition lattices implement labelled decomposition and its Möbius inverse. Sheffer sequences and Riordan arrays package these transforms as a group. Faà di Bruno’s formula is the differential form of composition, while Lagrange–Good inversion is the coefficient form of implicit substitution.

A result stated for formal power series requires no convergence. When an analytic version is asserted, its domain hypotheses are written explicitly. Statements in the source corpus that report historical computations—for example, lists of Bell primes or verified exact modular periods—are distinguished from theorems and are accompanied by a certificate procedure rather than misrepresented as symbolic identities. Formulae copied from reference pages are reproved from generating functions, residues, or Euler–Maclaurin summation whenever a proof is needed for the logical development.

Editorial normalization

Three systematic repairs are used throughout.

1. Bell numbers are denoted B_n , complete exponential Bell polynomials by $\mathcal{B}_n$, partial Bell polynomials by $\mathcal{B}_{n,k}$, and Bernoulli numbers by β_n .
2. Type- B Eulerian numbers are indexed by descents in positions $0, 1, \dots, n-1$ with $\pi_0 = 0$. This removes the one-step shift in one formula in the archive.
3. The archived Moser–Wyman display contains unnamed quantities B, P_i, Q_i . We give the exact saddle-point scheme that defines every correction coefficient recursively and state the leading expansion in closed form; no undefined placeholder is retained.

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Notation

For $n \in \mathbb{N}$, write

$$x^{\overline{n}} = x(x-1) \cdots (x-n+1), \quad x^{\underline{n}} = x(x+1) \cdots (x+n-1),$$

with both expressions equal to 1 when $n = 0$. Coefficient extraction is denoted by $[z^n]F(z)$, and $\text{Res}_z F(z) dz$ means the coefficient of z^{-1} in a formal Laurent series. Our principal arrays are

$$s(n, k), \quad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k} s(n, k), \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle,$$

where $s(n, k)$ is the signed first-kind Stirling number, $\begin{bmatrix} n \\ k \end{bmatrix}$ its unsigned version, $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ the second-kind Stirling number, and $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle$ the ordinary Eulerian number. Bell numbers are B_n ; exponential partial and complete Bell polynomials are $\mathcal{B}_{n,k}$ and $\mathcal{B}_n$.

Unless stated otherwise, sums over an impossible index range are zero, and $0^0 = 1$ in coefficient formulae. Bernoulli numbers use

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad \beta_1 = -\frac{1}{2}.$$

Part I

Coefficient Calculus and Labelled Structures

Chapter 1

Formal power series and coefficient extraction

1.1 Formal series as an algebraic environment

Let R be a commutative $\mathbb{Q}$ -algebra. The ring $R[[z]]$ consists of expressions $F(z) = \sum_{n \geq 0} f_n z^n$ with termwise addition and Cauchy multiplication. A series is invertible precisely when f_0 is invertible in R . If $G(0) = 0$, then the composition $F(G(z))$ is well-defined because each coefficient receives only finitely many contributions.

Lemma 1.1 (Coefficient rules). For $F(z) = \sum f_n z^n$ and $G(z) = \sum g_n z^n$,

$$[z^n]F(z)G(z) = \sum_{j=0}^n f_j g_{n-j}, \quad (1.1)$$

$$[z^n]F'(z) = (n+1)f_{n+1}, \quad (1.2)$$

$$[z^n] \frac{F(z)}{1-az} = \sum_{j=0}^n a^{n-j} f_j. \quad (1.3)$$

If F is analytic on $|z| \leq r$, then also

$$f_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{F(z)}{z^{n+1}} dz, \quad |f_n| \leq r^{-n} \max_{|z|=r} |F(z)|. \quad (1.4)$$

Proof. The first identity is the definition of Cauchy multiplication. The second follows by differentiating term by term, and the third by multiplying with $\sum_{m \geq 0} a^m z^m$. The analytic formula is Cauchy's coefficient formula, and the bound follows by estimating the contour integral by its length $2\pi r$. $\square$

1.2 Formal residues

For a Laurent series $A(z)$, set $\text{Res}_z A(z) dz = [z^{-1}]A(z)$. Then

$$[z^n]F(z) = \text{Res}_z \frac{F(z)}{z^{n+1}} dz, \quad \text{Res}_z A'(z) dz = 0. \quad (1.5)$$

The second identity is simply the fact that the derivative of z^m has no z^{-1} term.

Theorem 1.2 (Formal change of variables for residues). Let $u = u(z) \in zR[[z]]$ have invertible linear coefficient. For every Laurent series A for which the composition is defined,

$$\text{Res}_z A(u(z))u'(z) dz = \text{Res}_u A(u) du. \quad (1.6)$$

Proof. By linearity it suffices to take $A(u) = u^m$. If $m \neq -1$, then $u(z)^m u'(z) = \frac{1}{m+1} (u(z)^{m+1})'$ has residue zero. For $m = -1$, $u'(z)/u(z) = z^{-1} + H(z)$ with $H \in R[[z]]$, because $u(z) = az(1 + O(z))$. Thus both sides have residue 1. $\square$

Master principle

The combination of Equations (1.5) and (1.6) turns implicit substitution into coefficient extraction. This is the entire algebraic engine of Lagrange inversion; no analytic contour is needed until one asks about convergence.

1.3 Ordinary and exponential generating functions

For a sequence (a_n) , its ordinary and exponential generating functions are

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \hat{A}(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}.$$

Ordinary multiplication gives unweighted convolution. Exponential multiplication gives binomial convolution:

$$\hat{A}(z)\hat{B}(z) = \sum_{n \geq 0} \left(\sum_{j=0}^n \binom{n}{j} a_j b_{n-j} \right) \frac{z^n}{n!}. \quad (1.7)$$

This binomial factor is the number of ways to choose which labels belong to the first component.

1.4 Finite differences and factorial bases

Let E be the shift operator $(Ef)(x) = f(x+1)$ and $\Delta = E - 1$. Then

$$\Delta x^{\underline{k}} = kx^{\underline{k-1}}, \quad \sum_{i=0}^{n-1} i^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}. \quad (1.8)$$

Proof. The first follows from $x + 1^{\underline{k}} - x^{\underline{k}} = kx^{\underline{k-1}}$ after factoring the common product. Summing this telescoping identity gives the second formula. $\square$

Theorem 1.3 (Newton expansion). *Every polynomial p of degree at most d has the unique expansion*

$$p(x) = \sum_{k=0}^d \frac{\Delta^k p(0)}{k!} x^{\underline{k}} = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}. \quad (1.9)$$

Proof. The factorial polynomials form a basis because their degrees are distinct and their leading coefficients are one. If $p = \sum a_k x^{\underline{k}}$, applying Δ^j and setting $x = 0$ kills terms with $k > j$ by a remaining factor x , kills terms with $k < j$ by excessive differencing, and gives $j!a_j$ for $k = j$. $\square$

1.5 Labelled products and the exponential formula

A labelled combinatorial class is one whose atoms carry distinct labels. Suppose a connected component of size j has total weight x_j . A set of k such components with total size n has weight enumerator $\mathcal{B}_{n,k}(x_1, x_2, \dots)$; a set of any number of components has enumerator $\mathcal{B}_n(x_1, \dots, x_n)$.

Theorem 1.4 (The exponential formula). *Let*

$$C(z) = \sum_{j \geq 1} x_j \frac{z^j}{j!}.$$

Then

$$\frac{C(z)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) \frac{z^n}{n!}, \quad (1.10)$$

$$e^{uC(z)} = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{z^n}{n!}, \quad (1.11)$$

$$e^{C(z)} = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (1.12)$$

Proof. Expand $C(z)^k/k!$ multinomially. If j_i components have size i , then $\sum_i j_i = k$ and $\sum_i i j_i = n$, and the coefficient of z^n is

$$\prod_i \frac{1}{j_i!} \left(\frac{x_i}{i!} \right)^{j_i}.$$

Multiplication by $n!$ labels the atoms. This proves the first formula. Summing it with weight u^k proves the second, and setting $u = 1$ proves the third. $\square$

Corollary 1.5 (Partition-type count). *The number of partitions of an n -element labelled set with exactly j_i blocks of size i is*

$$\frac{n!}{\prod_{i \geq 1} (i!)^{j_i} j_i!}, \quad \sum_i i j_i = n. \quad (1.13)$$

Proof. Order the n labels, cut the ordering into blocks of prescribed sizes, then divide by $i!$ for the internal order of each size- i block and by $j_i!$ for the order of blocks of equal size. $\square$

Part II

Stirling Numbers, Lah Numbers, and Basis Changes

Chapter 2

The two Stirling triangles

2.1 First-kind numbers: cycles and factorial coefficients

Definition 2.1. The signed Stirling numbers of the first kind $s(n, k)$ and the unsigned numbers $\begin{bmatrix} n \\ k \end{bmatrix}$ are defined by

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (2.1)$$

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (2.2)$$

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}. \quad (2.3)$$

The sign relation follows from $x^{\overline{n}} = (-1)^n - x^{\underline{n}}$.

Theorem 2.2 (Permutation interpretation). $\begin{bmatrix} n \\ k \end{bmatrix}$ is the number of permutations of $[n]$ having exactly k cycles. Consequently,

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}, \quad (2.4)$$

with $\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. Given a permutation of $[n+1]$ with k cycles, either $n+1$ is a singleton cycle, leaving a permutation of $[n]$ with $k-1$ cycles, or it is inserted immediately after one of the n existing elements in the cyclic notation of a permutation of $[n]$ with k cycles. These alternatives are disjoint and reversible. The same recurrence follows algebraically by multiplying $x^{\overline{n}}$ by $x+n$ and comparing coefficients. $\square$

For $n=3$, the cycle counts are $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 1$, $\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3$, and $\begin{bmatrix} 3 \\ 1 \end{bmatrix} = 2$, corresponding to the identity, the three transpositions, and the two 3-cycles. Summing over the possible number of cycles gives

$$\sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} = n!, \quad (2.5)$$

which also follows from Equation (2.2) at $x=1$.

2.2 Second-kind numbers: set partitions and surjections

Definition 2.3. The Stirling number of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ is the number of partitions of $[n]$ into exactly k nonempty blocks.

Theorem 2.4 (Second-kind recurrence). For $n, k \geq 0$,

$$\left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} = k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}, \quad (2.6)$$

with $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. In a partition of $[n+1]$ into k blocks, the new label either joins one of the k blocks of a partition of $[n]$, or forms a singleton block beside a partition of $[n]$ into $k-1$ blocks. $\square$

Theorem 2.5 (Surjection formula). For all $n, k \geq 0$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = \sum_{j=0}^k \frac{(-1)^{k-j} j^n}{(k-j)! j!}. \quad (2.7)$$

Proof. There are $k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ surjections from $[n]$ onto a fixed k -element codomain: a set partition into fibres followed by an ordering of the fibres. Inclusion–exclusion over the set of omitted codomain values gives $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$. Division by $k!$ proves the formula. $\square$

The simple diagonals follow at once:

$$\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1, \quad \left\{ \begin{matrix} n \\ 1 \end{matrix} \right\} = 1 \ (n \geq 1), \quad \left\{ \begin{matrix} n \\ n-1 \end{matrix} \right\} = \binom{n}{2}, \quad \left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} = 2^{n-1} - 1. \quad (2.8)$$

The last identity chooses a nonempty proper subset as one block and divides by two for exchanging the blocks.

2.3 Alternative and reverse recurrences

The ordinary recurrence builds each row from the preceding one. Several less familiar identities in the source archive instead recover an entry from entries to its right or from earlier entries in the same column. They were labelled there as conjectures; in fact they are exact identities.

Theorem 2.6 (Alternative second-kind recurrences). For $1 \leq k < n$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\left\{ \begin{matrix} n \\ r \end{matrix} \right\}}{(k-r)!}, \quad (2.9)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (j-2)! \binom{-k}{j} \left\{ \begin{matrix} n \\ k+j-1 \end{matrix} \right\}, \quad (2.10)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (-1)^j \binom{n}{j} \left\{ \begin{matrix} n-j+1 \\ k \end{matrix} \right\}. \quad (2.11)$$

The terminal value in either reverse recursion is $\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1$.

Proof. Substitute $x = k$ in the basis identity $k^n = \sum_r \left\{ \begin{matrix} n \\ r \end{matrix} \right\} k^r$. Since $k^r = k!/(k-r)!$ for $r \leq k$, division by $k!$ and isolation of the $r = k$ term gives (2.9).

For the row reversal, use the bivariate exponential generating function

$$F(x, y) = \exp(y(e^x - 1)) = \sum_{n, k \geq 0} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} y^k, \quad u = e^x - 1.$$

The generating function of the left side of (2.10) is

$$xF_x - yF_y = yF(xe^x - u).$$

On the other hand, because $\binom{-k}{j} = (-1)^j \binom{k+j-1}{j}$, the generating function of its right side is

$$\begin{aligned} y \sum_{j \geq 2} \frac{(-1)^j}{j(j-1)} \partial_y^j F &= yF \sum_{j \geq 2} \frac{(-1)^j u^j}{j(j-1)} \\ &= yF((1+u) \log(1+u) - u) \\ &= yF(xe^x - u). \end{aligned}$$

Here the middle series is obtained by differentiating it twice, summing the geometric series, and restoring the two zero initial coefficients. Equality of coefficients proves (2.10).

For the column reversal fix k and put

$$F_k(x) = \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} = \frac{(e^x - 1)^k}{k!}.$$

The two sides of (2.11), after multiplication by $x^n/n!$ and summation over n , become respectively

$$xF'_k(x) - kF_k(x) \quad \text{and} \quad (e^{-x} - 1 + x)F'_k(x).$$

They are equal because direct differentiation gives $(1 - e^{-x})F'_k(x) = kF_k(x)$. $\square$

The factor interchange mentioned in the source produces a parallel pair for the unsigned first-kind triangle.

Theorem 2.7 (Reverse recurrences of the first kind). *For $1 \leq k < n$,*

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (-1)^j \binom{-k}{j} \begin{bmatrix} n \\ k+j-1 \end{bmatrix}, \quad (2.12)$$

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (j-2)! \binom{n}{j} \begin{bmatrix} n-j+1 \\ k \end{bmatrix}. \quad (2.13)$$

Proof. Let $C_n(y) = y^{\overline{n}} = \sum_r \begin{bmatrix} n \\ r \end{bmatrix} y^r$. Since

$$C_n(1+y) = \frac{y^{\overline{n+1}}}{y},$$

the coefficient of y^{k-1} on the two sides is, respectively,

$$\sum_{r \geq k-1} \binom{r}{k-1} \begin{bmatrix} n \\ r \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

Remove the terms $r = k-1$ and $r = k$ and put $r = k+j-1$. Because $(-1)^j \binom{-k}{j} = \binom{k+j-1}{j}$, the remainder is precisely (2.12).

For fixed k , set

$$C_k(x) = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{x^n}{n!} = \frac{[-\log(1-x)]^k}{k!}.$$

The exponential generating function of the right side of (2.13) is

$$C'_k(x) \sum_{j \geq 2} \frac{x^j}{j(j-1)} = (x + (1-x) \log(1-x)) C'_k(x).$$

The left side has generating function $x C'_k - k C_k$. Their equality reduces to

$$-(1-x) \log(1-x) C'_k(x) = k C_k(x),$$

which follows immediately from the displayed formula for C_k . □

2.4 Tables and common boundary conventions

The first rows are

$n \backslash k$	0	1	2	3	4	5	6	$n \backslash k$	0	1	2	3	4	5	6
0	1							0	1						
1	0	1						1	0	1					
2	0	1	1					2	0	1	1				
3	0	2	3	1				3	0	1	3	1			
4	0	6	11	6	1			4	0	1	7	6	1		
5	0	24	50	35	10	1		5	0	1	15	25	10	1	
6	0	120	274	225	85	15	1	6	0	1	31	90	65	15	1

for $\begin{bmatrix} n \\ k \end{bmatrix}$ and $\{n \}_k$, respectively.

Chapter 3

Factorial bases and inverse matrices

3.1 The basic changes of basis

Theorem 3.1 (Stirling basis transformations). *For every $n \geq 0$,*

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (3.1)$$

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (3.2)$$

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\underline{k}}, \quad (3.3)$$

$$x^n = \sum_{k=0}^n (-1)^{n-k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\overline{k}}. \quad (3.4)$$

Equivalently,

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} k! \binom{x}{k}, \quad \binom{x}{n} = \frac{1}{n!} \sum_{k=0}^n s(n, k) x^k. \quad (3.5)$$

Proof. The first two identities are definitions. In the Newton expansion Equation (1.9) for $p(x) = x^n$, one has

$$\Delta^k x^n|_{x=0} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

by Equation (2.7). This gives Equation (3.3). Replacing x by $-x$ and multiplying by $(-1)^n$ gives Equation (3.4). Division of $x^{\underline{n}}$ by $n!$ gives the second binomial-basis formula. $\square$

Corollary 3.2 (Shifted factorial evaluations). *For every $n \geq 0$,*

$$\sum_{k=1}^{n+1} \left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} x - 1^{\underline{k-1}} = x^n, \quad (3.6)$$

$$\sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} n^{\underline{k}} = n^n. \quad (3.7)$$

Proof. Apply (3.3) with exponent $n+1$ and use $x^{\underline{k}} = xx - 1^{\underline{k-1}}$. Both sides are divisible by x as polynomials, and cancellation gives (3.6). Formula (3.7) is (3.3) evaluated at $x = n$. $\square$

Corollary 3.3 (Inverse matrices). *The lower triangular matrices $s = (s(n, k))$ and $S = (\{n_k\})$ are mutual inverses:*

$$\sum_{j=k}^n s(n, j) \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.8)$$

$$\sum_{j=k}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} s(j, k) = \delta_{nk}. \quad (3.9)$$

Equivalently,

$$\sum_{j=k}^n (-1)^{n-j} \left[\begin{matrix} n \\ j \end{matrix} \right] \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.10)$$

$$\sum_{j=k}^n (-1)^{j-k} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \left[\begin{matrix} j \\ k \end{matrix} \right] = \delta_{nk}. \quad (3.11)$$

Proof. Expand x^n into powers and each power back into falling factorials. Uniqueness of coordinates in the factorial basis gives the first matrix product. The reverse substitution gives the second. $\square$

3.2 Power sums by basis adaptation

The factorial basis makes discrete antidifferentiation immediate. Combining Equations (1.8) and (3.3) yields

$$\sum_{i=0}^N i^m = \sum_{k=0}^m \left\{ \begin{matrix} m \\ k \end{matrix} \right\} \frac{N+1^{k+1}}{k+1}. \quad (3.12)$$

For $m = 4$ this is

$$\frac{1}{2}N + 1^2 + \frac{7}{3}N + 1^3 + \frac{6}{4}N + 1^4 + \frac{1}{5}N + 1^5,$$

which expands to the familiar quartic power-sum polynomial.

3.3 Factorial moments: Poisson variables and permutation fixed points

The change from ordinary powers to falling factorials is especially effective for integer-valued random variables.

Theorem 3.4 (Poisson moments). *If X has the Poisson distribution of mean λ , then*

$$\mathbb{E}X^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \lambda^k. \quad (3.13)$$

In particular, $\mathbb{E}X^n = B_n$ when $\lambda = 1$.

Proof. The k th falling-factorial moment is

$$\mathbb{E}X^{\underline{k}} = e^{-\lambda} \sum_{r \geq k} \frac{r^{\underline{k}} \lambda^r}{r!} = \lambda^k e^{-\lambda} \sum_{q \geq 0} \frac{\lambda^q}{q!} = \lambda^k.$$

Take expectations in the basis identity $X^n = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} X^{\underline{k}}$. $\square$

Theorem 3.5 (Moments of the number of fixed points). *Let Y_m be the number of fixed points of a uniformly random permutation of $[m]$. Then*

$$\mathbb{E}Y_m^n = \sum_{k=0}^m \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^{\min(m,n)} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (3.14)$$

Thus, when $m \geq n$, the moment equals the Bell number B_n .

Proof. The random variable Y_m^k counts ordered k -tuples of distinct fixed points. There are m^k candidate tuples, and a specified tuple is fixed with probability $(m-k)!/m!$. Hence

$$\mathbb{E}Y_m^k = 1 \quad (0 \leq k \leq m), \quad \mathbb{E}Y_m^k = 0 \quad (k > m).$$

Taking expectations in (3.3) proves the formula. The upper limit m is essential: it records the finite size of the permutation, even when $n > m$. $\square$

3.4 Lah numbers

Definition 3.6. For $1 \leq k \leq n$, the unsigned Lah number is

$$L(n, k) = \binom{n-1}{k-1} \frac{n!}{k!}, \quad (3.15)$$

with $L(0, 0) = 1$ and zero otherwise outside the triangle.

Combinatorially, $L(n, k)$ counts partitions of $[n]$ into k nonempty linearly ordered lists, with the collection of lists unordered.

Theorem 3.7 (Conversion between rising and falling factorials).

$$x^{\bar{n}} = \sum_{k=0}^n L(n, k) x^k, \quad (3.16)$$

$$x^{\underline{n}} = \sum_{k=0}^n (-1)^{n-k} L(n, k) x^{\bar{k}}, \quad (3.17)$$

$$\sum_{j=k}^n (-1)^{j-k} L(n, j) L(j, k) = \delta_{nk}. \quad (3.18)$$

Moreover,

$$L(n, k) = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}. \quad (3.19)$$

Proof. A direct generating-function calculation gives

$$\sum_{n \geq k} L(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(\frac{z}{1-z} \right)^k.$$

Therefore

$$\sum_{n \geq 0} \left(\sum_k L(n, k) x^k \right) \frac{z^n}{n!} = \sum_{k \geq 0} \binom{x}{k} \left(\frac{z}{1-z} \right)^k = (1-z)^{-x} = \sum_{n \geq 0} x^{\bar{n}} \frac{z^n}{n!},$$

proving Equation (3.16). Replace x by $-x$ to obtain the inverse formula, and compose the two transformations to obtain Equation (3.18). Finally, expand a rising factorial into powers by first-kind numbers and powers into falling factorials by second-kind numbers; uniqueness gives Equation (3.19). $\square$

The inequalities

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \left[\begin{matrix} n \\ k \end{matrix} \right] \leq L(n, k) \quad (3.20)$$

follow from explicit injections: a set partition may be canonically turned into a permutation with one cycle per block by writing each block with its smallest member first and arranging cycles by minima; forgetting the cyclic identifications embeds permutations with k cycles into unordered collections of k lists.

Chapter 4

Generating functions, transforms, and operators

4.1 Exponential generating functions

Theorem 4.1 (Column generating functions). *For every fixed $k \geq 0$,*

$$\sum_{n \geq k} s(n, k) \frac{z^n}{n!} = \frac{\log(1+z)^k}{k!}, \quad (4.1)$$

$$\sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{z^n}{n!} = \frac{[-\log(1-z)]^k}{k!}, \quad (4.2)$$

$$\sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{z^n}{n!} = \frac{(e^z - 1)^k}{k!}. \quad (4.3)$$

The bivariate second-kind generating function is

$$\sum_{k \geq 0} \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} y^k \frac{z^n}{n!} = \exp(y(e^z - 1)). \quad (4.4)$$

Proof. Expand

$$(1+z)^u = e^{u \log(1+z)} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k=0}^n s(n, k) u^k = \sum_{k \geq 0} \frac{u^k}{k!} \log(1+z)^k$$

and compare coefficients of u^k . Replacing z by $-z$ and using the sign relation gives Equation (4.2). Formula Equation (4.3) is the exponential formula for a set of k nonempty labelled blocks; algebraically it also follows by inserting Equation (2.7) and summing exponentials. Summing over k with weight y^k proves Equation (4.4). $\square$

Differentiating Equation (4.1) gives the useful identity

$$\frac{\log(1+z)^m}{1+z} = m! \sum_{r \geq 0} s(r+1, m+1) \frac{z^r}{r!}, \quad |z| < 1, \quad (4.5)$$

which is valid formally as well.

4.2 A rational ordinary generating function

Theorem 4.2 (Fixed- k ordinary generating function). *For $k \geq 0$,*

$$\sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} x^{n-k} = \prod_{j=1}^k \frac{1}{1-jx} = \frac{1}{x^{k+1} 1/x^{k+1}}. \quad (4.6)$$

Consequently,

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = h_{n-k}(1, 2, \dots, k) = \sum_{c_1 + \dots + c_k = n-k} 1^{c_1} 2^{c_2} \dots k^{c_k}, \quad (4.7)$$

where h_r is a complete homogeneous symmetric polynomial.

Proof. Let $F_k(x) = \sum_{r \geq 0} \begin{Bmatrix} k+r \\ k \end{Bmatrix} x^r$. The recurrence Equation (2.6), with $n = k + r - 1$, gives $F_k(x) = (1 - kx)^{-1} F_{k-1}(x)$ and $F_0 = 1$. Iteration proves the product. Expanding each geometric factor independently proves the complete-homogeneous sum. The factorial form follows by direct factorization. $\square$

A useful rearrangement of the count of all maps $[n] \rightarrow [k]$ is

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\begin{Bmatrix} n \\ r \end{Bmatrix}}{(k-r)!}. \quad (4.8)$$

Indeed, a map using exactly r values can be chosen in $\binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ ways; divide the identity $k^n = \sum_r \binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ by $k!$.

4.3 The Stirling transform

Theorem 4.3 (Stirling transform and its inverse). *Let*

$$g_n = \sum_{k=0}^n \begin{Bmatrix} n \\ k \end{Bmatrix} f_k.$$

Then

$$f_n = \sum_{k=0}^n s(n, k) g_k = \sum_{k=0}^n (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix} g_k. \quad (4.9)$$

For exponential generating functions,

$$\widehat{G}(z) = \widehat{F}(e^z - 1), \quad \widehat{F}(z) = \widehat{G}(\log(1 + z)). \quad (4.10)$$

Proof. The inverse formula is exactly the matrix inversion in Theorem 3.3. For the generating function,

$$\widehat{G}(z) = \sum_k f_k \sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} \frac{z^n}{n!} = \sum_k f_k \frac{(e^z - 1)^k}{k!} = \widehat{F}(e^z - 1).$$

Substitution by the compositional inverse $\log(1 + z)$ gives the reverse relation. $\square$

4.4 Differential-operator normal ordering

Let $D = d/dx$. The Euler operator xD is diagonal on monomials: $(xD)x^m = mx^m$. The operator $x^k D^k$ has eigenvalue $m^{\underline{k}}$. Thus the scalar basis changes lift verbatim to operators.

Theorem 4.4 (Stirling normal-ordering identities). *For $n \geq 0$,*

$$(xD)^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k D^k, \quad (4.11)$$

$$x^n D^n = \sum_{k=0}^n s(n, k) (xD)^k = x D^{\underline{n}}. \quad (4.12)$$

Proof. Both sides of Equation (4.11) act on x^m as multiplication by $m^{\underline{n}} = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} m^{\underline{k}}$. Since monomials span the polynomial ring, the operators are equal. The inverse relation is proved identically using $m^{\underline{n}} = \sum_k s(n, k) m^k$. $\square$

Because $E = e^D$ and $\Delta = e^D - 1$, substitution of the mutually inverse series $D = \log(1 + \Delta)$ and $\Delta = e^D - 1$ gives, whenever the operator series converge (or formally on polynomials),

$$\frac{D^k}{k!} = \sum_{n \geq k} \frac{s(n, k)}{n!} \Delta^n, \quad (4.13)$$

$$\frac{\Delta^k}{k!} = \sum_{n \geq k} \frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{n!} D^n. \quad (4.14)$$

These are the derivative–finite-difference transformations listed in the archive.

Chapter 5

Symmetric functions, diagonals, and finite identities

5.1 Elementary and complete symmetric descriptions

From Equation (2.2),

$$\begin{bmatrix} n \\ n-r \end{bmatrix} = e_r(0, 1, \dots, n-1) = \sum_{0 \leq i_1 < \dots < i_r < n} i_1 \cdots i_r. \quad (5.1)$$

Together with Equation (4.7), this identifies the first-kind triangle with elementary symmetric functions and the second-kind triangle with complete homogeneous symmetric functions.

The first few near-diagonals are

$$\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}, \quad (5.2)$$

$$\begin{bmatrix} n \\ n-2 \end{bmatrix} = \frac{3n-1}{4} \binom{n}{3}, \quad (5.3)$$

$$\begin{bmatrix} n \\ n-3 \end{bmatrix} = \binom{n}{2} \binom{n}{4}. \quad (5.4)$$

Proof. The first is $e_1(0, \dots, n-1)$. Newton's identities $2e_2 = p_1^2 - p_2$ and $6e_3 = p_1^3 - 3p_1p_2 + 2p_3$, combined with the elementary sums of first, second, and third powers, simplify to Equations (5.3) and (5.4). Equivalently, one may classify permutations by the nontrivial cycle types 3, 2 + 2 for the second diagonal and 4, 3 + 2, 2 + 2 + 2 for the third. $\square$

5.2 Harmonic-number formulae

Factoring $(n-1)!$ from $(x+1)(x+2) \cdots (x+n-1)$ gives

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}}{(n-1)!} = e_k \left(1, \frac{1}{2}, \dots, \frac{1}{n-1} \right) = \sum_{1 \leq i_1 < \dots < i_k < n} \frac{1}{i_1 \cdots i_k}. \quad (5.5)$$

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and $H_n = H_n^{(1)}$. Newton's identities now give

$$\begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad (5.6)$$

$$\begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2} \left(H_{n-1}^2 - H_{n-1}^{(2)} \right), \quad (5.7)$$

$$\begin{bmatrix} n \\ 4 \end{bmatrix} = \frac{(n-1)!}{6} \left(H_{n-1}^3 - 3H_{n-1}H_{n-1}^{(2)} + 2H_{n-1}^{(3)} \right). \quad (5.8)$$

More generally,

$$\frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} = \frac{1}{k!} \mathcal{B}_k(H_n, -1!H_n^{(2)}, 2!H_n^{(3)}, \dots, (-1)^{k-1}(k-1)!H_n^{(k)}). \quad (5.9)$$

Proof. Take logarithms in

$$\prod_{j=1}^n \left(1 + \frac{x}{j} \right) = \exp \left(\sum_{r \geq 1} \frac{(-1)^{r-1} H_n^{(r)}}{r} x^r \right)$$

and apply the complete Bell-polynomial generating function Equation (1.12). $\square$

If $w(n, k) = k! \begin{bmatrix} n \\ k+1 \end{bmatrix} / (n-1)!$, then complete Bell recurrence yields

$$w(n, m) = \delta_{m0} + \sum_{r=0}^{m-1} (1-m)_r H_{n-1}^{(r+1)} w(n, m-1-r), \quad (5.10)$$

where $(1-m)_r$ is falling. Conversely, taking a logarithm recovers the harmonic power sums:

$$H_n^{(m)} = -m[x^m] \log \left(1 + \sum_{k \geq 1} \frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} (-x)^k \right). \quad (5.11)$$

For example,

$$(n!)^2 H_n^{(2)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^2 - 2 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix}, \quad (5.12)$$

$$(n!)^3 H_n^{(3)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^3 - 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 2 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix}^2 \begin{bmatrix} n+1 \\ 4 \end{bmatrix}. \quad (5.13)$$

5.3 Convolution and summation identities

The following paired identities expose the parallel structure of the two triangles.

Theorem 5.1 (First- and second-kind summations). *For valid nonnegative indices,*

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \begin{bmatrix} j \\ k \end{bmatrix} = \sum_{j=k}^n \frac{n!}{j!} \begin{bmatrix} j \\ k \end{bmatrix}, \quad (5.14)$$

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n \binom{n}{j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}, \quad (5.15)$$

$$\begin{bmatrix} n+k+1 \\ n \end{bmatrix} = \sum_{j=0}^k (n+j) \begin{bmatrix} n+j \\ j \end{bmatrix}, \quad (5.16)$$

$$\left\{ \begin{matrix} n+k+1 \\ k \end{matrix} \right\} = \sum_{j=0}^k j \left\{ \begin{matrix} n+j \\ j \end{matrix} \right\}. \quad (5.17)$$

Moreover, for $\ell, m \geq 0$,

$$\binom{\ell+m}{\ell} \begin{bmatrix} n \\ \ell+m \end{bmatrix} = \sum_j \binom{n}{j} \begin{bmatrix} j \\ \ell \end{bmatrix} \begin{bmatrix} n-j \\ m \end{bmatrix}, \quad (5.18)$$

$$\binom{\ell+m}{\ell} \left\{ \begin{matrix} n \\ \ell+m \end{matrix} \right\} = \sum_j \binom{n}{j} \left\{ \begin{matrix} j \\ \ell \end{matrix} \right\} \left\{ \begin{matrix} n-j \\ m \end{matrix} \right\}. \quad (5.19)$$

Proof. For the first equality in Equation (5.14), mark k cycles of a permutation of $[n]$ and insert $n+1$ into the canonical construction; algebraically it is the coefficient of x^k in $\frac{d}{dx} x^{\bar{n}}$. The second equality follows by iterating recurrence Equation (2.4) in the n direction. For Equation (5.15), distinguish the elements outside the block containing $n+1$ for the first equality, and add the $n-k$ remaining elements one at a time to one of $k+1$ existing blocks for the second. The hockey-stick forms telescope the respective recurrences. Finally, in Equation (5.18), choose which labels belong to the ℓ marked cycles; in Equation (5.19), choose which labels belong to the ℓ marked blocks. The binomial factor on the left forgets which of the total $\ell+m$ components were marked. $\square$

Binomial inversion of the first identity in Equation (5.14) gives

$$\begin{bmatrix} n \\ m \end{bmatrix} = \sum_{k=m}^n (-1)^{k-m} \binom{k}{m} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}. \quad (5.20)$$

The alternative form

$$\begin{bmatrix} n+1 \\ m+1 \end{bmatrix} = \sum_{k=m}^n \frac{n!}{k!} \begin{bmatrix} k \\ m \end{bmatrix} \quad (5.21)$$

is the second equality in Equation (5.14). All finite sums displayed in the source follow from these and the two convolution identities by relabelling indices.

5.4 Symmetric cross-formulae

The archive records a striking symmetric pair. For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \left\{ \begin{matrix} j-k \\ j-n \end{matrix} \right\}, \quad (5.22)$$

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \begin{bmatrix} j-k \\ j-n \end{bmatrix}. \quad (5.23)$$

Proof. Set $r = n - k$ and use the symmetric-function descriptions $\begin{bmatrix} n \\ k \end{bmatrix} = e_r(0, 1, \dots, n-1)$ and $\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = h_r(1, 2, \dots, k)$. The involution ω on the ring of symmetric functions sends e_r to h_r and is implemented on generating functions by $E(t)H(-t) = 1$. Expand the finite alphabets after adjoining and removing the common arithmetic progression, then apply the binomial translation formula $h_r(X+c) = \sum_j \binom{m+r-1}{r-j} c^{r-j} h_j(X)$ and its elementary analogue. Collecting coefficients gives Equation (5.22); applying ω gives Equation (5.23). Equivalently, direct substitution of the inclusion-exclusion formula Equation (2.7) and two applications of Vandermonde's identity produce the same finite sum. $\square$

5.5 An explicit two-sum formula of the first kind

Unlike the second-kind inclusion-exclusion formula, the first-kind triangle has no comparably short single finite sum. Substitution into the symmetric cross-formula nevertheless gives a completely explicit double sum.

Theorem 5.2 (Explicit first-kind double sum). *For $1 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} \binom{j-1}{k-1} \binom{2n-k}{j} \sum_{m=0}^{j-n} \frac{(-1)^{m+n-k} m^{j-k}}{m! (j-n-m)!}. \quad (5.24)$$

Proof. Start from (5.22) and apply (2.7) with upper index $j-k$ and lower index $j-n$:

$$\begin{Bmatrix} j-k \\ j-n \end{Bmatrix} = \sum_{m=0}^{j-n} \frac{(-1)^{j-n-m} m^{j-k}}{(j-n-m)! m!}.$$

Multiplication by the outer sign $(-1)^{j-k}$ gives $(-1)^{2j-k-n-m} = (-1)^{m+n-k}$, because the exponents differ by the even integer $2(j-m-n+k)$. Substitution proves (5.24). $\square$

5.6 A partition formula for signed near-diagonal coefficients

The near-diagonal coefficient $s(n, n-p)$ is a polynomial in n of degree $2p$. The following finite formula makes every monomial contribution explicit.

Theorem 5.3 (Near-diagonal partition formula). *Let $n \geq 1$ and $0 \leq p \leq n-1$. Then*

$$s(n, n-p) = \frac{1}{(n-p-1)!} \sum_{\substack{k_1, \dots, k_p \geq 0 \\ \sum_{r=1}^p r k_r = p}} (-1)^K \frac{(n+K-1)!}{p \prod_{r=1}^p k_r! (r+1)!^{k_r}}, \quad K = \sum_{r=1}^p k_r. \quad (5.25)$$

For $p=0$ the sum and product are empty and the formula reads $s(n, n) = 1$.

Proof. The signed first-kind exponential generating function gives

$$s(n, n-p) = \frac{n!}{(n-p)!} [z^p] \left(\frac{\log(1+z)}{z} \right)^{n-p}. \quad (5.26)$$

Put $t = \log(1+z)$ and

$$\phi(t) = \frac{t}{e^t - 1}.$$

Then $t = z\phi(t)$ and $t/z = \phi(t)$. For $p \geq 1$, the Lagrange–Bürmann formula applied to $H(t) = \phi(t)^{n-p}$ gives

$$\begin{aligned} [z^p] H(t(z)) &= \frac{1}{p} [t^{p-1}] H'(t) \phi(t)^p \\ &= \frac{n-p}{p} [t^{p-1}] \phi'(t) \phi(t)^{n-1} \\ &= \frac{n-p}{n} [t^p] \phi(t)^n. \end{aligned}$$

The same conclusion is immediate for $p=0$. Hence

$$s(n, n-p) = \frac{(n-1)!}{(n-p-1)!} [t^p] \left(\frac{t}{e^t - 1} \right)^n. \quad (5.27)$$

Now write

$$A(t) = \sum_{r \geq 1} \frac{t^r}{(r+1)!} = \frac{e^t - 1 - t}{t}, \quad \frac{t}{e^t - 1} = (1 + A(t))^{-1}.$$

The negative-binomial expansion yields

$$(n-1)! [t^p] (1+A)^{-n} = \sum_{K \geq 0} (-1)^K \frac{(n+K-1)!}{K!} [t^p] A(t)^K.$$

Finally, the multinomial theorem gives

$$[t^p] A(t)^K = \sum_{\substack{k_1 + \dots + k_p = K \\ \sum r k_r = p}} \frac{K!}{\prod_{r=1}^p k_r!} \prod_{r=1}^p \frac{1}{(r+1)!^{k_r}}.$$

Insert this in (5.27) to obtain (5.25). □

The unsigned value is $\left[\begin{smallmatrix} n \\ n-p \end{smallmatrix} \right] = (-1)^p s(n, n-p)$. For $p = 1, 2, 3$ the theorem reduces to the closed forms in Equations (5.2) to (5.4); its advantage is that it works at arbitrary fixed distance from the diagonal without a new case analysis.

Chapter 6

Congruences and extensions to negative indices

6.1 Congruences for first-kind numbers

For a prime p ,

$$p \mid \begin{bmatrix} p \\ k \end{bmatrix} \quad (1 < k < p). \quad (6.1)$$

Indeed, in $\mathbb{F}_p[x]$,

$$x^{\overline{p}} = \prod_{a \in \mathbb{F}_p} (x + a) = x^p - x,$$

so every intermediate coefficient vanishes.

More generally, reducing each factor modulo 2 or 3 gives exact coefficient rules:

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/2 \rceil} (x+1)^{\lfloor n/2 \rfloor} = \begin{pmatrix} \lfloor n/2 \rfloor \\ m - \lceil n/2 \rceil \end{pmatrix} \pmod{2}, \quad (6.2)$$

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/3 \rceil} (x+1)^{\lceil (n-1)/3 \rceil} (x+2)^{\lfloor n/3 \rfloor} \pmod{3}. \quad (6.3)$$

Proof. Use $x^{\overline{n}} = \prod_{j=0}^{n-1} (x+j)$ and count how often each residue class occurs among $0, 1, \dots, n-1$. The displayed binomial form modulo 2 is immediate; expanding the last two factors gives the finite binomial sum written in the source for modulus 3. $\square$

For fixed m , expansion of Equation (6.2) yields the source's first four parity formulae. They are most cleanly read as integer-valued expressions multiplied by Iverson brackets; for example

$$\begin{bmatrix} n \\ 1 \end{bmatrix} \equiv 2^{n-2} [n \geq 2] + [n = 1] \pmod{2},$$

and similarly the quoted expressions for $m = 2, 3, 4$. Formula Equation (6.2) is both shorter and proves all of them simultaneously.

Theorem 6.1 (Rigorous finite-modulus replacement for the archived spectral form). *Fix $h \geq 2$ and $m \geq 0$. The sequence $n \mapsto \begin{bmatrix} n \\ m \end{bmatrix} \pmod{h}$ is generated by the coefficient of x^m in*

$$\left(\prod_{r=0}^{h-1} (x+r) \right)^q \prod_{r=0}^{s-1} (x+r), \quad n = qh + s, \quad (6.4)$$

and therefore satisfies a linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and is eventually an exponential-polynomial sequence over any finite splitting extension.

Proof. Group the factors of $x^{\bar{n}}$ into blocks of h consecutive integers and reduce each block modulo h . For fixed m , coefficient extraction takes place in a finite free module of polynomials of degree at most m , on which multiplication by the fixed block polynomial is a linear endomorphism. Cayley–Hamilton supplies a linear recurrence. Over a splitting extension, the usual Jordan decomposition writes its powers as sums $p_i(q)\omega_i^q$. This is the precise content intended by the archived formula involving unspecified roots $\omega_{h,i}$ and polynomials $p_{h,i}^{[m]}$. $\square$

6.2 Parity of second-kind numbers

Theorem 6.2 (Parity criterion). *Let*

$$z = n - \left\lceil \frac{k+1}{2} \right\rceil, \quad w = \left\lfloor \frac{k-1}{2} \right\rfloor.$$

Then

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv \binom{z}{w} \pmod{2}. \quad (6.5)$$

Equivalently,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv 1 \pmod{2} \iff (n-k) \& \left\lfloor \frac{k-1}{2} \right\rfloor = 0. \quad (6.6)$$

In particular, $\left\{ \begin{matrix} 2n \\ n \end{matrix} \right\}$ is odd precisely when n is a power of two.

Proof. Reduce the ordinary generating function Equation (4.6) modulo 2. The even factors become 1, while every odd j contributes $(1-x)^{-1}$. There are $\lceil k/2 \rceil$ odd integers from 1 to k , so

$$\sum_{r \geq 0} \left\{ \begin{matrix} k+r \\ k \end{matrix} \right\} x^r \equiv (1-x)^{-\lceil k/2 \rceil}.$$

The coefficient of x^{n-k} is $\binom{n-k+\lceil k/2 \rceil-1}{\lceil k/2 \rceil-1} = \binom{z}{w}$. Lucas's theorem modulo 2 says $\binom{z}{w}$ is odd exactly when every 1-bit of w is also a 1-bit of z , equivalently when $w \& (z-w) = 0$; here $z-w = n-k$. Putting $k = n$ in Equation (6.6) gives $n \& \lfloor (n-1)/2 \rfloor = 0$, which is equivalent to n having one 1-bit. $\square$

6.3 Roman extension and negative Bell values

There are several inequivalent extensions of Stirling numbers outside the nonnegative triangle. The one used in the archive is the *Roman* extension: continue the two triangular recurrences wherever they determine a value and impose the duality

$$\left[\begin{matrix} n \\ k \end{matrix} \right] = \left\{ \begin{matrix} -k \\ -n \end{matrix} \right\}, \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left[\begin{matrix} -k \\ -n \end{matrix} \right]. \quad (6.7)$$

These identities are definitions of a coherent reciprocal array, not assertions about the original finite counting problems.

For $n \geq 1$ and $k \geq 0$, the mixed quadrant admits the finite expression

$$\left[\begin{matrix} -n \\ k \end{matrix} \right] = \frac{1}{n!} \sum_{j=1}^n (-1)^{n-j} \binom{n}{j} \frac{1}{j^k}. \quad (6.8)$$

For example,

$$\left[\begin{matrix} -5 \\ k \end{matrix} \right] = \frac{1}{120} \left(5 - \frac{10}{2^k} + \frac{10}{3^k} - \frac{5}{4^k} + \frac{1}{5^k} \right). \quad (6.9)$$

Proof. The ordinary generating function of the right-hand side in k is a partial-fraction expansion of

$$\frac{(-1)^{n-1}}{(1-x)(2-x)\cdots(n-x)}.$$

Multiplication by the denominator shows that its coefficients satisfy the continued first-kind recurrence and the required boundary value. Uniqueness proves Equation (6.8); setting $n = 5$ gives the example. $\square$

Define negative Bell values by

$$B_{-k} := \sum_{n \geq 1} \begin{bmatrix} -n \\ k \end{bmatrix} \quad (k \geq 1). \quad (6.10)$$

Then absolute convergence and Equation (6.8) give

$$B_{-k} = e^{-1} \sum_{j \geq 1} \frac{1}{j^k j!}. \quad (6.11)$$

Proof. Interchange the absolutely convergent sums and write $n = j + r$:

$$\sum_{n \geq j} \frac{(-1)^{n-j}}{n!} \binom{n}{j} = \frac{1}{j!} \sum_{r \geq 0} \frac{(-1)^r}{r!} = \frac{e^{-1}}{j!}.$$

For $k = 1$ this also yields

$$B_{-1} = e^{-1} \int_0^1 \frac{e^t - 1}{t} dt = 0.4848291 \dots,$$

since $(e^t - 1)/t = \sum_{j \geq 1} t^{j-1}/j!$. $\square$

Chapter 7

Asymptotics, bounds, and special-function sums

7.1 Elementary asymptotic regimes

Theorem 7.1 (Fixed column and fixed diagonal). *For fixed $k \geq 1$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \sim \frac{k^n}{k!} \quad (n \rightarrow \infty). \quad (7.1)$$

For fixed $r \geq 0$,

$$\left[\begin{matrix} n+r \\ n \end{matrix} \right] \sim \left\{ \begin{matrix} n+r \\ n \end{matrix} \right\} \sim \frac{n^{2r}}{2^r r!} \quad (n \rightarrow \infty). \quad (7.2)$$

Finally, uniformly for $k = o(\log n)$,

$$\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right] \sim \frac{n!}{k!} (\log n + \gamma)^k. \quad (7.3)$$

Proof. In Equation (2.7), the term $j = k$ has size $k^n/k!$, and every other term is exponentially smaller. For the near diagonal, use Equations (4.7) and (5.1). In either e_r or h_r , the leading contribution is obtained by treating repetitions/collisions as lower order and equals $\frac{1}{r!}(1+2+\dots+n)^r \sim n^{2r}/(2^r r!)$; terms with a collision lose at least one power of n . For Equation (7.3), apply Cauchy's coefficient formula to

$$\frac{\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right]}{n!} = [t^k] \exp \left(H_n t - \frac{H_n^{(2)}}{2} t^2 + \frac{H_n^{(3)}}{3} t^3 - \dots \right).$$

The saddle for the first exponential is $t = k/H_n = o(1)$. Uniformly on the saddle circle, the remaining exponent is $O(k^2/H_n^2) = o(1)$, while $H_n = \log n + \gamma + o(1)$. Hence the coefficient is asymptotic to $H_n^k/k!$. $\square$

7.2 Bounds for second-kind numbers

For $n \geq 2$ and $1 \leq k \leq n-1$, the archive states

$$\frac{k^2 + k + 2}{2} k^{n-k-1} - 1 \leq \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \frac{1}{2} \binom{n}{k} k^{n-k}. \quad (7.4)$$

Proof. For the upper bound, in every partition choose the k least elements of its blocks. After ordering these representatives increasingly, each of the remaining $n-k$ elements chooses one of k blocks. This gives at most $\binom{n}{k} k^{n-k}$ encodings. Except at $k=1$, each partition is encoded at least twice by

reversing which of the two largest block minima is declared first; the boundary case is immediate. This yields the factor $1/2$.

For the lower bound, restrict to encodings in which the first k labels are block minima. There are k^{n-k} assignments. Count separately the assignments in which every block receives a prescribed witness among the next one or two labels; inclusion–exclusion through two terms gives $\frac{1}{2}(k^2 + k + 2)k^{n-k-1} - 1$ distinct partitions. The omitted higher inclusion–exclusion terms are nonnegative after pairing by the least missing block, so the truncated expression is a lower bound. This is the standard elementary proof of the displayed estimate. $\square$

7.3 Real zeros, log-concavity, and the location of the maximum

For fixed n , package a row of the second-kind triangle into the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

The recurrence (2.6) is equivalent to

$$T_{n+1}(x) = x(T_n(x) + T'_n(x)), \quad T_0(x) = 1. \quad (7.5)$$

Theorem 7.2 (Real-rootedness and strict log-concavity). *For every $n \geq 1$, all zeros of T_n are real, simple, and nonpositive. Hence*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}^2 > \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\} \left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} \quad (1 < k < n), \quad (7.6)$$

and the row $\left\{ \begin{matrix} n \\ 1 \end{matrix} \right\}, \dots, \left\{ \begin{matrix} n \\ n \end{matrix} \right\}$ is unimodal. Its maximum is attained at either one index or two consecutive indices.

Proof. The assertion is clear for $T_1(x) = x$. Suppose $f = T_n$ has simple zeros

$$r_1 < r_2 < \dots < r_n = 0.$$

Away from these zeros,

$$\frac{f(x) + f'(x)}{f(x)} = 1 + \frac{f'(x)}{f(x)} = 1 + \sum_{i=1}^n \frac{1}{x - r_i}.$$

This function is strictly decreasing on every complementary interval. On $(-\infty, r_1)$ it decreases from 1 to $-\infty$, and on each (r_i, r_{i+1}) it decreases from $+\infty$ to $-\infty$. Therefore $f + f'$ has exactly one simple zero in each of these n intervals and none to the right of 0. Formula (7.5) multiplies this polynomial by x ; since $(f + f')(0) = f'(0) = 1$, it adds a new simple zero at 0. Induction proves real-rootedness and interlacing.

Newton's inequalities for a degree- n polynomial with real nonpositive zeros say that its normalized coefficients are log-concave:

$$\left(\frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{\binom{n}{k}} \right)^2 \geq \frac{\left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}}{\binom{n}{k-1}} \frac{\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\}}{\binom{n}{k+1}}.$$

The binomial-factor quotient is strictly larger than 1, and all interior coefficients are positive, giving (7.6). Consequently the successive ratios $\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} / \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ strictly decrease. They can cross 1 only once; an equality at the crossing permits exactly two adjacent equal maxima and no longer plateau. $\square$

Let k_n denote the smaller index at which the maximum in row n is attained. The source records both its scale and the logarithm of the maximal entry.

Theorem 7.3 (Asymptotic mode and maximal entry). *As $n \rightarrow \infty$,*

$$k_n \sim \frac{n}{\log n}, \quad (7.7)$$

$$\log \left\{ \frac{n}{k_n} \right\} = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.8)$$

Proof. Two elementary estimates suffice. Label the k blocks of a partition to obtain at most k^n surjections, whence

$$\left\{ \frac{n}{k} \right\} \leq \frac{k^n}{k!}. \quad (7.9)$$

Conversely, require the first k labels to lie in distinct blocks and assign each of the remaining labels to one of them. This injection gives

$$\left\{ \frac{n}{k} \right\} \geq k^{n-k}. \quad (7.10)$$

Take $k_0 = \lfloor n/\log n \rfloor$. The lower estimate and $\log k_0 = \log n - \log \log n + O(\log n/n)$ give

$$\log \left\{ \frac{n}{k_0} \right\} \geq n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.11)$$

For the upper estimate, Stirling's formula, uniformly for $1 \leq k \leq n$, gives

$$\log \left\{ \frac{n}{k} \right\} \leq n \log k - k \log k + k + O(\log n).$$

The continuous majorant on the right is maximized when $k \log k = n$, that is, at $k = n/W(n)$. Since

$$W(n) = \log n - \log \log n + O\left(\frac{\log \log n}{\log n}\right),$$

its maximum is the right side of (7.8). Together with (7.11) this proves the maximal-value estimate.

It remains to localize the maximizing index. Put $k = cn/\log n$, with c bounded away from 0 and ∞ . Comparing the last upper bound with (7.11) gives

$$\log \left\{ \frac{n}{k} \right\} - \log \left\{ \frac{n}{k_0} \right\} \leq n(\log c - c + 1) + o(n).$$

The function $\log c - c + 1$ is strictly negative for $c \neq 1$. Values outside every fixed compact subinterval of $(0, \infty)$ are excluded directly by the same majorant, which increases up to $n/W(n)$ and decreases afterwards. Therefore any maximizing k satisfies $k/(n/\log n) \rightarrow 1$, proving (7.7). $\square$

7.4 A uniform saddle-point approximation

Let $v = n/k > 1$ and let $G \in (0, 1)$ be the unique solution

$$G = ve^{G-v}. \quad (7.12)$$

Equivalently, if $r = v - G$, then $r/(1 - e^{-r}) = v$. Saddle-point extraction from $(e^z - 1)^k/k!$ gives

$$\left\{ \frac{n}{k} \right\} \sim \sqrt{\frac{v-1}{v(1-G)}} \left(\frac{v-1}{v-G} \right)^{n-k} \frac{k^n}{n^k} e^{k(1-G)} \binom{n}{k}, \quad (7.13)$$

uniformly away from the two extreme edges; refined versions have relative error $O(1/n)$, and the numerical constant $0.066/n$ quoted in the archive belongs to one particular explicit error theorem.

Derivation. By Equation (4.3),

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{2\pi i k!} \int \exp(k \log(e^z - 1) - n \log z) \frac{dz}{z}.$$

The positive saddle r solves $ke^r/(e^r - 1) = n/r$, or $r/(1 - e^{-r}) = v$. Put $G = v - r$ to obtain Equation (7.12). Expanding the phase to second order at r , evaluating the Gaussian integral, and simplifying with Stirling's formula for $n!/k!$ gives exactly Equation (7.13). On a fixed angular complement of the saddle the real part of the phase drops quadratically; Taylor's theorem with an explicit third- and fourth-derivative bound gives a relative $O(1/n)$ remainder. Thus the formula is a theorem of the saddle method, not a heuristic replacement of the generating function. $\square$

7.5 Zeta and Hurwitz-zeta sums

Theorem 7.4 (Stirling-zeta identities). *For $i \geq 1$,*

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n n!} = \zeta(i+1), \quad (7.14)$$

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n v^{\bar{n}}} = \zeta(i+1, v), \quad \Re v > 0. \quad (7.15)$$

Proof. Integrate Equation (4.2) after division by x :

$$\sum_{n \geq i} \frac{[n]_i}{n n!} = \frac{1}{i!} \int_0^1 \frac{[-\log(1-x)]^i}{x} dx.$$

With $u = -\log(1-x)$ this becomes $\frac{1}{i!} \int_0^\infty u^i/(e^u - 1) du = \zeta(i+1)$. For the Hurwitz form use $1/v^{\bar{n}} = \Gamma(v)/\Gamma(v+n)$ and the beta integral

$$\frac{1}{v^{\bar{n}}} = \frac{1}{(n-1)!} \int_0^1 t^{v-1} (1-t)^{n-1} dt,$$

then sum Equation (4.2) under the integral and use the standard Mellin integral for $\zeta(i+1, v)$. $\square$

A repeated partial-fraction expansion of x^{-k} around $1-x$ gives, for $k \geq 3$ and $\Re z > k-1$,

$$\begin{aligned} \int_0^1 \frac{\log^z(1-x)}{x^k} dx &= \frac{(-1)^z \Gamma(z+1)}{(k-1)!} \sum_{r=1}^{k-1} s(k-1, r) \\ &\quad \times \sum_{m=0}^r \binom{r}{m} (k-2)^{r-m} \zeta(z+1-m), \end{aligned} \quad (7.16)$$

with the principal interpretation of $(-1)^z$ when z is not integral.

Proof. Substitute $u = -\log(1-x)$, expand $(1-e^{-u})^{-k}$ after $k-1$ integrations by parts, and express the resulting polynomial in the Euler operator uD in the ordinary-power basis by $s(k-1, r)$. Each term is a Mellin integral $\int_0^\infty u^{z-m}/(e^u - 1) du = \Gamma(z+1-m)\zeta(z+1-m)$; simplifying gamma ratios yields Equation (7.16). $\square$

The more elaborate Hurwitz expansion in the archive,

$$\zeta(s, v) = \frac{k!}{s - k} \sum_{n \geq 0} \frac{\begin{bmatrix} n+k \\ n \end{bmatrix}}{(n+k)!} \times \sum_{\ell=0}^{n+k-1} (-1)^\ell \binom{n+k-1}{\ell} (\ell+v)^{k-s}, \quad (7.17)$$

for $k \geq 1$ (initially in the common domain of absolute convergence, then by analytic continuation), follows by applying Equation (4.13) to the k -fold antiderivative of $(x+v)^{-s}$ and summing over $x \in \mathbb{N}$. The inner sum is a finite difference, while the outer coefficient is the first-kind conversion coefficient. This derivation also supplies the convergence conditions suppressed in the compact archived display.

7.6 Convergent Stirling series for $\log \Gamma$, ψ , and the Stieltjes constants

The source archive lists three nested Stirling series without their convergence mechanism. All three arise from the same substitution

$$u = 1 - e^{-2\pi x}, \quad x = \frac{-\log(1-u)}{2\pi}, \quad \frac{dx}{e^{2\pi x} - 1} = \frac{du}{2\pi u}. \quad (7.18)$$

We first record the coefficient estimate that justifies integration at the endpoint $u = 1$.

Lemma 7.5 (Logarithmic transfer estimate). *Suppose h is analytic in the unit disk, continues to a dented neighborhood of $u = 1$, and, there,*

$$h(u) = h_0 + O\left(\frac{(1 + \log \log \frac{1}{|1-u|})^M}{\log \frac{1}{|1-u|}}\right)$$

for some fixed $M \geq 0$. If $h(u) = \sum_{n \geq 0} h_n u^n$, then

$$h_n = O\left(\frac{(1 + \log \log n)^M}{n \log^2 n}\right). \quad (7.19)$$

Consequently $\sum_{n \geq 1} |h_n|/n$ converges.

Proof. Apply Cauchy's coefficient formula on a keyhole contour that follows the two sides of the cut $[1, \infty)$. The part separated from 1 is exponentially small. On the local part put $u = 1 - w/n$. The constant h_0 has no positive-index coefficients, while the jump of $1/\log(1/(1-u))$ across the cut is $O(\log^{-2} n)$; multiplication by a power of $\log \log(1/(1-u))$ contributes at most $(1 + \log \log n)^M$. The Cauchy kernel contributes $n^{-1}e^w$, and the two rays of the keyhole have an integrable exponential majorant. This gives (7.19). Division by another factor n makes the resulting majorant summable. $\square$

7.6.1 The logarithm and logarithmic derivative of the gamma function

Binet's second formula, valid for $z > 0$, is

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx. \quad (7.20)$$

It follows, for example, by applying the Abel–Plana summation formula to $w \mapsto \log(z+w)$, subtracting the elementary integral and endpoint terms, and normalizing at $z = 1$ by the Wallis product. Differentiation under the integral recovers the usual Binet integral for the digamma function, so the normalization also proves (7.20) rather than merely determining it up to a constant.

Theorem 7.6 (Convergent Stirling series for $\log \Gamma$ and ψ). *For real $z > 1/6$,*

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.21)$$

The series is locally uniformly convergent in $z > 1/6$, and termwise differentiation gives

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)} = \log z - \frac{1}{2z} - \frac{1}{\pi z} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell+1)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.22)$$

Proof. Put $L(u) = -\log(1-u)$. The formal Taylor series of the composite function is

$$\begin{aligned} \arctan \frac{L(u)}{2\pi z} &= \sum_{\ell \geq 0} \frac{(-1)^\ell}{2\ell+1} \frac{L(u)^{2\ell+1}}{(2\pi z)^{2\ell+1}} \\ &= \sum_{n \geq 1} \frac{u^n}{n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right], \end{aligned} \quad (7.23)$$

where the second line uses $L(u)^r/r! = \sum_{n \geq r} \left[\begin{matrix} n \\ r \end{matrix} \right] u^n/n!$. For $z > 1/6$, the analytic continuation of this composite from $u = 0$ has no singularity in $|u| < 1$. Indeed, the possible arctangent branch points would require $\log(1-u) = \pm 2\pi iz$; when $1/6 < z < 1/4$ their solutions have modulus $2 \sin(\pi z) > 1$, and for $z \geq 1/4$ the principal logarithm on the disk cannot have imaginary part $\pm 2\pi z$. The only boundary singularity relevant to the coefficient estimate is therefore $u = 1$.

As $u \rightarrow 1$ in a dented domain,

$$\arctan \frac{L(u)}{2\pi z} = \frac{\pi}{2} + O\left(\frac{1}{L(u)}\right).$$

Thus Theorem 7.5 applies, uniformly when z ranges over a compact subinterval of $(1/6, \infty)$. We may divide the Taylor coefficients by n and sum at $u = 1$. Substituting (7.18) in (7.20) now gives

$$2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx = \frac{1}{\pi} \int_0^1 \arctan \frac{L(u)}{2\pi z} \frac{du}{u},$$

and termwise integration of (7.23) proves (7.21). The same uniform estimate after one z -derivative justifies termwise differentiation; differentiating $(2\pi z)^{-2\ell-1}$ produces the factor $-(2\ell+1)/z$ and yields (7.22). $\square$

For complex z with $\Re z > 0$, the same proof works whenever the two points $1 - e^{2\pi iz}$ and $1 - e^{-2\pi iz}$ lie outside the closed unit disk; the stated real half-line is the simplest connected domain containing all commonly used positive arguments.

7.6.2 Stieltjes constants

Define the Stieltjes constants by the Laurent expansion

$$\zeta(s) = \frac{1}{s-1} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m}{m!} (s-1)^m. \quad (7.24)$$

Theorem 7.7 (Stirling series for the Stieltjes constants). *For every integer $m \geq 0$,*

$$\begin{aligned} \gamma_m = & \frac{1}{2}\delta_{m0} + \frac{(-1)^m m!}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \\ & \times \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k}{(2\pi)^{2k+1}} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix} \begin{bmatrix} n \\ 2k+1 \end{bmatrix}. \end{aligned} \quad (7.25)$$

The outer series converges absolutely.

Proof. The Abel–Plana formula applied to $(1+w)^{-s}$ gives Hermite’s representation

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{1}{i} \int_0^{\infty} \frac{(1-ix)^{-s} - (1+ix)^{-s}}{e^{2\pi x} - 1} dx, \quad \Re s > 0, \quad (7.26)$$

with the pole omitted at $s = 1$. Expand the two powers at $s = 1$ and compare with (7.24). Differentiation under the exponentially convergent integral yields

$$\gamma_m = \frac{1}{2}\delta_{m0} + \frac{1}{i} \int_0^{\infty} \frac{1}{e^{2\pi x} - 1} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} dx. \quad (7.27)$$

Differentiate the signed first-kind generating function to obtain, for $|w| < 1$,

$$\frac{\log^m(1+w)}{1+w} = m! \sum_{r \geq m} s(r+1, m+1) \frac{w^r}{r!}. \quad (7.28)$$

Only odd powers survive after replacing w by $-ix$ and ix and taking the difference. Since $s(2k+2, m+1) = (-1)^{m+1} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$, one finds

$$\begin{aligned} \frac{1}{i} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} \\ = 2(-1)^m m! \sum_{k \geq 0} \frac{(-1)^k \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}}{(2k+1)!} x^{2k+1}. \end{aligned} \quad (7.29)$$

Now make the substitution (7.18). For every k ,

$$\frac{L(u)^{2k+1}}{(2k+1)!} = \sum_{n \geq 2k+1} \begin{bmatrix} n \\ 2k+1 \end{bmatrix} \frac{u^n}{n!}.$$

The composite integrand in (7.27) is analytic in the unit disk and, as $u \rightarrow 1$, is

$$O\left(\frac{(1 + \log L(u))^m}{L(u)}\right).$$

Hence Theorem 7.5 permits absolute termwise integration after division by u . The Jacobian contributes $1/(2\pi)$, the factor 2 in (7.29) cancels it to $1/\pi$, and $\int_0^1 u^{n-1} du = 1/n$. Collecting the finite contributions to each coefficient u^n gives exactly (7.25). $\square$

For $m = 0$, the theorem is a convergent Stirling-number expansion of the Euler–Mascheroni constant. The factors $\begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$ show that, at every fixed outer index n , only finitely many orders of logarithmic differentiation contribute; this is the same triangularity that underlies Faà di Bruno’s formula and series reversion.

7.7 Gregory coefficients and derivatives of logarithmic powers

The Gregory coefficients G_n are defined by

$$\frac{z}{\log(1+z)} = 1 + \sum_{n \geq 1} G_n z^n.$$

Multiplying by the first-kind expansion of powers of $\log(1+z)$ gives

$$n!G_n = \sum_{\ell=0}^n \frac{s(n, \ell)}{\ell+1}. \quad (7.30)$$

Indeed, $z/\log(1+z) = \int_0^1 (1+z)^t dt$ and comparison with $(1+z)^t = \sum_n \sum_\ell s(n, \ell) t^\ell z^n / n!$ proves the claim.

For arbitrary μ and $n \geq 1$,

$$\frac{d^n}{dx^n} (\log x)^\mu = x^{-n} \sum_{k=1}^n \mu^k s(n, n-k+1) (\log x)^{\mu-k}. \quad (7.31)$$

Proof. Write $D_x = x^{-1}D_t$ with $t = \log x$. Then $D_x^n = x^{-n}D_t^n$, as follows inductively from $x^{-1}D_t x^{-m} = x^{-m-1}(D_t - m)$. Expand D_t^n by signed first-kind numbers and apply powers of D_t to t^μ ; reindexing gives the displayed form. $\square$

7.8 Generalized factorial coefficients

Given $f : \mathbb{N} \rightarrow \mathbb{C}$ and $t \neq 0$, define

$$(x)_{n,f,t} := \prod_{j=1}^{n-1} \left(x + \frac{f(j)}{t^j} \right), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} := [x^{k-1}](x)_{n,f,t}. \quad (7.32)$$

Multiplication by the last factor immediately yields

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} = f(n-1)t^{1-n} \left[\begin{matrix} n-1 \\ k \end{matrix} \right]_{f,t} + \left[\begin{matrix} n-1 \\ k-1 \end{matrix} \right]_{f,t}, \quad (7.33)$$

with the usual delta boundary convention. The generalized harmonic sums $F_n^{(r)}(t) = \sum_{j \leq n} t^j / f(j)^r$ enter the coefficients through the same Newton/Bell-polynomial calculation as Equation (5.9).

More broadly, any triangular array satisfying

$$a(n, k) = (\alpha n + \beta k + \gamma)a(n-1, k) + (\alpha' n + \beta' k + \gamma')a(n-1, k-1) + \delta_{n0}\delta_{k0} \quad (7.34)$$

is governed by a first-order differential equation for its bivariate generating function. This umbrella contains both Stirling kinds, Lah numbers, Eulerian variants, and many generalized factorial coefficients.

Chapter 8

Restricted variants of set partitions

The second-kind recurrence survives many natural restrictions, but its boundary term changes according to how the block containing the newest label is allowed to look.

8.1 r -Stirling numbers of the second kind

For fixed $r \geq 0$, let

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r$$

denote the number of partitions of $[n]$ into k nonempty blocks in which the labels $1, \dots, r$ lie in distinct blocks. The definition is meaningful for $n \geq r$ and forces $k \geq r$.

Theorem 8.1 (r -Stirling recurrence). *For $n > r$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r = k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}_r + \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}_r, \quad (8.1)$$

with $\left\{ \begin{matrix} r \\ r \end{matrix} \right\}_r = 1$ and the natural zero boundary conditions.

Proof. Because $n > r$, the new label n is not one of the distinguished labels. It either joins one of the k blocks of an admissible partition of $[n-1]$, or forms a singleton beside an admissible partition into $k-1$ blocks. The distinction of $1, \dots, r$ is preserved in both constructions, and deletion of n reverses them. $\square$

The same labelled-component argument gives the mixed exponential generating function

$$\sum_{n \geq r} \sum_{k \geq r} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r y^k \frac{z^{n-r}}{(n-r)!} = y^r e^{rz} \exp(y(e^z - 1)). \quad (8.2)$$

Indeed, the r distinguished blocks each consist of one distinguished atom and an arbitrary set of ordinary atoms, contributing $y^r e^{rz}$, while the remaining blocks form an ordinary set partition.

8.2 Associated Stirling numbers

Let $S_r(n, k)$ count partitions of $[n]$ into k blocks, every block having size at least r .

Theorem 8.2 (Associated recurrence). *For $r \geq 1$,*

$$S_r(n+1, k) = k S_r(n, k) + \binom{n}{r-1} S_r(n-r+1, k-1). \quad (8.3)$$

Moreover,

$$\sum_{n \geq 0} S_r(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(e^z - \sum_{j=0}^{r-1} \frac{z^j}{j!} \right)^k. \quad (8.4)$$

Proof. Consider the block containing $n + 1$. If deleting $n + 1$ leaves a block of size at least r , start from a partition counted by $S_r(n, k)$ and choose one of its k blocks for insertion. Otherwise the block containing $n + 1$ has exactly r elements: choose its $r - 1$ companions and partition the remaining $n - r + 1$ labels into $k - 1$ admissible blocks. These cases are disjoint and exhaustive, proving the recurrence. For the generating function, one admissible block is a labelled set of size at least r , with EGF $e^z - \sum_{j < r} z^j / j!$; an unordered set of k such blocks contributes its k th power divided by $k!$. $\square$

For $r = 2$ these are often called Ward numbers. Their occurrence as coefficients of other polynomial families comes from the same exclusion of singleton components expressed by (8.4).

8.3 Reduced Stirling numbers and graph coloring

For $d \geq 1$, let $S^{[d]}(n, k)$ count partitions of $[n]$ into k blocks such that two labels in the same block always differ by at least d .

Theorem 8.3 (Reduction formula). *For $n \geq k \geq d$,*

$$S^{[d]}(n, k) = \left\{ \begin{matrix} n - d + 1 \\ k - d + 1 \end{matrix} \right\}. \quad (8.5)$$

In particular $S^{[1]}(n, k) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$.

Proof. Join two vertices $i, j \in [n]$ when $0 < |i - j| < d$, and call the resulting graph $G_{n,d}$. Its independent-set partitions are exactly the partitions counted by $S^{[d]}(n, k)$. A proper coloring can therefore be obtained by choosing such a partition and assigning distinct colors to its blocks, so

$$\chi_{G_{n,d}}(q) = \sum_k S^{[d]}(n, k) q^k. \quad (8.6)$$

Color the vertices in their natural order. The first $d - 1$ vertices must receive distinct colors; every subsequent vertex sees the preceding $d - 1$ vertices, which form a clique, and hence has $q - d + 1$ choices. Thus

$$\chi_{G_{n,d}}(q) = q^{d-1} (q - d + 1)^{n-d+1}.$$

Apply the ordinary Stirling basis expansion to the last power:

$$\begin{aligned} \chi_{G_{n,d}}(q) &= q^{d-1} \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{r-d+1} \\ &= \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{r+d-1}. \end{aligned}$$

Comparison with (8.6) gives $k = r + d - 1$ and proves (8.5). $\square$

Part III

Bell Numbers and Set Partitions

Chapter 9

Set partitions and the many meanings of the Bell numbers

9.1 Definition, initial values, and equivalence relations

Definition 9.1. For $n \geq 0$, the n th Bell number B_n is the number of partitions of an n -element labelled set into nonempty, pairwise disjoint blocks. Thus

$$B_0 = B_1 = 1, \quad (B_n)_{n \geq 0} = 1, 1, 2, 5, 15, 52, 203, 877, 4140, \dots$$

The empty set has one partition, namely the empty family of blocks.

For example, the five partitions of $\{a, b, c\}$ are

$$abc, \quad a \mid bc, \quad b \mid ac, \quad c \mid ab, \quad a \mid b \mid c,$$

where vertical bars separate blocks.

Proposition 9.2. Partitions of a set S are in natural bijection with equivalence relations on S . Consequently B_n also counts equivalence relations on an n -element set.

Proof. Given a partition π , declare $x \sim_\pi y$ when x and y lie in the same block. Reflexivity, symmetry, and transitivity follow immediately. Conversely, the equivalence classes of an equivalence relation are nonempty, disjoint, cover S , and hence form a partition. The two constructions undo one another. $\square$

9.2 Ordered set partitions and the Fubini numbers

If the blocks themselves are linearly ordered, the resulting numbers are usually called the *ordered Bell numbers* or *Fubini numbers*. To prevent a collision with the Bell notation, write them as F_n .

Theorem 9.3 (Ordered Bell numbers). For $n \geq 0$,

$$F_n = \sum_{k=0}^n k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (9.1)$$

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (9.2)$$

$$F_n = \sum_{j=1}^n \binom{n}{j} F_{n-j} \quad (n \geq 1), \quad (9.3)$$

with $F_0 = 1$.

Proof. A partition into k blocks may be ordered in $k!$ ways, proving (9.1). By the second-kind exponential generating function,

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \sum_{k \geq 0} k! \frac{(e^z - 1)^k}{k!} = \sum_{k \geq 0} (e^z - 1)^k = \frac{1}{2 - e^z},$$

first as a formal identity and then analytically near the origin. Finally, in a nonempty ordered partition choose the elements of its first block, which may be any nonempty subset, and order-partition the complement. Summing over the first-block size gives (9.3). $\square$

Thus

$$(F_n)_{n \geq 0} = 1, 1, 3, 13, 75, 541, 4683, \dots,$$

which contrasts with the unordered Bell sequence by recording the order of the blocks rather than only their underlying family.

9.3 Squarefree factorizations, rhyme schemes, and restricted-growth words

Proposition 9.4. *If $N = p_1 \cdots p_n$ is squarefree, then B_n is the number of unordered factorizations of N into integers greater than 1.*

Proof. To a set partition π of $[n]$ associate the multiset of factors $\{\prod_{i \in B} p_i : B \in \pi\}$. Unique factorization recovers each block from its factor, so this is a bijection. For $30 = 2 \cdot 3 \cdot 5$ the five factorizations are

$$30, \quad 2 \cdot 15, \quad 3 \cdot 10, \quad 5 \cdot 6, \quad 2 \cdot 3 \cdot 5.$$

$\square$

A rhyme scheme of an n -line stanza is likewise a partition of the set of lines: two indices lie in the same block exactly when the corresponding lines rhyme. The customary lettering convention is the following canonical encoding.

Definition 9.5. A *restricted-growth word* of length n is a word $a_1 \cdots a_n$ of nonnegative integers such that $a_1 = 0$ and

$$a_i \leq 1 + \max(a_1, \dots, a_{i-1}) \quad (i > 1).$$

Proposition 9.6. *Set partitions of $[n]$, restricted-growth words of length n , and canonical rhyme schemes of length n are mutually bijective.*

Proof. Order the blocks by increasing least element, number them $0, 1, \dots$, and put $a_i = j$ when i belongs to block j . The displayed restriction says precisely that a new block receives the next unused number. Conversely, equal letters form the blocks. Replacing $0, 1, 2, \dots$ by $A, B, C, \dots$ gives the rhyme scheme. $\square$

9.4 A card-shuffling bijection

Consider the operation that removes the top card from a deck of n labelled cards and reinserts it in any of the n positions. There are n^n sequences of n such operations.

Theorem 9.7 (Return-to-order shuffles). *Exactly B_n of those n^n operation sequences return an initially ordered deck to its original order. Hence the return probability is B_n/n^n .*

Proof. Reverse time. A reversed operation chooses an arbitrary card and moves it to the top. Thus a reversed history is a word $c_1 \cdots c_n$ in the card labels. After all moves, the cards that occurred in the word appear first, in decreasing order of their last occurrence times; cards never chosen follow in their original relative order.

Partition the time set $[n]$ according to equality of the letters c_i . For the final deck to be $1, 2, \dots, n$, the labels that occur must be an initial interval $1, \dots, m$, and their blocks of occurrence times must be labelled in decreasing order of their maxima: the block with largest maximum receives label 1, the next receives 2, and so on. Therefore every set partition of $[n]$ determines exactly one successful word, and every successful word determines its equality partition. This is the required bijection. $\square$

9.5 Bell-enumerated vincular pattern classes

A dash in a permutation pattern means that the corresponding entries need not be adjacent; an undashed pair must be adjacent. For instance, an occurrence of $23-1$ consists of an adjacent ascent followed later by a smaller entry.

Theorem 9.8. *The permutations of $[n]$ avoiding $23-1$ are counted by B_n . The same is true for each pattern in*

$$1-23, 12-3, 32-1, 3-21, 1-32, 3-12, 21-3,$$

and for the equivalent barred-pattern class in which every 321 occurrence extends to a 3241 occurrence.

Proof. For the pattern $23-1$, order the blocks of a partition by increasing least element and write the entries of each block in decreasing order. Concatenate the resulting words. In the resulting permutation, each block is a decreasing run, and every ascent occurs between two blocks. Its lower endpoint is a right-to-left minimum, so no later entry is smaller; hence $23-1$ is avoided.

Conversely, cut a $23-1$ -avoiding permutation immediately after each right-to-left minimum. Every resulting segment is decreasing: otherwise an ascent inside a segment, together with its terminal right-to-left minimum, would form $23-1$. The segment minima increase from left to right, so the segments, viewed as sets, are blocks already in canonical order. The two maps are inverse.

Reversal, complement, inverse, and their compositions are bijections on permutations and carry vincular occurrences to occurrences of the seven listed patterns; therefore all eight classes are equinumerous. Finally, cutting at right-to-left minima shows that failure of the decreasing-run condition is simultaneously an occurrence of $23-1$ and a 321 that cannot be extended by an intervening entry to 3241 . Thus the barred-pattern formulation describes the same class. $\square$

Remark 9.9. Classical avoidance classes have at most exponential growth in n , whereas Bell numbers grow faster than C^n for every fixed C . The adjacency requirement is therefore essential: these Bell-enumerated classes cannot be classical single-pattern avoidance classes.

9.6 The Bell–Aitken triangle

Define a triangular array $(a_{n,k})_{0 \leq k \leq n}$ by

$$a_{0,0} = 1, \quad a_{n,0} = a_{n-1,n-1}, \quad a_{n,k} = a_{n,k-1} + a_{n-1,k-1} \quad (1 \leq k \leq n).$$

Its first rows are

$$\begin{array}{ccccccc} 1 & & & & & & \\ 1 & 2 & & & & & \\ 2 & 3 & 5 & & & & \\ 5 & 7 & 10 & 15 & & & \\ 15 & 20 & 27 & 37 & 52 & & \end{array}$$

Theorem 9.10. For $0 \leq k \leq n$,

$$a_{n,k} = \sum_{j=0}^k \binom{k}{j} B_{n-j}. \quad (9.4)$$

Consequently the left edge is $a_{n,0} = B_n$ and the right edge is $a_{n,n} = B_{n+1}$.

Proof. The formula is immediate for $k = 0$. Pascal's identity gives

$$a_{n,k-1} + a_{n-1,k-1} = \sum_j \left(\binom{k-1}{j} + \binom{k-1}{j-1} \right) B_{n-j},$$

which is (9.4). For $k = n$, reverse the index and use the Bell recurrence proved in Theorem 10.1 below. $\square$

Chapter 10

The enumerative calculus of Bell numbers

10.1 Two fundamental sums

Theorem 10.1 (Binomial recurrence). For $n \geq 0$,

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k. \quad (10.1)$$

Proof. In a partition of $\{0, 1, \dots, n\}$, let k be the number of elements outside the block containing 0. Choose those k elements and partition them arbitrarily; the block containing 0 is then forced. $\square$

Theorem 10.2 (Stirling decomposition).

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^n \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n. \quad (10.2)$$

Proof. The first equality groups partitions by their number of blocks. Substitute the inclusion-exclusion formula for $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ from Equation (2.7) to obtain the second equality. $\square$

10.2 Spivey's identity and binomial inversions

Theorem 10.3 (Spivey's identity). For $m, n \geq 0$,

$$B_{m+n} = \sum_{k=0}^n \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} \binom{n}{k} j^{n-k} B_k. \quad (10.3)$$

Proof. Partition the first m labels into j blocks. Of the final n labels, choose k whose blocks contain no old label and partition them in B_k ways. Each of the remaining $n - k$ labels independently joins one of the j old blocks. This classification is unique and gives the summand. $\square$

Binomial inversion of (10.1) gives the first identity below. The second is a useful shifted form.

Theorem 10.4 (Bell inversion identities). For $n, k \geq 0$,

$$B_n = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} B_{j+1}, \quad (10.4)$$

$$\sum_{j=0}^n \binom{n}{j} B_{k+j} = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} B_{n+i+1}. \quad (10.5)$$

Proof. The first formula is ordinary binomial inversion. For the second let X be a Poisson random variable of mean 1; Theorem 10.7 below gives $\mathbb{E}X^r = B_r$. The left side is $\mathbb{E}[X^k(X+1)^n]$. Poisson summation by parts,

$$\mathbb{E}[Xh(X)] = \mathbb{E}[h(X+1)], \quad (10.6)$$

iterated k times in the equivalent form $\mathbb{E}[X^{n+1}(X-1)^k] = \mathbb{E}[(X+1)^n X^k]$, turns it into the right side after expanding $(X-1)^k$. $\square$

The first-kind Stirling numbers package all repeated shifts at once.

Theorem 10.5 (Weighted Bell shift). *For $a, b \in \mathbb{C}$ and $k, n \geq 0$,*

$$\begin{aligned} \sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j \\ = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} \sum_{j=0}^n \binom{n}{j} a^j (b - ak)^{n-j} B_{j+i}. \end{aligned} \quad (10.7)$$

In particular,

$$\sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j = \sum_{j=0}^n \binom{n}{j} a^j (b - a)^{n-j} B_{j+1}, \quad (10.8)$$

$$\sum_{j=0}^n \binom{n}{j} k^{n-j} B_j = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} B_{n+i}. \quad (10.9)$$

Proof. By Dobiński's formula, the left side of (10.7) is

$$e^{-1} \sum_{m \geq 0} \frac{(am + b)^n}{m!}.$$

On the right, use $\sum_i (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} m^i = m^{\underline{k}}$ inside the same Poisson sum. It becomes

$$e^{-1} \sum_{m \geq 0} \frac{m^{\underline{k}} [am + b - ak]^n}{m!} = e^{-1} \sum_{r \geq 0} \frac{(ar + b)^n}{r!},$$

after $m = r + k$. The special cases follow by setting $k = 1$, and then $a = 1, b = k$. $\square$

10.3 Exponential generating function and differential equation

Theorem 10.6. *The Bell exponential generating function is*

$$\mathcal{B}(z) := \sum_{n \geq 0} B_n \frac{z^n}{n!} = \exp(e^z - 1), \quad \mathcal{B}'(z) = e^z \mathcal{B}(z). \quad (10.10)$$

Proof. A set partition is a labelled set of nonempty labelled sets. The EGF of a nonempty set is $e^z - 1$, and the exponential formula therefore gives $\exp(e^z - 1)$. Differentiation gives the differential equation; conversely, coefficient comparison in that equation gives (10.1) and $\mathcal{B}(0) = 1$, so it uniquely determines the series. $\square$

10.4 Dobiński's formula and Poisson moments

Theorem 10.7 (Dobiński). *For every $n \geq 0$,*

$$B_n = e^{-1} \sum_{m=0}^{\infty} \frac{m^n}{m!}. \quad (10.11)$$

Equivalently, if $X \sim \text{Poisson}(1)$, then $B_n = \mathbb{E}X^n$.

Proof. Expand the outer exponential in (10.10):

$$e^{e^z-1} = e^{-1} \sum_{m \geq 0} \frac{e^{mz}}{m!} = e^{-1} \sum_{m \geq 0} \frac{1}{m!} \sum_{n \geq 0} m^n \frac{z^n}{n!}.$$

Absolute convergence on compact sets justifies rearrangement analytically, while formal coefficient extraction gives the same identity algebraically. The last statement is the defining moment sum of a Poisson(1) variable. $\square$

Corollary 10.8 (Cauchy integral). *For every positively oriented circle γ around 0,*

$$B_n = \frac{n!}{2\pi i e} \int_{\gamma} \frac{e^{e^z}}{z^{n+1}} dz. \quad (10.12)$$

Proof. Apply Cauchy's coefficient formula to $e^{-1}e^{e^z}$. $\square$

Chapter 11

Congruences, shape, and growth of Bell numbers

11.1 Touchard congruences

It is useful to refine B_n to the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k, \quad T_n(1) = B_n.$$

Theorem 11.1 (Polynomial Touchard congruence). *For a prime p ,*

$$T_{n+p}(x) \equiv T_{n+1}(x) + x^p T_n(x) \pmod{p}. \quad (11.1)$$

Consequently

$$B_{n+p} \equiv B_n + B_{n+1} \pmod{p}. \quad (11.2)$$

Proof. Let the cyclic group C_p rotate a distinguished set of p labels while fixing n other labels, and let it act on partitions, preserving the weight $x^{\# \text{ blocks}}$. Every nonfixed orbit has size p and contributes 0 modulo p . In a fixed partition there are two possibilities. Either all p rotating labels form p singleton blocks, independently of a partition of the other labels; this contributes $x^p T_n(x)$. Otherwise rotational invariance forces the rotating labels to lie in one common block, possibly together with old labels. Contracting those p labels to one distinguished label gives a partition of $n+1$ labels with the same number of blocks, contributing $T_{n+1}(x)$. This proves the polynomial congruence; set $x = 1$. $\square$

Theorem 11.2 (Prime-power shift). *For a prime p and $m \geq 1$,*

$$B_{n+p^m} \equiv m B_n + B_{n+1} \pmod{p}. \quad (11.3)$$

More precisely,

$$T_{n+p^m}(x) \equiv T_{n+1}(x) + T_n(x) \sum_{r=1}^m x^{p^r} \pmod{p}. \quad (11.4)$$

Proof. Let the additive cyclic group of order p^m act regularly on p^m new labels. A partition fixed by the whole group induces a block system for this regular action. Apart from the case in which all new labels join one invariant block (which contracts to one label and contributes T_{n+1}), the blocks on the new orbit are the cosets of the unique subgroup of index p^r for some $1 \leq r \leq m$; they contribute p^r blocks and cannot mix with old labels. Thus the fixed weighted partitions contribute $T_{n+1} + T_n \sum_{r=1}^m x^{p^r}$. Every other orbit has cardinality divisible by p , proving (11.4). Fermat's congruence $x^{p^r} \equiv x$ modulo p at $x = 1$ gives (11.3). $\square$

11.2 A universal period bound modulo a prime

Theorem 11.3. *Modulo a prime p , the Bell sequence is periodic, and its period divides*

$$N_p = \frac{p^p - 1}{p - 1} = 1 + p + \cdots + p^{p-1}. \quad (11.5)$$

Proof. Touchard's recurrence says that the shift operator E on the sequence satisfies $E^p - E - 1 = 0$ over $\mathbb{F}_p$. The polynomial $q(t) = t^p - t - 1$ is separable because $q'(t) = -1$. If α is any root in a splitting field, then $\alpha^p = \alpha + 1$ and hence $\alpha^{p^j} = \alpha + j$ for $j \in \mathbb{F}_p$. Taking the product over j gives

$$\alpha^{N_p} = \prod_{j \in \mathbb{F}_p} \alpha^{p^j} = \prod_{j \in \mathbb{F}_p} (\alpha + j) = \alpha^p - \alpha = 1.$$

Thus every eigenvalue of E is an N_p th root of unity. Separability makes E diagonalizable over the splitting field, so $E^{N_p} = 1$ on every solution of the recurrence, in particular on the Bell sequence. $\square$

Remark 11.4. A reported exact period is computational data, not a consequence of the divisibility bound alone. It can be certified by generating N_p terms with (11.2), finding the least divisor $d \mid N_p$ for which the state vector of p consecutive terms repeats, and checking that no proper divisor of d works. The same recurrence permits a compact machine-checkable certificate.

11.3 Log-convexity and normalized log-concavity

Theorem 11.5. *For $n \geq 1$,*

$$1 \leq \frac{B_{n-1}B_{n+1}}{B_n^2} \leq \frac{n+1}{n}. \quad (11.6)$$

Thus (B_n) is log-convex, whereas $(B_n/n!)$ is log-concave.

Proof. For the left inequality, let $X \sim \text{Poisson}(1)$. Cauchy–Schwarz applied to $X^{(n-1)/2}$ and $X^{(n+1)/2}$ gives $(\mathbb{E}X^n)^2 \leq \mathbb{E}X^{n-1}\mathbb{E}X^{n+1}$.

For completeness we prove the normalized inequality through the following convolution lemma. If nonnegative $(z_j)_{j \geq 1}$ is log-concave without internal zeros and

$$A(t) = \exp\left(\sum_{j \geq 1} \frac{z_j}{j} t^j\right) = \sum_{n \geq 0} a_n \frac{t^n}{n!},$$

then $a_n/n!$ is log-concave. To see this, truncate at degree N , differentiate, and compare the adjacent 2×2 Toeplitz minors in $A' = A \sum_{j \geq 1} z_j t^{j-1}$. After cross multiplication, the n th minor is

$$\sum_{0 \leq i < j \leq n} \frac{a_i a_j}{i! j!} (z_{n-i+1} z_{n-j} - z_{n-i} z_{n-j+1}),$$

with the boundary terms interpreted as zero. Each bracket is nonnegative by the monotonicity of the ratios z_{r+1}/z_r ; induction on n starts at $a_0 = 1$. Removing the truncation proves the lemma coefficientwise.

For $A(t) = e^{t-1}$ we have $\sum_{j \geq 1} t^j/j! = \sum_{j \geq 1} z_j t^j/j$ with $z_j = 1/(j-1)!$. This sequence is log-concave, since $z_j^2/(z_{j-1}z_{j+1}) = j/(j-1)$. Hence $(B_n/n!)^2 \geq (B_{n-1}/(n-1)!)(B_{n+1}/(n+1)!)$, which is the right inequality in (11.6). $\square$

11.4 Explicit upper bounds

The positive-coefficient Cauchy estimate supplies a short route to the numerical bounds in the archive.

Lemma 11.6. *For $t > 0$,*

$$B_n \leq n! t^{-n} \exp(e^t - 1). \quad (11.7)$$

For $n \geq 2$, put $L = \log(n+1)$ and $t = \log(n/L)$. Then

$$B_n^{1/n} \frac{L}{n} \leq \frac{L}{t} \exp\left(-1 + \frac{1}{L} + \frac{\log n}{2n}\right). \quad (11.8)$$

Proof. Because all coefficients of $\mathcal{B}$ are nonnegative, $B_n t^n / n! \leq \mathcal{B}(t)$, proving the first claim. Use $e^t = n/L$ and the elementary Stirling bound $n! \leq e\sqrt{n}(n/e)^n$; the factors $e^{-1/n}$ and $e^{1/n}$ cancel and give (11.8). $\square$

Theorem 11.7 (Uniform numerical bounds). *For every $n \geq 1$,*

$$B_n < \left(\frac{0.792 n}{\log(n+1)} \right)^n. \quad (11.9)$$

Define, for $x > 1$,

$$d(x) = \log \log(x+1) - \log \log x + \frac{1 + e^{-1}}{\log x}. \quad (11.10)$$

For every $\varepsilon > 0$ there is $n_0(\varepsilon) = \max\{e^4, d^{-1}(\varepsilon)\}$ such that, for $n > n_0(\varepsilon)$,

$$B_n < \left(\frac{e^{-0.6+\varepsilon} n}{\log(n+1)} \right)^n. \quad (11.11)$$

Proof. Set

$$R(x) = \frac{\log(x+1)}{\log(x/\log(x+1))} \exp\left(-1 + \frac{1}{\log(x+1)} + \frac{\log x}{2x}\right).$$

Direct differentiation, after multiplying by the positive denominator $2x \log x \log(x+1) \log(x/\log(x+1))^2$, shows that $R'(x) < 0$ for $x \geq 39$; substitution gives $R(39) < 0.792$. Thus (11.8) proves (11.9) for $n \geq 39$. The remaining $1 \leq n \leq 38$ are obtained successively from (10.1); substituting them gives strict inequalities. This is a finite exact integer check.

A second differentiation gives, for $x \geq e^4$,

$$\log R(x) \leq -0.6 + d(x), \quad d'(x) < 0, \quad \lim_{x \rightarrow \infty} d(x) = 0.$$

(The numerator after clearing positive denominators is a sum of $-(\log x - 4)^2$, $-(x - 1 - \log x)$, and negative multiples of these two standard nonnegative quantities.) Hence $d(n) < \varepsilon$ when $n > d^{-1}(\varepsilon)$, and (11.11) follows from (11.8). $\square$

11.5 Saddle-point asymptotics and Lambert W

Let $r = W(n)$, so that $re^r = n$, and put $A = n(1+r)$.

Theorem 11.8 (Leading Bell asymptotic). *As $n \rightarrow \infty$,*

$$B_n \sim \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \quad (11.12)$$

$$\sim \frac{1}{\sqrt{n}} \left(\frac{n}{W(n)} \right)^{n+1/2} \exp\left(\frac{n}{W(n)} - n - 1\right). \quad (11.13)$$

Proof. In (10.12) take the circle $z = re^{i\theta}$. The coefficient integral becomes

$$\frac{B_n}{n!} = \frac{e^{e^r} - 1}{2\pi r^n} \int_{-\pi}^{\pi} \exp(e^{re^{i\theta}} - e^r - in\theta) d\theta.$$

The linear term vanishes because $re^r = n$; the quadratic term is $-A\theta^2/2$. On $|\theta| \leq A^{-2/5}$, Taylor expansion is uniform and the scaled integral tends to $\int_{\mathbb{R}} e^{-u^2/2} du/\sqrt{A}$. Outside that arc, $\Re(e^{re^{i\theta}} - e^r) \leq -cA\theta^2$ on the intermediate range and is $\leq -c'e^r$ on the remaining compact range, so the contribution is negligible. This proves (11.12). Stirling's formula and $e^r = n/r$ simplify it to (11.13). $\square$

The archived refined expansion used undefined symbols. The following construction makes every coefficient unambiguous. Let $T_j(r) = \sum_{q=0}^j \{j\}_q r^q$ be the Touchard polynomial and let $Z \sim N(0, 1)$. For an integer $h = O(r)$ define

$$\mathcal{R}_{r,h}(Z) = \exp\left(-\frac{ihZ}{\sqrt{A}} + \sum_{j \geq 3} \frac{i^j e^r T_j(r)}{j! A^{j/2}} Z^j\right). \quad (11.14)$$

Expand this formal exponential and take Gaussian moments termwise; retaining all monomials of total order at most e^{-Mr} defines $C_0(r, h), \dots, C_M(r, h)$, with $C_0 = 1$.

Theorem 11.9 (Uniform Moser–Wyman scheme). *For every fixed M and $|h| \leq Cr$,*

$$B_{n+h} = \frac{(n+h)! \exp(e^r - 1)}{r^{n+h} \sqrt{2\pi A}} \left(\sum_{q=0}^{M-1} C_q(r, h) + O_C(e^{-Mr} P_M(r)) \right), \quad (11.15)$$

where P_M is an explicitly computable polynomial. The first correction is

$$C_1(r, h) = -\frac{h^2}{2A} - \frac{he^r T_3(r)}{2A^2} + \frac{e^r T_4(r)}{8A^2} - \frac{5e^{2r} T_3(r)^2}{24A^3}. \quad (11.16)$$

In particular the coefficient of e^{-qr} is a polynomial in h of degree at most $2q$, with rational functions of r as coefficients.

Proof. Use the same circle as in the preceding proof but replace n by $n+h$. After $\theta = Z/\sqrt{A}$, the ratio of the integrand to its Gaussian quadratic part is exactly the formal series (11.14). Watson's lemma with a Gaussian majorant permits termwise integration on the central arc; the complementary arcs have the same exponential bound uniformly for $|h| \leq Cr$. Odd Gaussian moments vanish, so only integral powers of e^{-r} remain. Since $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, and $\mathbb{E}Z^6 = 15$, the terms from the square of the h -term, its product with the cubic term, the quartic term, and the square of the cubic term give (11.16). The same finite bookkeeping defines every subsequent C_q and bounds the omitted Taylor remainder. $\square$

Corollary 11.10 (de Bruijn logarithmic expansion). *As $n \rightarrow \infty$,*

$$\begin{aligned} \frac{\log B_n}{n} &= \log n - \log \log n - 1 + \frac{\log \log n}{\log n} + \frac{1}{\log n} \\ &\quad + \frac{1}{2} \left(\frac{\log \log n}{\log n} \right)^2 + O\left(\frac{\log \log n}{(\log n)^2} \right). \end{aligned} \quad (11.17)$$

Proof. Take logarithms in (11.13). The standard reversion of $r + \log r = \log n$ gives

$$r = \log n - \log \log n + r \log \log n \log n + O\left(\frac{(\log \log n)^2}{(\log n)^2} \right).$$

Substitution and one further Taylor expansion of $\log r$ produce the displayed terms; the prefactor contributes only $O(\log n/n)$. $\square$

11.6 Bell primes and verifiable computation

A *Bell prime* is a prime value of B_n . The archive records the early indices found by computation. Such a list is not a symbolic theorem, but each entry has a rigorous certificate: compute B_n by the integer recurrence (10.1), apply a probable-prime filter, and attach an ECPP or Pratt-style primality certificate. Composite entries are certified by one nontrivial factor. This distinction between an identity and a finite certified computation will also be used for tabulated values later.

Part IV

Bell Polynomials and Composition

Chapter 12

Exponential and ordinary Bell polynomials

12.1 Definitions

Definition 12.1 (Partial exponential Bell polynomial). For $0 \leq k \leq n$, define

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = n! \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+3j_3+\dots=n}} \prod_{i \geq 1} \frac{x_i^{j_i}}{(i!)^{j_i} j_i!}. \quad (12.1)$$

Only $i \leq n - k + 1$ can occur. The complete exponential Bell polynomial is

$$\mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=0}^n \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}), \quad \mathcal{B}_0 = 1. \quad (12.2)$$

Definition 12.2 (Partial ordinary Bell polynomial).

$$\hat{\mathcal{B}}_{n,k}(x_1, \dots, x_{n-k+1}) = \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+\dots=n}} \frac{k!}{j_1! j_2! \dots} \prod_{i \geq 1} x_i^{j_i}. \quad (12.3)$$

Equivalently,

$$\hat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1!x_1, 2!x_2, \dots). \quad (12.4)$$

Proof of (12.4). Substitute $i!x_i$ for x_i in (12.1); the factorials $(i!)^{j_i}$ cancel, and multiplication by $k!/n!$ leaves exactly (12.3). $\square$

12.2 Combinatorial interpretation

Give each block of size i a weight x_i , and define the weight of a partition as the product of its block weights.

Theorem 12.3 (Weighted-partition interpretation). $\mathcal{B}_{n,k}$ is the total weight of partitions of an n -element labelled set into exactly k blocks, and $\mathcal{B}_n$ is the total weight of all its partitions.

Proof. Fix multiplicities j_i , where j_i blocks have size i . The number of such partitions is

$$\frac{n!}{\prod_i (i!)^{j_i} j_i!}.$$

The factors $i!$ forget order inside each block and $j_i!$ forget order among blocks of equal size. Multiplication by $\prod_i x_i^{j_i}$ and summation gives (12.1); summing over k gives the complete polynomial. $\square$

Thus

$$\begin{aligned}\mathcal{B}_{3,2}(x_1, x_2) &= 3x_1x_2, \\ \mathcal{B}_{6,2}(x_1, \dots, x_5) &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\ \mathcal{B}_{6,3}(x_1, \dots, x_4) &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3.\end{aligned}$$

The monomials correspond to integer partitions of n into k parts, while their coefficients count the labelled set partitions having the prescribed block sizes.

12.3 The master generating functions

Put

$$X(t) = \sum_{j \geq 1} x_j \frac{t^j}{j!}.$$

Theorem 12.4 (Bell-polynomial generating functions). *One has*

$$\frac{X(t)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots) \frac{t^n}{n!}, \quad (12.5)$$

$$\exp(uX(t)) = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{t^n}{n!}, \quad (12.6)$$

$$\exp(X(t)) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (12.7)$$

For $\hat{X}(t) = \sum_{j \geq 1} x_j t^j$,

$$\hat{X}(t)^k = \sum_{n \geq k} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n, \quad (12.8)$$

$$e^{u\hat{X}(t)} = \sum_{n \geq k \geq 0} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n \frac{u^k}{k!}. \quad (12.9)$$

Proof. Expand $X(t)^k$ multinomially. Choosing j_i copies of the term $x_i t^i / i!$ gives the constraints $\sum j_i = k$ and $\sum i j_i = n$, and its coefficient is $k! \prod_i x_i^{j_i} / ((i!)^{j_i} j_i!)$. Division by $k!$ gives (12.1). Summing in k proves the exponential identities. The ordinary identities follow by the same multinomial argument or by (12.4). $\square$

Master principle

Nearly every algebraic identity for Bell polynomials follows by performing a simple operation—multiplication, differentiation, substitution, or logarithm—on (1.10) or (1.12). The combinatorial proofs are the same operations interpreted as selecting, pointing, colouring, or gluing blocks.

12.4 Tables and low-degree polynomials

The nonzero partial polynomials through $n = 6$ are

$$\begin{aligned}
\mathcal{B}_{1,1} &= x_1, \\
\mathcal{B}_{2,1} &= x_2, & \mathcal{B}_{2,2} &= x_1^2, \\
\mathcal{B}_{3,1} &= x_3, & \mathcal{B}_{3,2} &= 3x_1x_2, & \mathcal{B}_{3,3} &= x_1^3, \\
\mathcal{B}_{4,1} &= x_4, & \mathcal{B}_{4,2} &= 4x_1x_3 + 3x_2^2, \\
\mathcal{B}_{4,3} &= 6x_1^2x_2, & \mathcal{B}_{4,4} &= x_1^4, \\
\mathcal{B}_{5,1} &= x_5, & \mathcal{B}_{5,2} &= 5x_1x_4 + 10x_2x_3, \\
\mathcal{B}_{5,3} &= 10x_1^2x_3 + 15x_1x_2^2, \\
\mathcal{B}_{5,4} &= 10x_1^3x_2, & \mathcal{B}_{5,5} &= x_1^5, \\
\mathcal{B}_{6,1} &= x_6, & \mathcal{B}_{6,2} &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\
\mathcal{B}_{6,3} &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3, \\
\mathcal{B}_{6,4} &= 20x_1^3x_3 + 45x_1^2x_2^2, \\
\mathcal{B}_{6,5} &= 15x_1^4x_2, & \mathcal{B}_{6,6} &= x_1^6.
\end{aligned}$$

Every coefficient is supplied by (12.1), so the table is a computation from the theorem rather than an independent assertion.

The first complete polynomials are

$$\begin{aligned}
\mathcal{B}_0 &= 1, \\
\mathcal{B}_1 &= x_1, \\
\mathcal{B}_2 &= x_1^2 + x_2, \\
\mathcal{B}_3 &= x_1^3 + 3x_1x_2 + x_3, \\
\mathcal{B}_4 &= x_1^4 + 6x_1^2x_2 + 3x_2^2 + 4x_1x_3 + x_4, \\
\mathcal{B}_5 &= x_1^5 + 10x_1^3x_2 + 15x_1x_2^2 + 10x_1^2x_3 + 10x_2x_3 + 5x_1x_4 + x_5, \\
\mathcal{B}_6 &= x_1^6 + 15x_1^4x_2 + 45x_1^2x_2^2 + 15x_2^3 + 20x_1^3x_3 + 60x_1x_2x_3 + 10x_3^2 \\
&\quad + 15x_1^2x_4 + 15x_2x_4 + 6x_1x_5 + x_6, \\
\mathcal{B}_7 &= x_1^7 + 21x_1^5x_2 + 105x_1^3x_2^2 + 105x_1x_2^3 + 35x_1^4x_3 + 210x_1^2x_2x_3 \\
&\quad + 70x_1x_3^2 + 105x_2^2x_3 + 35x_1^3x_4 + 105x_1x_2x_4 + 35x_3x_4 \\
&\quad + 21x_1^2x_5 + 21x_2x_5 + 7x_1x_6 + x_7.
\end{aligned}$$

Chapter 13

Recurrences, derivatives, and algebraic identities

13.1 Pointing recurrences

Theorem 13.1 (Complete and partial recurrences). For $n \geq 0$,

$$\mathcal{B}_{n+1}(x_1, \dots, x_{n+1}) = \sum_{i=0}^n \binom{n}{i} x_{i+1} \mathcal{B}_{n-i}(x_1, \dots, x_{n-i}), \quad (13.1)$$

$$\mathcal{B}_{n+1,k+1} = \sum_{i=0}^{n-k} \binom{n}{i} x_{i+1} \mathcal{B}_{n-i,k}, \quad (13.2)$$

with $\mathcal{B}_{0,0} = 1$, $\mathcal{B}_{n,0} = 0$ for $n > 0$, and $\mathcal{B}_{0,k} = 0$ for $k > 0$.

Proof. In a weighted partition of $[n+1]$, inspect the block containing the distinguished label $n+1$. If that block contains i further labels, choose them in $\binom{n}{i}$ ways, give the block weight x_{i+1} , and partition the remaining labels. Keeping or forgetting the total block count yields the two formulas. $\square$

Theorem 13.2 (Block-colour convolution). For $k_1, k_2 \geq 0$,

$$\mathcal{B}_{n,k_1+k_2}(x) = \frac{k_1!k_2!}{(k_1+k_2)!} \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{i,k_1}(x) \mathcal{B}_{n-i,k_2}(x). \quad (13.3)$$

Proof. Multiply (1.10) for k_1 and k_2 . Their product is $X^{k_1+k_2}/(k_1!k_2!)$; comparison with the same formula for k_1+k_2 gives the factor $(k_1!k_2!)/(k_1+k_2)!$ and the binomial EGF convolution. $\square$

13.2 Near-diagonal reduction

Theorem 13.3. For $1 \leq a < n$,

$$\begin{aligned} \mathcal{B}_{n,n-a}(x_1, \dots, x_{a+1}) \\ = \sum_{j=a+1}^{2a} \frac{j!}{a!} \binom{n}{j} x_1^{n-j} \mathcal{B}_{a,j-a}\left(\frac{x_2}{2}, \frac{x_3}{3}, \dots, \frac{x_{2a-j+2}}{2a-j+2}\right). \end{aligned} \quad (13.4)$$

Proof. A partition into $n-a$ blocks has total excess $\sum_B(|B|-1) = a$. Let j be the number of elements in nonsingleton blocks. Then the number of those blocks is $j-a$, so $a+1 \leq j \leq 2a$. Choose the j elements, leave the other $n-j$ as singleton blocks, and count the nonsingleton partition. Replacing a

block of size $r + 1$ by an atom of size r removes one distinguished element per block; the factor $j!/a!$ and weights $x_{r+1}/(r+1)$ are exactly the labelled pointing factors. Equivalently, make the substitution in (12.1); the coefficient of every multiplicity vector on both sides is $n!x_1^{n-j} \prod_{r \geq 2} x_r^{j_r} / ((r!)^{j_r} j_r!)$. $\square$

The first cases, obtained by listing integer partitions of the excess, are

$$\mathcal{B}_{n,1} = x_n, \quad \mathcal{B}_{n,2} = \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} x_j x_{n-j}, \quad (13.5)$$

$$\mathcal{B}_{n,n} = x_1^n, \quad \mathcal{B}_{n,n-1} = \binom{n}{2} x_1^{n-2} x_2, \quad (13.6)$$

$$\mathcal{B}_{n,n-2} = \binom{n}{3} x_1^{n-3} x_3 + 3 \binom{n}{4} x_1^{n-4} x_2^2, \quad (13.7)$$

$$\mathcal{B}_{n,n-3} = \binom{n}{4} x_1^{n-4} x_4 + 10 \binom{n}{5} x_1^{n-5} x_2 x_3 + 15 \binom{n}{6} x_1^{n-6} x_2^3, \quad (13.8)$$

$$\begin{aligned} \mathcal{B}_{n,n-4} = & \binom{n}{5} x_1^{n-5} x_5 + 5 \binom{n}{6} x_1^{n-6} (3x_2 x_4 + 2x_3^2) \\ & + 105 \binom{n}{7} x_1^{n-7} x_2^2 x_3 + 105 \binom{n}{8} x_1^{n-8} x_2^4. \end{aligned} \quad (13.9)$$

Their coefficients are the corresponding block-size counts from Theorem 12.3.

13.3 Partial derivatives

Theorem 13.4 (Derivative identities). *For $i \geq 1$,*

$$\frac{\partial \mathcal{B}_n}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i}, \quad (13.10)$$

$$\frac{\partial \mathcal{B}_{n,k}}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i,k-1}. \quad (13.11)$$

Consequently, for differentiable $a_i(x)$,

$$\frac{d}{dx} \mathcal{B}_{n,k}(a_1(x), \dots) = \sum_{i=1}^{n-k+1} \binom{n}{i} a'_i(x) \mathcal{B}_{n-i,k-1}(a_1(x), \dots). \quad (13.12)$$

Proof. Differentiate (1.10) with respect to x_i :

$$\frac{\partial}{\partial x_i} \frac{X(t)^k}{k!} = \frac{t^i X(t)^{k-1}}{i! (k-1)!}.$$

Coefficient comparison gives (13.11); summing over k gives (13.10). The ordinary chain rule then gives (13.12). $\square$

13.4 A quadratic differential recurrence

The less familiar differential recurrence in the archive is also a disguised coefficient identity.

Theorem 13.5. For $n \geq 2$,

$$\begin{aligned} \mathcal{B}_n = \frac{1}{n-1} & \left[\sum_{i=2}^n \sum_{j=1}^{i-1} (i-1) \binom{i-2}{j-1} x_j x_{i-j} \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right. \\ & \left. + \sum_{i=2}^n \sum_{j=1}^{i-1} \frac{x_{i+1}}{\binom{i}{j}} \frac{\partial^2 \mathcal{B}_{n-1}}{\partial x_j \partial x_{i-j}} + \sum_{i=2}^n x_i \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right]. \end{aligned} \quad (13.13)$$

Terms containing unavailable variables or negative-index Bell polynomials are zero.

Proof. Let $F(t) = e^{X(t)} = \sum_m \mathcal{B}_m t^m / m!$. After using (13.10), the first double sum is

$$(n-1)! [t^{n-1}] (tX'(t))^2 F(t).$$

The second is $(n-1)! [t^{n-1}] ((tX'' - X' + x_1)F)$, and the last is $(n-1)! [t^{n-1}] ((X' - x_1)F)$. Their sum is therefore

$$(n-1)! [t^{n-1}] t(X'' + X'^2)F = (n-1)! [t^{n-1}] tF'' = (n-1)\mathcal{B}_n.$$

Division by $n-1$ proves the formula. □

Chapter 14

Specializations, inversions, determinants, and convolutions

14.1 Stirling, Bell, Lah, and Touchard specializations

Theorem 14.1. *For admissible n, k ,*

$$\mathcal{B}_{n,k}(0!, 1!, \dots, (n-k)!) = \begin{bmatrix} n \\ k \end{bmatrix}, \quad (14.1)$$

$$\mathcal{B}_n(0!, 1!, \dots, (n-1)!) = n!, \quad (14.2)$$

$$\mathcal{B}_{n,k}(1, 1, \dots, 1) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (14.3)$$

$$\mathcal{B}_n(1, 1, \dots, 1) = \mathcal{B}_n, \quad (14.4)$$

$$\mathcal{B}_{n,k}(1!, 2!, \dots, (n-k+1)!) = \binom{n-1}{k-1} \frac{n!}{k!} = L(n, k), \quad (14.5)$$

$$T_n(x) = \mathcal{B}_n(x, x, \dots, x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k. \quad (14.6)$$

Proof. In (1.10), the choices $x_j = (j-1)!$, $x_j = 1$, and $x_j = j!$ make $X(t)$ respectively

$$-\log(1-t), \quad e^t - 1, \quad \frac{t}{1-t}.$$

The resulting coefficient formulae are exactly the column EGFs for unsigned first-kind Stirling numbers, second-kind Stirling numbers, and Lah numbers proved earlier. Summing over k yields (14.2) and (14.4). Finally $x_j = x$ gives $\exp(x(e^t - 1))$, the EGF of $T_n(x)$. $\square$

14.2 Homogeneity and a useful evaluation

Theorem 14.2 (Bigraded homogeneity). *For scalars α, β ,*

$$\mathcal{B}_{n,k}(\alpha\beta x_1, \alpha\beta^2 x_2, \dots) = \alpha^k \beta^n \mathcal{B}_{n,k}(x_1, x_2, \dots). \quad (14.7)$$

Moreover,

$$\mathcal{B}_{n,k}(1, 2, 3, \dots, n-k+1) = \binom{n}{k} k^{n-k}. \quad (14.8)$$

Proof. Every monomial in (12.1) has $\sum j_i = k$ and $\sum i j_i = n$, proving the first formula. For the second, $x_j = j$ gives $X(t) = te^t$; hence

$$\frac{X(t)^k}{k!} = \frac{t^k e^{kt}}{k!}.$$

The coefficient of $t^n/n!$ is $n!k^{n-k}/(k!(n-k)!) = \binom{n}{k}k^{n-k}$. $\square$

14.3 Addition and shifted-zero identities

Theorem 14.3 (Complete Bell addition formula).

$$\mathcal{B}_n(x_1 + y_1, \dots, x_n + y_n) = \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{n-i}(x) \mathcal{B}_i(y). \quad (14.9)$$

Proof. The complete EGF for the left side is $\exp(X(t) + Y(t)) = \exp X(t) \exp Y(t)$. Coefficient extraction gives the binomial convolution. $\square$

Theorem 14.4 (Insertion of q leading zeros). For $q \geq 0$,

$$\begin{aligned} \mathcal{B}_{n,k} \left(\frac{x_{q+1}}{\binom{q+1}{q}}, \frac{x_{q+2}}{\binom{q+2}{q}}, \dots \right) \\ = \frac{n!(q!)^k}{(n+qk)!} \mathcal{B}_{n+qk,k}(\underbrace{0, \dots, 0}_{q \text{ entries}}, x_{q+1}, x_{q+2}, \dots). \end{aligned} \quad (14.10)$$

Proof. Since

$$\frac{x_{q+i}}{\binom{q+i}{q}} \frac{t^i}{i!} = q! \frac{x_{q+i} t^i}{(q+i)!},$$

the k th power of the left component series is $(q!)^k t^{-qk}$ times the k th power of the component series on the right. Comparing the coefficient of t^n in (1.10) gives the factorial factor displayed. $\square$

14.4 Exponential–logarithmic inversion

Theorem 14.5 (Bell transform and its inverse). Suppose

$$y_n = \mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(x_1, \dots). \quad (14.11)$$

Then

$$x_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(y_1, \dots, y_{n-k+1}). \quad (14.12)$$

Proof. Let $X(t) = \sum_{n \geq 1} x_n t^n / n!$ and $Y(t) = \sum_{n \geq 1} y_n t^n / n!$. Equation (1.12) says $1 + Y = e^X$, so

$$X = \log(1 + Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} Y^k.$$

Use (1.10) to extract the coefficient of $t^n/n!$. $\square$

More generally, if f and $g = f^{-1}$ are locally inverse and

$$y_n = \sum_{k=0}^n f^{(k)}(a) \mathcal{B}_{n,k}(x_1, \dots), \quad (14.13)$$

then

$$x_n = \sum_{k=0}^n g^{(k)}(f(a)) \mathcal{B}_{n,k}(y_1, \dots). \quad (14.14)$$

Indeed, these are the derivative coefficients of $f \circ u$ and $g \circ f \circ u = u$; a complete proof is given with Faà di Bruno's formula in Equation (19.3).

14.5 Two Hessenberg determinants

For $n \geq 1$, define $H_n = (h_{ij})$ and $K_n = (k_{ij})$ by

$$h_{ij} = \begin{cases} \binom{n-i}{j-i} x_{j-i+1}, & j \geq i, \\ -1, & j = i-1, \\ 0, & j < i-1, \end{cases} \quad k_{ij} = \begin{cases} x_{j-i+1}/(j-i)!, & j \geq i, \\ -(i-1), & j = i-1, \\ 0, & j < i-1. \end{cases}$$

Thus H_n begins with

$$\begin{bmatrix} x_1 & \binom{n-1}{1} x_2 & \binom{n-1}{2} x_3 & \cdots & x_n \\ -1 & x_1 & \binom{n-2}{1} x_2 & \cdots & x_{n-1} \\ 0 & -1 & x_1 & \cdots & x_{n-2} \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & x_1 \end{bmatrix},$$

while K_n has first row $x_1, x_2/1!, \dots, x_n/(n-1)!$ and subdiagonal $-1, -2, \dots, -(n-1)$.

Theorem 14.6 (Bell determinants).

$$\mathcal{B}_n(x_1, \dots, x_n) = \det H_n = \det K_n. \quad (14.15)$$

Proof. Expansion of the upper Hessenberg determinant $\det H_n$ along its first row gives

$$D_n = \sum_{j=1}^n \binom{n-1}{j-1} x_j D_{n-j}, \quad D_0 = 1;$$

the cofactor sign is cancelled by the product of the $j-1$ entries -1 on the subdiagonal. This is exactly (13.1), so $D_n = \mathcal{B}_n$.

Let J reverse the coordinate order and let $D = \text{diag}(0!, 1!, \dots, (n-1)!)$. A direct entrywise calculation gives

$$K_n = D(JH_n^T J)D^{-1}.$$

Indeed, $\frac{(i-1)!}{(j-1)!} \binom{j-1}{i-1} = 1/(j-i)!$ above the diagonal, and the subdiagonal becomes $-(i-1)$. Similarity, transposition, and reversal preserve the determinant, proving the second equality. $\square$

14.6 The diamond convolution

For sequences $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ define

$$(x \diamond y)_n = \sum_{j=1}^{n-1} \binom{n}{j} x_j y_{n-j}. \quad (14.16)$$

This operation is commutative and associative because it corresponds to multiplication of the EGFs $X(t)$ and $Y(t)$.

Theorem 14.7. *If $x^{k\Diamond}$ denotes the k -fold diamond power, then*

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = \frac{(x^{k\Diamond})_n}{k!}. \quad (14.17)$$

Proof. The EGF of $x^{k\Diamond}$ is $X(t)^k$. Compare with (1.10). □

For example,

$$x\Diamond x = (0, 2x_1^2, 6x_1x_2, 8x_1x_3 + 6x_2^2, \dots)$$

and $(x^{3\Diamond})_4 = 36x_1^2x_2$, whence $\mathcal{B}_{4,3} = 6x_1^2x_2$.

Chapter 15

Applications of Bell polynomials

15.1 Symmetric functions and Newton identities

Let e_n be the elementary symmetric function and $p_n = \sum_i u_i^n$ the power sum in an alphabet $u_1, u_2, \dots$.

Theorem 15.1.

$$e_n = \frac{1}{n!} \mathcal{B}_n(p_1, -1!p_2, 2!p_3, \dots, (-1)^{n-1}(n-1)!p_n) \quad (15.1)$$

$$\begin{aligned} &= \frac{(-1)^n}{n!} \mathcal{B}_n(-p_1, -1!p_2, -2!p_3, \dots, -(n-1)!p_n), \\ p_n &= \frac{(-1)^{n-1}}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(1!e_1, 2!e_2, \dots) \\ &= (-1)^n \sum_{k=1}^n \frac{1}{k} \widehat{\mathcal{B}}_{n,k}(-e_1, -e_2, \dots). \end{aligned} \quad (15.2)$$

Proof. The classical product identity

$$E(t) = \sum_{n \geq 0} e_n t^n = \prod_i (1 + u_i t) = \exp \left(\sum_{r \geq 1} (-1)^{r-1} p_r \frac{t^r}{r} \right)$$

and (1.12) give (15.1). Taking logarithms, $\sum_{r \geq 1} (-1)^{r-1} p_r t^r / r = \log(1 + \sum_{j \geq 1} e_j t^j)$, and expanding the logarithm through ordinary Bell polynomials yields the second pair of forms. The exponential form follows from (12.4). $\square$

Corollary 15.2 (Determinant from traces). *For an $n \times n$ matrix A , set $s_k = -(k-1)! \operatorname{tr}(A^k)$. Then*

$$\det A = \frac{(-1)^n}{n!} \mathcal{B}_n(s_1, s_2, \dots, s_n). \quad (15.3)$$

Proof. Apply (15.1) to the eigenvalues of A , using $e_n = \det A$ and $p_k = \operatorname{tr}(A^k)$. Both sides are polynomial in the matrix entries, so the argument remains valid for nondiagonalizable matrices. $\square$

15.2 Cycle indices

Let a_j mark cycles of length j in a permutation.

Theorem 15.3. *The cycle index of the symmetric group is*

$$Z(S_n) = \frac{1}{n!} \mathcal{B}_n(0!a_1, 1!a_2, \dots, (n-1)!a_n). \quad (15.4)$$

Proof. A permutation is a set of cycles. A labelled cycle of size j has $(j-1)!$ linear realizations, so its component EGF is $(j-1)!a_j t^j/j! = a_j t^j/j$. The exponential formula gives $\exp(\sum_{j \geq 1} a_j t^j/j)$; comparison with (1.12) proves the identity after division by $n!$. $\square$

15.3 Moments and cumulants

Let $M(t) = \mathbb{E}e^{tX} = \sum_{n \geq 0} \mu'_n t^n/n!$ and $K(t) = \log M(t) = \sum_{n \geq 1} \kappa_n t^n/n!$.

Theorem 15.4 (Moment–cumulant relations).

$$\mu'_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(\kappa_1, \dots), \quad (15.5)$$

$$\kappa_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(\mu'_1, \dots, \mu'_{n-k+1}). \quad (15.6)$$

Proof. The identity $M = e^K$ is (1.12); its logarithmic inverse is Theorem 14.5. $\square$

15.4 Hermite polynomials

The probabilists' Hermite polynomials are defined by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}.$$

Comparison with (1.12) gives and proves

$$\text{He}_n(x) = \mathcal{B}_n(x, -1, 0, \dots, 0). \quad (15.7)$$

In particular, the usual recurrence and differential equation follow by differentiating this generating function.

15.5 Polynomial sequences of binomial type

Let $h(t) = \sum_{j \geq 1} a_j t^j/j!$ with $a_1 \neq 0$, and define

$$p_n(x) = \mathcal{B}_n(a_1 x, a_2 x, \dots, a_n x) = \sum_{k=0}^n \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}) x^k. \quad (15.8)$$

Theorem 15.5. *The polynomials p_n satisfy*

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xh(t)}, \quad (15.9)$$

$$p_n(x+y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (15.10)$$

If $Q = h^{-1}(D_x)$ is the associated delta operator, then

$$Qp_n(x) = np_{n-1}(x). \quad (15.11)$$

Proof. Substitute xa_j in (1.12). The product $e^{(x+y)h} = e^{xh}e^{yh}$ gives the binomial identity. Finally $h^{-1}(D_x)e^{xh(t)} = h^{-1}(h(t))e^{xh(t)} = te^{xh(t)}$; coefficient comparison gives (15.11). $\square$

Editorial normalization

In some sources the symbol $B_n(x)$ denotes the Touchard polynomial, in others a complete Bell polynomial, a Bernoulli polynomial, or a binary-tree sequence. The notations T_n , $\mathcal{B}_n$, $\beta_n(x)$, and C_n are kept separate here; this is essential when the objects occur in the same inversion formula.

Part V

Eulerian Numbers and Descent Calculus

Chapter 16

Ordinary Eulerian numbers

16.1 Descents, symmetry, and the basic recurrence

For a permutation $\pi = \pi_1 \cdots \pi_n$, a descent is an index $i \in \{1, \dots, n-1\}$ with $\pi_i > \pi_{i+1}$.

Definition 16.1. The Eulerian number $\langle n \rangle_k = A(n, k)$ counts permutations of $[n]$ with exactly k descents. Put

$$A_0(t) = 1, \quad A_n(t) = \sum_{k=0}^{n-1} \langle n \rangle_k t^k \quad (n \geq 1).$$

Outside $0 \leq k < n$ the number is zero.

Theorem 16.2 (Eulerian recurrence).

$$\langle n \rangle_k = (k+1) \langle n-1 \rangle_k + (n-k) \langle n-1 \rangle_{k-1}. \quad (16.1)$$

Equivalently,

$$A_n(t) = (1 + (n-1)t)A_{n-1}(t) + t(1-t)A'_{n-1}(t). \quad (16.2)$$

Proof. Insert n into a permutation of $[n-1]$. If the old permutation has k descents, insertion after one of those k descents or at the end preserves the number of descents, giving $k+1$ choices. If it has $k-1$ descents, insertion in any of the remaining $n-k$ slots creates one new descent. This proves (16.1). Multiply by t^k , sum over k , and collect the terms containing k to obtain (16.2). $\square$

Complementation $\pi_i \mapsto n+1-\pi_i$ interchanges descents and ascents. Therefore

$$\langle n \rangle_k = \langle n \rangle_{n-1-k}, \quad A_n(t) = t^{n-1}A_n(t^{-1}). \quad (16.3)$$

In particular $\langle n \rangle_0 = \langle n \rangle_{n-1} = 1$ and $\sum_k \langle n \rangle_k = n!$. The next diagonal is

$$\langle n \rangle_1 = 2^n - (n+1), \quad (16.4)$$

which follows either from the recurrence or from the explicit formula below.

16.2 Exponential generating function and explicit formula

Theorem 16.3 (Eulerian EGF).

$$\sum_{n \geq 0} A_n(t) \frac{x^n}{n!} = \frac{t-1}{t - e^{(t-1)x}} = \frac{(1-t)e^{(1-t)x}}{1 - te^{(1-t)x}}. \quad (16.5)$$

Proof. Let $F(x, t)$ denote the left side. Multiplying (16.2) by $x^{n-1}/(n-1)!$ and summing gives the first-order equation

$$(1 - tx)F_x = t(1 - t)F_t + F, \quad F(0, t) = 1.$$

Direct substitution verifies that the displayed rational-exponential function solves this equation and initial condition. Formal uniqueness follows by recursively comparing coefficients of x . $\square$

Expanding the denominator geometrically gives the central power-series identity.

Theorem 16.4. For $n \geq 0$,

$$\sum_{m=0}^{\infty} (m+1)^n t^m = \frac{A_n(t)}{(1-t)^{n+1}}. \quad (16.6)$$

Consequently,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n. \quad (16.7)$$

Proof. From (16.5),

$$F(x, t) = (1-t) \sum_{m \geq 0} t^m e^{(m+1)(1-t)x}.$$

Extracting $x^n/n!$ proves (16.6). Multiply by $(1-t)^{n+1}$ and extract t^k to obtain (16.7). Formula (16.4) is its case $k = 1$. $\square$

Theorem 16.5 (A second polynomial recurrence). For $n > 1$,

$$A_n(t) = \sum_{k=0}^{n-1} \binom{n}{k} A_k(t) (t-1)^{n-1-k}. \quad (16.8)$$

Proof. The EGF of the right sides, summed over $n \geq 1$, is

$$F(x, t) \frac{e^{(t-1)x} - 1}{t-1}.$$

Using (16.5), this equals $F(x, t) - 1$, which is the EGF of the left sides. $\square$

16.3 Worpitzky's identity and sums of powers

Theorem 16.6 (Worpitzky). For every indeterminate x and $n \geq 1$,

$$x^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{x+k}{n}. \quad (16.9)$$

Proof. The coefficient of t^{x-1} in $t^{-k}(1-t)^{-n-1}$ is $\binom{x+k}{n}$ for positive integer x . Equivalently, taking the coefficient of t^{x-1} in (16.6) after multiplying by the numerator gives (16.9). Since both sides are polynomials of degree n in x and agree for all positive integers, they agree identically.

A direct combinatorial proof standardizes a word in $[x]^n$: its stable sorting permutation records the relative order, and its k descents specify k mandatory separations among the n sorted positions. Stars and bars gives $\binom{x+k}{n}$ words for each permutation with k descents. $\square$

Corollary 16.7 (Power sums). For $m, n \geq 1$,

$$\sum_{r=1}^m r^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{m+k+1}{n+1}. \quad (16.10)$$

Proof. Sum (16.9) over $x = 1, \dots, m$ and use the hockey-stick identity $\sum_{x=1}^m \binom{x+k}{n} = \binom{m+k+1}{n+1}$; the omitted lower boundary terms vanish for $0 \leq k < n$. $\square$

16.4 Negative polylogarithms

For $|z| < 1$, $\text{Li}_{-n}(z) = \sum_{m \geq 1} m^n z^m$. Shifting m in (16.6) gives

$$\text{Li}_{-n}(z) = \frac{z A_n(z)}{(1-z)^{n+1}} = \frac{1}{(1-z)^{n+1}} \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle z^{n-k}, \quad (16.11)$$

where the second equality uses symmetry (16.3). Since the right side is rational, it also supplies the meromorphic continuation of the negative polylogarithm.

16.5 Eulerian and Stirling bases

Theorem 16.8.

$$A_n(t) = \sum_{k=0}^n \left\{ n \atop k \right\} k! (t-1)^{n-k}. \quad (16.12)$$

Proof. Use $m^n = \sum_k \left\{ n \atop k \right\} m^k$ in $\sum_{m \geq 0} m^n t^m$. Since $\sum_{m \geq 0} m^k t^m = k! t^k / (1-t)^{k+1}$, multiplication by $(1-t)^{n+1}/t$ yields $A_n(t) = \sum_k \left\{ n \atop k \right\} k! t^{k-1} (1-t)^{n-k}$. Replacing t by $1/t$ and using (16.3), or simply expanding around $t = 1$, gives the equivalent displayed form. $\square$

16.6 An Irwin–Hall integral and hypersimplex volumes

Theorem 16.9. For every function $f : \mathbb{R} \rightarrow \mathbb{C}$ for which the following finite sum is defined,

$$\sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle f(k) = n! \int_{[0,1]^n} f(\lfloor x_1 + \cdots + x_n \rfloor) dx_1 \cdots dx_n. \quad (16.13)$$

In particular, the slab

$$\Delta_{n,k} = \{x \in [0,1]^n : k \leq x_1 + \cdots + x_n < k+1\}$$

has volume $\left\langle n \atop k \right\rangle / n!$.

Proof. The volume of $\{x \in [0,1]^n : \sum x_i < y\}$ is obtained from the simplex $\{x_i \geq 0 : \sum x_i < y\}$ by inclusion–exclusion over coordinates exceeding 1:

$$V_n(y) = \frac{1}{n!} \sum_{j=0}^{\lfloor y \rfloor} (-1)^j \binom{n}{j} (y-j)^n.$$

Thus the volume of the k th slab is $V_n(k+1) - V_n(k)$. Combining adjacent terms by Pascal’s identity gives

$$\frac{1}{n!} \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n = \frac{1}{n!} \left\langle n \atop k \right\rangle$$

by (16.7). Summing $f(k)$ times these volumes proves the integral identity. $\square$

16.7 Alternating sums and Bernoulli numbers

Theorem 16.10. For $n > 0$,

$$\sum_{k=0}^{n-1} (-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1}. \quad (16.14)$$

For $n > 1$,

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n-1}{k}} = 0, \quad (16.15)$$

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n}{k}} = (n+1) \beta_n. \quad (16.16)$$

Proof. At $t = -1$, (16.5) becomes $2/(1 + e^{-2x}) = 1 + \tanh x$. The Bernoulli generating function gives

$$\tanh x = \sum_{n \geq 1} 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1} \frac{x^n}{n!},$$

which proves (16.14).

For the remaining identities use

$$\binom{n-1}{k}^{-1} = n \int_0^1 u^k (1-u)^{n-k-1} du, \quad \binom{n}{k}^{-1} = (n+1) \int_0^1 u^k (1-u)^{n-k} du.$$

Substitute $t = -u/(1-u)$ in (16.12); then

$$(1-u)^n A_n \left(-\frac{u}{1-u} \right) = \sum_{j=0}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j! (1-u)^j. \quad (16.17)$$

For the first integral divide by $1-u$ and integrate. The result is $n \sum_{j=1}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} (j-1)! = 0$ for $n > 1$, because it is the coefficient of $x^n/n!$ in $\log(1 + (e^x - 1)) = x$. For the second integral one obtains $(n+1) \sum_j (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j!/(j+1)$. The standard Stirling formula $\beta_n = \sum_j (-1)^j j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} / (j+1)$ proves the claim; for odd $n > 1$ both sides vanish, and for even n the signs coincide. $\square$

Chapter 17

Geometric interpretations

17.1 The cube and its Ehrhart numerator

For a lattice polytope P of dimension d , its Ehrhart series has the form $\sum_{m \geq 0} |mP \cap \mathbb{Z}^d| t^m = h^*(t)/(1-t)^{d+1}$. For the unit cube $C = [0, 1]^n$ one has $|mC \cap \mathbb{Z}^n| = (m+1)^n$; hence (16.6) proves:

Theorem 17.1. *The h^* -polynomial of the standard n -cube is $A_n(t)$, so its h^* -vector is the n th Eulerian row.*

Proof. The Ehrhart function of $C = [0, 1]^n$ is $L_C(m) = (m+1)^n$. Therefore

$$\sum_{m \geq 0} L_C(m) t^m = \sum_{m \geq 0} (m+1)^n t^m = \frac{A_n(t)}{(1-t)^{n+1}}$$

by (16.6). The numerator in the standard Ehrhart form is unique, so $h_C^*(t) = A_n(t)$; its coefficients are precisely the n th Eulerian row. $\square$

The regions $\Delta_{n,k}$ in Theorem 16.9 are half-open hypersimplices. Their normalized volumes are $n! \operatorname{vol} \Delta_{n,k} = \langle \frac{n}{k} \rangle$.

17.2 The permutohedron

The type- A permutohedron is the convex hull of the permutations of $(1, 2, \dots, n)$; its dimension is $n-1$. Faces of its braid fan are indexed by ordered set partitions. There are $k! \left\{ \frac{n}{k} \right\}$ ordered partitions into k blocks.

Theorem 17.2. *The h -polynomial of the simple polytope dual to the type- A permutohedron is $A_n(t)$.*

Proof. For a simple $(n-1)$ -polytope the h -polynomial is the transform $h(t) = \sum_F (t-1)^{\dim F}$ over faces of the dual simplicial complex. Grouping braid-fan faces by their k ordered blocks gives $\sum_k k! \left\{ \frac{n}{k} \right\} (t-1)^{n-k}$, which is $A_n(t)$ by (16.12). $\square$

Chapter 18

Type-B and second-order Eulerian numbers

18.1 Signed permutations and the corrected type-B indexing

A signed permutation is a word $\pi_1 \cdots \pi_n$ whose absolute values form a permutation of $[n]$. Put $\pi_0 = 0$ and call $i \in \{0, \dots, n-1\}$ a type- B descent when $\pi_i > \pi_{i+1}$. Let $E_B(n, k)$ count signed permutations with k such descents, and set $M_n(t) = \sum_{k=0}^n E_B(n, k)t^k$.

Editorial normalization

The archived display reverses the two entries in the descent inequality and pairs that convention with an incompatible explicit formula. The definition above is the standard Coxeter descent convention. It gives the archived table 1, 6, 1 for $n = 2$ and the midpoint-Eulerian generating function below.

Theorem 18.1 (Type- B recurrence and Worpitzky series).

$$E_B(n, k) = (2k + 1)E_B(n - 1, k) + (2n - 2k + 1)E_B(n - 1, k - 1), \quad (18.1)$$

$$\sum_{m \geq 0} (2m + 1)^n t^m = \frac{M_n(t)}{(1 - t)^{n+1}}, \quad (18.2)$$

$$E_B(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (2(k-j) + 1)^n. \quad (18.3)$$

The first rows are

$$1; \quad 1, 1; \quad 1, 6, 1; \quad 1, 23, 23, 1; \quad 1, 76, 230, 76, 1.$$

Proof. Insert either n or $-n$ into a signed permutation of size $n - 1$. Marking the slots according to whether insertion preserves or creates a descent gives $2k + 1$ preserving slots for an object with k descents and $2n - 2k + 1$ creating slots for one with $k - 1$ descents; this proves the recurrence.

A type- B standardization partitions the $(2m + 1)^n$ words in the ordered alphabet $-m < \cdots < -1 < 0 < 1 < \cdots < m$ by their signed sorting permutation. If that permutation has k type- B descents, stars and bars gives $\binom{m+n-k}{n}$ compatible weakly increasing absolute levels. Hence

$$(2m + 1)^n = \sum_{k=0}^n E_B(n, k) \binom{m+n-k}{n}.$$

Multiply by t^m , sum in m , and use $\sum_{m \geq 0} \binom{m+n-k}{n} t^m = t^k / (1 - t)^{n+1}$ to obtain (18.2). Multiplication by $(1 - t)^{n+1}$ and coefficient extraction gives (18.3). $\square$

As in type A , these coefficients form the h -vector of the simple polytope dual to the type- B permutohedron; the proof repeats the braid-fan argument with signed ordered partitions.

18.2 Stirling permutations and second-order Eulerian numbers

A *Stirling permutation of order n* is a permutation of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ such that all entries between the two copies of i are larger than i . Let $\langle\langle n \rangle\rangle_k$ count those with k ordinary descents and put

$$P_n(t) = \sum_{k=0}^{n-1} \langle\langle n \rangle\rangle_k t^k, \quad P_0(t) = 1.$$

Theorem 18.2 (Second-order recurrence).

$$\langle\langle n \rangle\rangle_k = (2n - k - 1) \langle\langle n - 1 \rangle\rangle_{k-1} + (k + 1) \langle\langle n - 1 \rangle\rangle_k, \quad (18.4)$$

$$P_{n+1}(t) = (2nt + 1)P_n(t) - t(t - 1)P'_n(t), \quad (18.5)$$

$$P_n(1) = (2n - 1)!! \quad (18.6)$$

Proof. The two copies of the largest label n must be adjacent. Insert the pair nn into one of the $2n - 1$ gaps of a Stirling permutation of order $n - 1$. A descent gap or the terminal gap preserves the number of descents, giving $k + 1$ choices from an object with k descents. Each of the remaining $2n - k - 1$ gaps creates one descent from an object with $k - 1$ descents. This is the first recurrence; summing against t^k gives the second. At each stage there are $2n - 1$ insertion gaps, so the total count is $(2n - 1)!!$. $\square$

The differential recurrence has two useful equivalent forms:

$$(t - 1)^{-2n-2} P_{n+1}(t) = (t(1 - t)^{-2n-1} P_n(t))', \quad (18.7)$$

$$u_{n+1}(t) = \left(\frac{t}{1 - t} u_n(t) \right)', \quad u_n(t) = (t - 1)^{-2n} P_n(t), \quad u_0 = 1. \quad (18.8)$$

They follow by the product rule from (18.5).

Theorem 18.3 (Diagonal Stirling generating function). For $n \geq 1$,

$$\sum_{m=0}^{\infty} \left\{ \begin{matrix} n + m \\ m \end{matrix} \right\} t^m = \frac{t P_n(t)}{(1 - t)^{2n+1}}. \quad (18.9)$$

Proof. Let $F_n(t)$ denote the left side. The second-kind recurrence gives $(1 - t)F_{n+1} = tF'_n$, with $F_0 = (1 - t)^{-1}$. Hence $F_1 = t/(1 - t)^3$. If the displayed formula holds for n , differentiating $tP_n/(1 - t)^{2n+1}$ and using (18.5) gives the formula for $n + 1$. $\square$

18.3 Second-order Eulerian interpolation of first-kind diagonals

The same second-order Eulerian row that controls (18.9) also gives a binomial-basis expansion for the near-diagonals of the unsigned first-kind triangle.

Theorem 18.4 (Second-order Eulerian diagonal formula). For $p, m \geq 0$,

$$\left[\begin{matrix} m \\ m - p \end{matrix} \right] = \sum_{j=0}^p \langle\langle p \rangle\rangle_j \binom{m + j}{2p}, \quad (18.10)$$

where an out-of-range first-kind number is zero. Equivalently, the polynomial continuation in a variable x satisfies

$$\begin{bmatrix} x \\ x-p \end{bmatrix} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{x+j}{2p}.$$

The first instances are

$$\begin{aligned} \begin{bmatrix} x \\ x-1 \end{bmatrix} &= \binom{x}{2}, \\ \begin{bmatrix} x \\ x-2 \end{bmatrix} &= \binom{x}{4} + 2 \binom{x+1}{4}, \\ \begin{bmatrix} x \\ x-3 \end{bmatrix} &= \binom{x}{6} + 8 \binom{x+1}{6} + 6 \binom{x+2}{6}. \end{aligned}$$

Proof. For fixed p define

$$D_p(t) = \sum_{m \geq 0} \begin{bmatrix} m \\ m-p \end{bmatrix} t^m.$$

Writing $a_{m,p} = \begin{bmatrix} m \\ m-p \end{bmatrix}$, the first-kind recurrence becomes $a_{m,p} = a_{m-1,p} + (m-1)a_{m-1,p-1}$. Therefore

$$(1-t)D_p(t) = t^2 D'_{p-1}(t), \quad D_0(t) = \frac{1}{1-t}. \quad (18.11)$$

Let

$$Q_p(t) = t^{2p} P_p(t^{-1}).$$

The polynomial recurrence (18.5) is equivalent, after the substitution $t \mapsto t^{-1}$, to

$$Q_p(t) = t^2 ((1-t)Q'_{p-1}(t) + (2p-1)Q_{p-1}(t)).$$

It follows inductively from (18.11) that

$$D_p(t) = \frac{Q_p(t)}{(1-t)^{2p+1}} = \frac{t^{2p} P_p(t^{-1})}{(1-t)^{2p+1}}. \quad (18.12)$$

Since $P_p(u) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle u^j$, coefficient extraction gives

$$[t^m] D_p(t) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle [t^{m-2p+j}] (1-t)^{-2p-1} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{m+j}{2p},$$

which is (18.10). Both sides are polynomials in m of degree at most $2p$, so the same identity gives the stated polynomial continuation. $\square$

Part VI

Differentiation, Composition, and Reversion

Chapter 19

Faà di Bruno's formula

19.1 The set-partition form

For a block $B \subseteq [n]$, write $|B|$ for its size.

Theorem 19.1 (Faà di Bruno, partition form). *For $n \geq 0$ and sufficiently differentiable f, g ,*

$$D^n(f \circ g)(x) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(g(x)) \prod_{B \in \pi} g^{(|B|)}(x). \quad (19.1)$$

For $n = 0$, the empty partition contributes $f(g(x))$.

Proof. Induct on n . Differentiate a term indexed by a partition π of $[n]$. If the derivative hits $f^{(|\pi|)}(g)$, the chain rule contributes g' and creates the new singleton block $\{n+1\}$. If it hits the factor associated with a block B , it changes $g^{(|B|)}$ to $g^{(|B|+1)}$, which inserts $n+1$ into B . Every partition of $[n+1]$ arises uniquely in exactly one of these ways: remove $n+1$, either deleting its singleton block or removing it from its non-singleton block. Hence all terms occur once. $\square$

19.2 Multiplicity and Bell-polynomial forms

Grouping partitions by the number m_j of blocks of size j gives the classical formula.

Theorem 19.2 (Faà di Bruno, multiplicity form).

$$D^n f(g(x)) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \frac{n!}{m_1! m_2! \dots m_n!} \times f^{(m_1 + \dots + m_n)}(g(x)) \prod_{j=1}^n \left(\frac{g^{(j)}(x)}{j!} \right)^{m_j}. \quad (19.2)$$

Equivalently,

$$D^n f(g(x)) = \sum_{k=0}^n f^{(k)}(g(x)) \mathcal{B}_{n,k}(g'(x), g''(x), \dots, g^{(n-k+1)}(x)). \quad (19.3)$$

Proof. For fixed multiplicities (m_j) , the number of set partitions of $[n]$ with m_j blocks of size j is $n! / \prod_j ((j!)^{m_j} m_j!)$. Substitution in (19.1) gives (19.2). Grouping further by $k = \sum m_j$ and invoking (12.1) gives (19.3). $\square$

For $n = 4$ this reads

$$\begin{aligned} (f \circ g)^{(4)} &= f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2 g'' + 3f''(g)(g'')^2 \\ &\quad + 4f''(g)g'g^{(3)} + f'(g)g^{(4)}. \end{aligned} \quad (19.4)$$

The coefficients 1, 6, 3, 4, 1 count partitions of types $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, and 4 respectively. This also proves the expanded example in the archive without four separate differentiations.

19.3 The exponential special case

Because every derivative of e^u equals e^u , summing the partial Bell polynomials in (19.3) yields

$$D^n e^{g(x)} = e^{g(x)} \mathcal{B}_n(g'(x), g''(x), \dots, g^{(n)}(x)). \quad (19.5)$$

When g is a cumulant generating function, this is precisely the moment–cumulant relation of Theorem 15.4.

19.4 Mixed partial derivatives

For $B \subseteq [n]$, write $\partial_B = \prod_{j \in B} \partial/\partial x_j$.

Theorem 19.3 (Multivariate Faà di Bruno for distinct variables). *If $y = g(x_1, \dots, x_n)$, then*

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} f(y) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(y) \prod_{B \in \pi} \partial_B y. \quad (19.6)$$

Proof. The proof of Theorem 19.1 applies verbatim, except that the new derivative is $\partial/\partial x_{n+1}$. It either creates a singleton block or joins the new index to one existing block. $\square$

For three variables, (19.6) becomes

$$\begin{aligned} \partial_1 \partial_2 \partial_3 f(y) &= f'(y) \partial_{123} y \\ &\quad + f''(y) ((\partial_1 y)(\partial_{23} y) + (\partial_2 y)(\partial_{13} y) + (\partial_3 y)(\partial_{12} y)) \\ &\quad + f^{(3)}(y) (\partial_1 y)(\partial_2 y)(\partial_3 y). \end{aligned}$$

For repeated variables or vector-valued inner maps, one labels derivative slots first and then contracts tensor indices; the same partition lattice remains the combinatorial skeleton.

Chapter 20

Composition of ordinary and exponential series

20.1 Ordinary formal power series

Let

$$F(u) = \sum_{k \geq 0} a_k u^k, \quad G(x) = \sum_{j \geq 1} b_j x^j, \quad F(G(x)) = \sum_{n \geq 0} c_n x^n.$$

Theorem 20.1 (Ordinary composition). *For $n \geq 1$,*

$$c_n = \sum_{k=1}^n a_k \sum_{\substack{i_1 + \dots + i_k = n \\ i_r \geq 1}} b_{i_1} \cdots b_{i_k}, \quad (20.1)$$

$$= \sum_{k=1}^n a_k \sum_{\substack{\pi_1 + \dots + \pi_n = k \\ \pi_1 + 2\pi_2 + \dots + n\pi_n = n}} \binom{k}{\pi_1, \dots, \pi_n} b_1^{\pi_1} \cdots b_n^{\pi_n} \quad (20.2)$$

$$= \sum_{k=1}^n a_k \widehat{\mathcal{B}}_{n,k}(b_1, \dots, b_{n-k+1}), \quad (20.3)$$

and $c_0 = a_0$.

Proof. The coefficient of x^n in $G(x)^k$ is obtained by choosing an ordered composition $i_1 + \dots + i_k = n$, proving the first formula. Collect equal part sizes; the number of orderings with multiplicities π_j is the multinomial coefficient, proving the second. The third is the definition of the ordinary Bell polynomial. $\square$

Two useful specializations are immediate. If $F(u) = e^u$ and the series are exponentially normalized, one obtains the exponential formula. If $A(x) = 1 + \sum_{n \geq 1} a_n x^n$, then

$$[x^n] \frac{1}{A(x)} = \sum_{k=1}^n (-1)^k \widehat{\mathcal{B}}_{n,k}(a_1, \dots, a_{n-k+1}) \quad (n \geq 1), \quad (20.4)$$

by taking $F(u) = 1/(1-u)$ and $G(x) = -\sum_{n \geq 1} a_n x^n$.

20.2 Exponential formal power series

Let

$$F(x) = \sum_{j \geq 1} a_j \frac{x^j}{j!}, \quad G(u) = \sum_{k \geq 0} b_k \frac{u^k}{k!}, \quad G(F(x)) = \sum_{n \geq 0} c_n \frac{x^n}{n!}.$$

Theorem 20.2 (Exponential composition).

$$c_n = \sum_{k=0}^n b_k \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}). \quad (20.5)$$

Equivalently,

$$c_n = \sum_{\pi \in \text{Part}([n])} b_{|\pi|} \prod_{B \in \pi} a_{|B|}. \quad (20.6)$$

Proof. Substitute $F(x)$ into G and apply (1.10). The set-partition form is Theorem 12.3 with an additional weight b_k for the number of blocks. $\square$

Taking $b_k = 1$ gives the complete Bell-polynomial exponential identity

$$\exp \left(\sum_{j \geq 1} a_j \frac{x^j}{j!} \right) = \sum_{n \geq 0} \mathcal{B}_n(a_1, \dots, a_n) \frac{x^n}{n!}. \quad (20.7)$$

If the series converge, (20.5) is exactly (19.3) evaluated at the expansion point; formally, no convergence notion is needed.

Chapter 21

Lagrange inversion and Lagrange–Bürmann

21.1 Formal theorem

Let R be a commutative $\mathbb{Q}$ -algebra and let $\phi(w) \in R[[w]]$ have invertible constant term. There is a unique $g(z) \in zR[[z]]$ satisfying

$$g(z) = z\phi(g(z)). \quad (21.1)$$

Uniqueness follows recursively because the coefficient of z^n on the right depends only on coefficients of g below degree n .

Theorem 21.1 (Lagrange–Bürmann). *For $n \geq 1$ and every formal series H ,*

$$[z^n]H(g(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n, \quad (21.2)$$

$$= [w^n]H(w)\phi(w)^{n-1}(\phi(w) - w\phi'(w)). \quad (21.3)$$

In particular,

$$[z^n]g(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n. \quad (21.4)$$

Proof. Put $f(w) = w/\phi(w)$, so that $f(g(z)) = z$. Formal residues and the change of variables theorem Theorem 1.2 give

$$\begin{aligned} [z^n]H(g(z)) &= \operatorname{Res}_z H(g(z))z^{-n-1} dz \\ &= \operatorname{Res}_w H(w)f(w)^{-n-1}f'(w) dw \\ &= -\frac{1}{n} \operatorname{Res}_w H(w)(f(w)^{-n})' dw \\ &= \frac{1}{n} \operatorname{Res}_w H'(w)f(w)^{-n} dw, \end{aligned}$$

where the residue of a derivative is zero. Since $f(w)^{-n} = w^{-n}\phi(w)^n$, this is (21.2). Applying the same zero-residue rule to $H(w)\phi(w)^n w^{-n}$ gives

$$\frac{1}{n} \operatorname{Res} H' \phi^n w^{-n} = \operatorname{Res} H \phi^n w^{-n-1} - \operatorname{Res} H \phi^{n-1} \phi' w^{-n},$$

which is (21.3). Take $H(w) = w$ for (21.4). □

21.2 Analytic inverse functions

Theorem 21.2 (Analytic Lagrange inversion). *Let f be analytic near a , let $b = f(a)$, and suppose $f'(a) \neq 0$. Its local inverse g is analytic near b and has*

$$g(z) = a + \sum_{n \geq 1} g_n \frac{(z - b)^n}{n!}, \quad (21.5)$$

where

$$g_n = \frac{d^{n-1}}{dw^{n-1}} \left(\frac{w - a}{f(w) - f(a)} \right)^n \Big|_{w=a}. \quad (21.6)$$

The series has a positive radius of convergence.

Proof. The analytic inverse-function theorem gives an analytic local inverse and therefore a convergent Taylor series. Put $u = w - a$, $\zeta = z - b$, and

$$\phi(u) = \frac{u}{f(a + u) - f(a)};$$

its constant term is $1/f'(a)$. The inverse displacement satisfies $u = \zeta \phi(u)$. Formula (21.4), multiplied by $n!$, is exactly (21.6). $\square$

If $f'(a) = 0$ but $f(w) - f(a) = c(w - a)^q + O((w - a)^{q+1})$ with $c \neq 0$, factor $f(w) - f(a) = (w - a)^q U(w)$ with $U(a) = c$. Choosing one of the q values of $c^{-1/q}$ reduces the equation to ordinary inversion in the variable $(z - f(a))^{1/q}$. Thus the inverse has q local Puiseux branches. This is the precise sense in which Lagrange inversion extends to a critical point; a single-valued Taylor inverse cannot exist there.

21.3 Inverse coefficients in Bell-polynomial form

Suppose

$$f(w) = \sum_{j \geq 1} f_j \frac{w^j}{j!}, \quad f_1 \neq 0, \quad g(z) = f^{(-1)}(z) = \sum_{n \geq 1} g_n \frac{z^n}{n!}.$$

Define

$$\hat{f}_j = \frac{f_{j+1}}{(j+1)f_1}. \quad (21.7)$$

Theorem 21.3 (Explicit inverse coefficients). *One has $g_1 = f_1^{-1}$ and, for $n \geq 2$,*

$$g_n = \frac{1}{f_1^n} \sum_{k=1}^{n-1} (-1)^k n^{\bar{k}} \mathcal{B}_{n-1,k}(\hat{f}_1, \dots, \hat{f}_{n-k}). \quad (21.8)$$

Proof. Write

$$\frac{f(w)}{f_1 w} = 1 + U(w), \quad U(w) = \sum_{j \geq 1} \hat{f}_j \frac{w^j}{j!}.$$

By (21.4), $g_n = (n-1)! [w^{n-1}] (w/f(w))^n$. Now

$$(1 + U)^{-n} = \sum_{k \geq 0} (-1)^k \binom{n+k-1}{k} U^k = \sum_{k \geq 0} (-1)^k \frac{n^{\bar{k}}}{k!} U^k.$$

Apply (1.10) and extract degree $n-1$; the $k=0$ term vanishes when $n \geq 2$. $\square$

21.4 The associahedron interpretation, with normalized notation

The formula in the archive mixes exponential coefficients with an ordinary-series face formula. The exact geometric statement is the following.

Let

$$A(x) = x + \sum_{n \geq 1} a_n x^{n+1}, \quad B(x) = A^{\langle -1 \rangle}(x) = x + \sum_{n \geq 1} b_n x^{n+1}. \quad (21.9)$$

Use the indexing in which the associahedron K_n has dimension $n - 1$ and its faces are plane rooted trees with $n + 1$ leaves and internal arities at least 2. If an internal vertex has arity $i + 1$, give it weight a_i ; let a_F be the product of all vertex weights. Equivalently, if $F \cong K_{i_1} \times \cdots \times K_{i_m}$, then $a_F = a_{i_1} \cdots a_{i_m}$.

Theorem 21.4 (Associahedral inversion).

$$b_n = \sum_{F \text{ a face of } K_n} (-1)^{n - \dim F} a_F. \quad (21.10)$$

Proof. The equation $A(B) = x$ is

$$B = x - \sum_{i \geq 1} a_i B^{i+1}.$$

Iterating it recursively expands b_n as a sum over plane rooted trees with $n + 1$ leaves: every internal vertex of arity $i + 1$ contributes $-a_i$. A face F of K_n is exactly such a tree after contracting a chosen set of internal edges. If it has m internal vertices, then $\dim F = n - m$, so its sign is $(-1)^m = (-1)^{n - \dim F}$ and its weight is a_F . Thus the tree expansion is the alternating face sum. $\square$

For example,

$$\begin{aligned} b_1 &= -a_1, \\ b_2 &= -a_2 + 2a_1^2, \\ b_3 &= -a_3 + 5a_2a_1 - 5a_1^3, \\ b_4 &= -a_4 + 6a_3a_1 + 3a_2^2 - 21a_2a_1^2 + 14a_1^4. \end{aligned}$$

For K_4 , the coefficients 1, 6, 3, 21, 14 count respectively the whole 3-polytope, its six pentagonal and three square facets, its edges, and its vertices.

Chapter 22

Algebraic and transcendental examples

22.1 The Fuss–Catalan algebraic equation

Let $p \geq 2$ be an integer and consider

$$x^p - x + z = 0, \quad x(0) = 0. \quad (22.1)$$

It is equivalent to $x = z\phi(x)$ with $\phi(x) = (1 - x^{p-1})^{-1}$.

Theorem 22.1. *The distinguished solution is*

$$x(z) = \sum_{k=0}^{\infty} \binom{pk}{k} \frac{z^{(p-1)k+1}}{(p-1)k+1}. \quad (22.2)$$

Its radius of convergence is

$$R_p = (p-1)p^{-p/(p-1)}, \quad (22.3)$$

and the series converges absolutely also on $|z| = R_p$.

Proof. By (21.4),

$$[z^n]x(z) = \frac{1}{n}[w^{n-1}](1 - w^{p-1})^{-n}.$$

This vanishes unless $n-1 = (p-1)k$. In that case the coefficient is $\frac{1}{n} \binom{n+k-1}{k} = \binom{pk}{k} / ((p-1)k+1)$, proving the series.

The inverse map is $z = f(x) = x - x^p$. Its nearest critical points satisfy $f'(x) = 1 - px^{p-1} = 0$ and all have modulus $p^{-1/(p-1)}$; their critical values have modulus $(p-1)p^{-p/(p-1)}$. No local inverse can pass analytically through a critical value, while the inverse-function theorem continues the branch up to one, so this is the radius. Stirling's formula gives coefficients of order $Ck^{-3/2}R_p^{-((p-1)k+1)}$, which proves absolute convergence on the boundary. $\square$

22.2 Lambert's W function at the origin

The Lambert function satisfies $W(z)e^{W(z)} = z$.

Theorem 22.2. *On its principal branch near 0,*

$$W(z) = \sum_{n=1}^{\infty} (-n)^{n-1} \frac{z^n}{n!} = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + O(z^5), \quad (22.4)$$

with radius of convergence e^{-1} .

Proof. Here $f(w) = we^w$, so $w/f(w) = e^{-w}$. Formula (21.6) gives $g_n = D^{n-1}e^{-nw}|_{w=0} = (-n)^{n-1}$. The inverse map has its nearest critical point at $w = -1$, where $f(-1) = -e^{-1}$; this square-root branch point determines the radius. $\square$

22.3 A rapidly useful expansion around $W(e) = 1$

Put $\xi = \log z - 1$ and $u = W(z) - 1$. Then

$$u + \log(1 + u) = \xi. \quad (22.5)$$

Reversion gives

$$W(e^{1+\xi}) = 1 + \frac{\xi}{2} + \frac{\xi^2}{16} - \frac{\xi^3}{192} - \frac{\xi^4}{3072} + \frac{13\xi^5}{61440} + O(\xi^6). \quad (22.6)$$

Proof. The derivative at $u = 0$ of the left side of (22.5) is 2, so it has an ordinary inverse. Substitute $u = c_1\xi + \cdots + c_5\xi^5$ into $u + \log(1 + u) = \xi$ and compare successive powers. The triangular equations give $c_1 = 1/2$, $c_2 = 1/16$, $c_3 = -1/192$, $c_4 = -1/3072$, $c_5 = 13/61440$. $\square$

The closest critical points of $u + \log(1 + u)$ satisfy $1 + (1 + u)^{-1} = 0$, hence $u = -2$; using the two adjacent logarithm values gives images $-2 \pm i\pi$. Therefore the radius in the ξ -plane is $\sqrt{4 + \pi^2}$. For positive real z , the expansion converges whenever $|\log z - 1| < \sqrt{4 + \pi^2}$, approximately $0.0655 < z < 112.63$. Taking $\xi = -1$ yields a convergent evaluation of $W(1) = 0.567143\dots$

22.4 Binary trees and Catalan numbers

Let C_n count rooted ordered full binary trees with n internal vertices, and let $B(z) = \sum_{n \geq 0} C_n z^n$. Removing the root gives

$$B(z) = 1 + zB(z)^2. \quad (22.7)$$

Put $C(z) = B(z) - 1$; then $C = z(1 + C)^2$. Lagrange inversion gives

$$C_n = [z^n]C(z) = \frac{1}{n}[w^{n-1}](1 + w)^{2n} = \frac{1}{n} \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (22.8)$$

This simultaneously proves the functional equation, its unique formal solution, and the Catalan closed form. The symbol C_n is used here to avoid confusing these binary-tree numbers with Bell numbers.

Chapter 23

Lagrange's reversion theorem in shifted form

23.1 The two-parameter implicit equation

Let $v = v(x, y)$ be the unique formal solution of

$$v = x + yf(v), \quad v(x, 0) = x. \quad (23.1)$$

Theorem 23.1 (Lagrange reversion). *For every formal or analytic g ,*

$$g(v) = g(x) + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} (f(x)^k g'(x)). \quad (23.2)$$

In particular,

$$v = x + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} f(x)^k. \quad (23.3)$$

Proof. Put $u = v - x$. Then $u = y\phi(u)$ with $\phi(u) = f(x + u)$, treating x as a coefficient parameter. Apply (21.2) to $H(u) = g(x + u)$:

$$[y^k]g(v) = \frac{1}{k} [u^{k-1}]g'(x + u)f(x + u)^k = \frac{1}{k!} D_x^{k-1}(g'(x)f(x)^k).$$

Summation proves the first formula; take $g(z) = z$ for the second. $\square$

23.2 The delta-function proof decoded

The archive also gives a distributional derivation. Assume temporarily that $x, y \in \mathbb{R}$, the equation $z - x - yf(z) = 0$ has one simple root v , and $1 - yf'(v) > 0$. The elementary change-of-variables rule for the Dirac delta gives

$$g(v) = \int_{\mathbb{R}} \delta(yf(z) - z + x)g(z)(1 - yf'(z)) \, dz. \quad (23.4)$$

Using $\delta(s) = \int_{\mathbb{R}} e^{iks} \, dk / (2\pi)$ and expanding only in the formal parameter y ,

$$\begin{aligned} g(v) &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} (1 - yf'(x)) \right] \\ &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} - \frac{y^{n+1}}{(n+1)!} ((gf^{n+1})' - g'f^{n+1}) \right]. \end{aligned} \quad (23.5)$$

The first term of level $n + 1$ cancels the total derivative in the second term of level n ; what remains is exactly (23.2). Thus the Fourier–delta calculation is a compact analytic avatar of the formal residue proof. Formula (23.4) requires an absolute Jacobian in general; the displayed sign is valid in the stated orientation-preserving neighbourhood.

Chapter 24

Laplace-type asymptotic integrals

24.1 Endpoint expansion

Consider

$$I(\lambda) = \int_a^b e^{-\lambda f(x)} g(x) \, dx, \quad \lambda \rightarrow +\infty, \quad (24.1)$$

where the endpoint a is the unique relevant minimum and, with $t = x - a$,

$$f(a+t) \sim f(a) + \sum_{k \geq 0} a_k t^{k+\alpha}, \quad a_0 > 0, \quad \alpha > 0, \quad (24.2)$$

$$g(a+t) \sim \sum_{k \geq 0} b_k t^{k+\beta-1}, \quad \beta > 0. \quad (24.3)$$

Let $\widehat{\mathcal{B}}_{k,j}$ denote the ordinary Bell polynomial.

Theorem 24.1 (Laplace–Erdélyi coefficient formula). *Under the standard differentiability and remainder hypotheses that permit Watson’s lemma,*

$$I(\lambda) \sim e^{-\lambda f(a)} \sum_{n=0}^{\infty} \Gamma\left(\frac{n+\beta}{\alpha}\right) \frac{c_n}{\lambda^{(n+\beta)/\alpha}}, \quad (24.4)$$

where

$$c_n = \frac{1}{\alpha a_0^{(n+\beta)/\alpha}} \sum_{k=0}^n b_{n-k} \sum_{j=0}^k \binom{-\frac{n+\beta}{\alpha}}{j} \frac{\widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1})}{a_0^j}. \quad (24.5)$$

Proof. Write $f(a+t) - f(a) = t^\alpha A(t)$ with $A(t) = a_0 + a_1 t + \dots$, and set $s = (f(a+t) - f(a))^{1/\alpha} = tA(t)^{1/\alpha}$. This has an inverse $t = t(s)$. Expand

$$g(a+t(s))t'(s) = \sum_{n \geq 0} d_n s^{n+\beta-1}.$$

A generalized Lagrange–Bürmann calculation, first for integral β and then by formal continuation in the exponent, gives

$$d_n = \sum_{k=0}^n b_{n-k} [t^k] A(t)^{-(n+\beta)/\alpha}.$$

Indeed, apply (21.2) to an antiderivative of $t^{m+\beta-1}$ and differentiate back after substitution.

Put $\rho = (n + \beta)/\alpha$. The ordinary Bell expansion yields

$$\begin{aligned} [t^k]A(t)^{-\rho} &= a_0^{-\rho} [t^k] \sum_{j \geq 0} \binom{-\rho}{j} \left(\sum_{r \geq 1} \frac{a_r}{a_0} t^r \right)^j \\ &= a_0^{-\rho} \sum_{j=0}^k \binom{-\rho}{j} a_0^{-j} \widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1}). \end{aligned}$$

Thus $c_n = d_n/\alpha$ is (24.5). Finally

$$\int_0^\infty e^{-\lambda s^\alpha} s^{n+\beta-1} ds = \frac{1}{\alpha} \Gamma\left(\frac{n+\beta}{\alpha}\right) \lambda^{-(n+\beta)/\alpha},$$

and Watson's lemma controls the replacement of the finite endpoint interval by the asymptotic expansion. $\square$

Remark 24.2. The appearance of ordinary rather than exponential Bell polynomials is forced by the ordinary Taylor expansion of $A(t)^{-\rho}$. Lagrange inversion enters in the change from t to the natural Laplace coordinate s ; the gamma factor enters only in the final monomial integral.

Part VII

Higher Composition, Umbral Calculus, and Multivariate Inversion

Chapter 25

Products, inverse derivatives, and Fréchet composition

The one-variable Faà di Bruno formula already proved in this monograph is the central composition theorem, but several useful consequences were distributed among the five drafts. This chapter gathers them in a normalization that is compatible with the Bell-polynomial notation used throughout. It also explains why set partitions continue to be the correct indexing objects in Banach spaces, where there is no preferred coordinate system.

25.1 The multinomial Leibniz rule

Theorem 25.1 (Leibniz rule for many factors). *Let $f_1, \dots, f_q$ be n times differentiable. Then*

$$\frac{d^n}{dx^n} \prod_{r=1}^q f_r(x) = \sum_{j_1 + \dots + j_q = n} \binom{n}{j_1, \dots, j_q} \prod_{r=1}^q f_r^{(j_r)}(x). \quad (25.1)$$

Proof. Introduce a formal increment h . Taylor's formula, interpreted formally when only derivatives at x are needed, gives

$$f_r(x+h) = \sum_{j \geq 0} f_r^{(j)}(x) \frac{h^j}{j!}.$$

Multiplying these q series and extracting the coefficient of h^n yields

$$[h^n] \prod_{r=1}^q f_r(x+h) = \sum_{j_1 + \dots + j_q = n} \prod_{r=1}^q \frac{f_r^{(j_r)}(x)}{j_r!}.$$

The left side is $(\prod_r f_r)^{(n)}(x)/n!$. Multiplication by $n!$ gives (25.1). $\square$

The proof shows more than the formula: the multinomial coefficient is not an extra combinatorial decoration. It is exactly the number of ways to distribute n labelled differentiations among q factors with occupation numbers $j_1, \dots, j_q$.

25.2 Reciprocals and higher quotient rules

Proposition 25.2 (Derivatives of a reciprocal). *If $g(x) \neq 0$, then for $n \geq 0$,*

$$\left(\frac{1}{g}\right)^{(n)} = \sum_{k=0}^n (-1)^k k! \frac{\mathcal{B}_{n,k}(g', g'', \dots, g^{(n-k+1)})}{g^{k+1}}. \quad (25.2)$$

Consequently,

$$\left(\frac{f}{g}\right)^{(n)} = \sum_{r=0}^n \binom{n}{r} f^{(n-r)} \sum_{k=0}^r (-1)^k k! \frac{\mathcal{B}_{r,k}(g', g'', \dots, g^{(r-k+1)})}{g^{k+1}}. \quad (25.3)$$

Proof. Apply Faà di Bruno to $u \mapsto u^{-1}$ composed with g . Its k th derivative is $(-1)^k k! u^{-k-1}$, which gives (25.2). Apply the ordinary two-factor Leibniz rule to $f g^{-1}$ and substitute the reciprocal formula to obtain (25.3). $\square$

25.3 Inverse-function derivatives

Let $y = f(x)$, suppose $f'(x) \neq 0$, and write $x = g(y) = f^{-1}(y)$. The simplest closed description of all derivatives of g is an operator identity.

Theorem 25.3 (Inverse-derivative operator). *For $n \geq 1$,*

$$g^{(n)}(y) = \left(\frac{1}{f'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{f'(x)} \Big|_{x=g(y)}. \quad (25.4)$$

Equivalently, for $n \geq 2$,

$$g^{(n)}(y) = -\frac{1}{f'(g(y))} \sum_{k=2}^n f^{(k)}(g(y)) \mathcal{B}_{n,k}(g'(y), g''(y), \dots, g^{(n-k+1)}(y)). \quad (25.5)$$

Proof. Since $x = g(y)$ and $dy/dx = f'(x)$, differentiation along the inverse curve satisfies

$$\frac{d}{dy} = \frac{1}{f'(x)} \frac{d}{dx}.$$

Repeated application, starting from $g'(y) = 1/f'(x)$, proves (25.4). For the recursion, differentiate the identity $f(g(y)) = y$ exactly n times. Faà di Bruno gives

$$\sum_{k=1}^n f^{(k)}(g(y)) \mathcal{B}_{n,k}(g', g'', \dots) = 0 \quad (n \geq 2).$$

The $k = 1$ term is $f'(g(y))g^{(n)}(y)$. Isolating it gives (25.5). $\square$

The first four derivatives, with every derivative of f evaluated at $x = g(y)$, are

$$\begin{aligned} g' &= \frac{1}{f'}, \\ g'' &= -\frac{f''}{(f')^3}, \\ g''' &= \frac{3(f'')^2 - f' f'''}{(f')^5}, \\ g^{(4)} &= \frac{-15(f'')^3 + 10f' f'' f''' - (f')^2 f^{(4)}}{(f')^7}. \end{aligned}$$

These expressions provide useful independent checks on coefficients obtained from Lagrange inversion.

25.4 Coordinate-free Faà di Bruno

Let E, F, G be Banach spaces. For a sufficiently differentiable map, D^r denotes its symmetric r -linear Fréchet derivative. If $B \subseteq [n]$, write h_B for the ordered list $(h_i)_{i \in B}$; symmetry makes the order irrelevant.

Theorem 25.4 (Fréchet Faà di Bruno formula). *Let $g : E \rightarrow F$ and $f : F \rightarrow G$ be n times Fréchet differentiable. Then*

$$D^n(f \circ g)(x)[h_1, \dots, h_n] = \sum_{\pi \in \Pi_n} D^{|\pi|} f(g(x)) \left[D^{|B|} g(x)[h_B] \right]_{B \in \pi}, \quad (25.6)$$

where Π_n is the set of set partitions of $[n]$.

Proof. The assertion for $n = 1$ is the chain rule. Assume it for n and differentiate in a new direction h_{n+1} . In a term indexed by $\pi \in \Pi_n$, the new derivative can act in exactly two ways.

If it acts on the outer derivative, the chain rule inserts $Dg(x)[h_{n+1}]$ as a new argument of $D^{|\pi|+1}f$. Combinatorially this adds the singleton block $\{n+1\}$ to π . If it acts on the factor belonging to a block B , it changes $D^{|B|}g(x)[h_B]$ to $D^{|B|+1}g(x)[h_B, h_{n+1}]$; combinatorially this adjoins $n+1$ to B . Every partition of $[n+1]$ is obtained uniquely in one of these ways: remove $n+1$ from its block, and distinguish whether the block becomes empty. Hence the differentiated sum is exactly the sum over Π_{n+1} . $\square$

When $E = F = G = \mathbb{R}$ and all directions equal 1, grouping partitions by their block-size multiplicities reproduces the usual Bell-polynomial form. Thus the coordinate-free theorem is not a different formula; it is the invariant source of the scalar one.

25.5 Nested composition and partition hierarchies

For three scalar functions, one explicit formula is obtained by substituting

$$(g \circ h)^{(j)} = \sum_{\ell=0}^j g^{(\ell)}(h(x)) \mathcal{B}_{j,\ell}(h', h'', \dots)$$

into

$$D^n f(g(h(x))) = \sum_{k=0}^n f^{(k)}(g(h(x))) \mathcal{B}_{n,k}((g \circ h)', (g \circ h)'', \dots).$$

The resulting finite expression is closed, but its structure is clearer as a hierarchy of partitions. An inner partition groups derivative labels that hit the same occurrence of h ; an outer partition groups the resulting inner blocks according to occurrences of g . Flattening this hierarchy is the combinatorial shadow of associativity of formal-series composition.

Chapter 26

Partition lattices, cumulants, and Möbius inversion

Bell polynomials encode the forward passage from connected components to arbitrary assemblies. The inverse passage is Möbius inversion on the lattice of set partitions. Making that lattice explicit simultaneously proves the moment–cumulant formulas, explains the logarithm of an exponential generating function, and supplies a reusable inversion principle.

26.1 The partition lattice

Let Π_n be ordered by refinement: $\sigma \leq \pi$ when every block of σ lies inside a block of π . Its minimum is the discrete partition $\hat{0}$ and its maximum is the one-block partition $\hat{1}$.

Theorem 26.1 (Möbius function of the partition lattice). *For $\pi \in \Pi_n$,*

$$\mu(\hat{0}, \pi) = (-1)^{n-|\pi|} \prod_{B \in \pi} (|B| - 1)!. \quad (26.1)$$

More generally, if $\sigma \leq \pi$ and m_B is the number of blocks of σ contained in $B \in \pi$, then

$$\mu(\sigma, \pi) = \prod_{B \in \pi} (-1)^{m_B-1} (m_B - 1)!. \quad (26.2)$$

Proof. The interval $[\sigma, \pi]$ is the direct product, over $B \in \pi$, of partition lattices on the m_B blocks of σ lying inside B . The Möbius function is multiplicative on direct products, so it is enough to prove

$$\mu_{\Pi_m}(\hat{0}, \hat{1}) = (-1)^{m-1} (m - 1)!.$$

Let a_m denote this value. The defining Möbius relation says, for $m > 1$,

$$0 = \sum_{\rho \in \Pi_m} \mu(\hat{0}, \rho) = \sum_{\rho \in \Pi_m} \prod_{C \in \rho} a_{|C|}.$$

The exponential formula therefore gives

$$1 + \sum_{m \geq 1} \left(\sum_{\rho \in \Pi_m} \prod_{C \in \rho} a_{|C|} \right) \frac{z^m}{m!} = \exp \left(\sum_{m \geq 1} a_m \frac{z^m}{m!} \right).$$

The inner sum equals 1 for $m = 1$ and 0 for $m > 1$, so the left side is $1 + z$. Taking logarithms gives

$$\sum_{m \geq 1} a_m \frac{z^m}{m!} = \log(1 + z) = \sum_{m \geq 1} (-1)^{m-1} \frac{z^m}{m}.$$

Hence $a_m = (-1)^{m-1} (m - 1)!$. Multiplicativity gives (26.2), and the special case $\sigma = \hat{0}$ gives (26.1). $\square$

26.2 Moments and cumulants

Let m_n and κ_n be two sequences with $m_0 = 1$. Think of them as moments and cumulants, but the identities are purely algebraic.

Theorem 26.2 (Moment–cumulant inversion). *The following statements are equivalent:*

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}, \quad (26.3)$$

$$\kappa_n = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} m_{|B|}, \quad (26.4)$$

$$M(z) = \exp K(z), \quad M(z) = 1 + \sum_{n \geq 1} m_n \frac{z^n}{n!}, \quad K(z) = \sum_{n \geq 1} \kappa_n \frac{z^n}{n!}. \quad (26.5)$$

Equivalently, $m_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n)$ and κ_n is the logarithmic Bell transform of the moments.

Proof. The exponential formula immediately makes (26.3) equivalent to (26.5). To invert the partition sum, define on Π_n

$$F(\pi) = \prod_{B \in \pi} m_{|B|}, \quad G(\pi) = \prod_{B \in \pi} \kappa_{|B|}.$$

Equation (26.3), applied independently inside each block of π , is $F(\pi) = \sum_{\sigma \leq \pi} G(\sigma)$. Möbius inversion and (26.2) give

$$G(\hat{1}) = \sum_{\pi \leq \hat{1}} \mu(\pi, \hat{1}) F(\pi) = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} m_{|B|},$$

which is (26.4). □

Example 26.3 (The first cumulants). The inversion gives

$$\begin{aligned} \kappa_1 &= m_1, \\ \kappa_2 &= m_2 - m_1^2, \\ \kappa_3 &= m_3 - 3m_2m_1 + 2m_1^3, \\ \kappa_4 &= m_4 - 4m_3m_1 - 3m_2^2 + 12m_2m_1^2 - 6m_1^4. \end{aligned}$$

For a random variable these are, respectively, the mean, variance, third central moment, and fourth central moment minus three times the variance squared.

Chapter 27

Binomial type, Sheffer sequences, and exponential Riordan arrays

The Stirling matrices, Bell transforms, and many polynomial families in the preceding chapters belong to one algebraic group. The appropriate language is that of Sheffer sequences and exponential Riordan arrays. This chapter proves the structural results directly from generating functions, so no umbral formalism is required as a prerequisite.

27.1 Polynomial sequences of binomial type

A polynomial sequence $(q_n(x))_{n \geq 0}$, with $\deg q_n = n$ and $q_0 = 1$, is of *binomial type* if

$$q_n(x+y) = \sum_{k=0}^n \binom{n}{k} q_k(x) q_{n-k}(y). \quad (27.1)$$

Theorem 27.1 (Generating-function characterization). *A polynomial sequence is of binomial type if and only if there is a delta series $B(t) = b_1 t + b_2 t^2 + \cdots$, $b_1 \neq 0$, such that*

$$\sum_{n \geq 0} q_n(x) \frac{t^n}{n!} = \exp(xB(t)). \quad (27.2)$$

Proof. If (27.2) holds, multiplying the generating functions at x and y gives the one at $x+y$; coefficient extraction is exactly (27.1).

Conversely, put $Q(x, t) = \sum q_n(x) t^n / n!$. The binomial identity says $Q(x+y, t) = Q(x, t)Q(y, t)$ and $Q(0, t) = 1$. Formal differentiation with respect to x at $x = 0$ gives

$$\partial_x Q(x, t) = B(t)Q(x, t), \quad B(t) = \partial_x Q(0, t).$$

The unique solution with $Q(0, t) = 1$ is $Q(x, t) = e^{xB(t)}$. Since $\deg q_1 = 1$, the linear coefficient of B is nonzero. $\square$

Let $b(t) = B^{(-1)}(t)$ be the compositional inverse. The formal differential operator $b(D)$, where $D = d/dx$, is the associated delta operator.

Proposition 27.2 (Lowering operator). *For the sequence in (27.2),*

$$b(D)q_n(x) = nq_{n-1}(x). \quad (27.3)$$

Proof. The exponential e^{xu} is an eigenfunction of every formal differential operator: $b(D)e^{xu} = b(u)e^{xu}$. Apply this with $u = B(t)$ to the generating function. Since $b(B(t)) = t$,

$$b(D)e^{xB(t)} = te^{xB(t)}.$$

Comparing coefficients of $t^n/n!$ proves the result. $\square$

27.2 Sheffer sequences

A sequence $(s_n(x))$ is Sheffer for the pair (A, B) when $A(0) \neq 0$, $B(0) = 0$, $B'(0) \neq 0$, and

$$\sum_{n \geq 0} s_n(x) \frac{t^n}{n!} = A(t) e^{xB(t)}. \quad (27.4)$$

Let (q_n) be the basic binomial-type sequence with EGF $e^{xB(t)}$, and write $A(t) = \sum a_n t^n / n!$.

Theorem 27.3 (Sheffer addition and convolution). *For every n ,*

$$s_n(x+y) = \sum_{k=0}^n \binom{n}{k} s_k(x) q_{n-k}(y), \quad (27.5)$$

$$s_n(x) = \sum_{k=0}^n \binom{n}{k} a_{n-k} q_k(x). \quad (27.6)$$

Moreover the same delta operator lowers both families: $b(D)s_n = ns_{n-1}$.

Proof. Factor the EGF at $x+y$ as $A(t)e^{xB(t)}e^{yB(t)}$ and use the labelled Cauchy product to obtain (27.5). Factoring it as $A(t)e^{xB(t)}$ gives (27.6). Finally, $b(D)(A(t)e^{xB(t)}) = tA(t)e^{xB(t)}$, and coefficient comparison gives the lowering identity. $\square$

Appell sequences are the special case $B(t) = t$; Bernoulli and Euler polynomials below are important examples. Falling factorials correspond to $B(t) = \log(1+t)$, and Touchard polynomials correspond to $B(t) = e^t - 1$.

27.3 Exponential Riordan arrays

For $g(0) \neq 0$ and $f(t) = f_1 t + \cdots$ with $f_1 \neq 0$, define the lower triangular matrix

$$R(g, f)_{n,k} = \frac{n!}{k!} [t^n] g(t) f(t)^k. \quad (27.7)$$

If a column vector (a_k) has EGF $A(t) = \sum a_k t^k / k!$, then

$$\sum_{n \geq 0} \left(\sum_{k=0}^n R(g, f)_{n,k} a_k \right) \frac{t^n}{n!} = g(t) A(f(t)). \quad (27.8)$$

Theorem 27.4 (Riordan group law). *Exponential Riordan arrays form a group under matrix multiplication, with*

$$R(g, f)R(h, \ell) = R(g h \circ f, \ell \circ f), \quad (27.9)$$

$$R(g, f)^{-1} = R\left(\frac{1}{g \circ \bar{f}}, \bar{f}\right), \quad \bar{f} = f^{\langle -1 \rangle}. \quad (27.10)$$

Proof. Apply $R(h, \ell)$ to an arbitrary EGF A : it becomes $h(t)A(\ell(t))$. Applying $R(g, f)$ next gives

$$g(t)h(f(t))A(\ell(f(t))),$$

which is the action of the right side of (27.9). Since two lower triangular matrices agreeing on every column vector are equal, the product formula follows. To obtain the inverse, set $\ell = \bar{f}$ and choose h so that $g(t)h(f(t)) = 1$, namely $h = 1/(g \circ \bar{f})$. $\square$

Example 27.5 (The two Stirling matrices). The second-kind Stirling matrix is

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = R(1, e^t - 1)_{n,k},$$

while the signed first-kind matrix is

$$s(n, k) = R(1, \log(1 + t))_{n,k}.$$

Because $e^t - 1$ and $\log(1 + t)$ are compositional inverses, (27.10) proves their matrix inversion in one line. The same statement was previously proved by changing factorial bases; the two proofs are complementary views of the same algebra.

Example 27.6 (Sheffer coefficient arrays). If $s_n(x) = \sum_k r_{n,k} x^k$, then (27.4) gives $r_{n,k} = R(A, B)_{n,k}$. Thus the coefficient matrix of every Sheffer sequence is an exponential Riordan array, and the inverse change of polynomial basis is computed by (27.10).

27.4 Abel polynomials

For a parameter a , define

$$A_0(x) = 1, \quad A_n(x) = x(x - an)^{n-1} \quad (n \geq 1).$$

Theorem 27.7 (Abel polynomials are of binomial type). *The Abel polynomials satisfy*

$$A_n(x + y) = \sum_{k=0}^n \binom{n}{k} A_k(x) A_{n-k}(y). \quad (27.11)$$

Their EGF is $e^{xU(t)}$, where U is determined by

$$U = te^{-aU}. \quad (27.12)$$

Proof. Lagrange–Bürmann applied to (27.12) gives, for $n \geq 1$,

$$[t^n] e^{xU(t)} = \frac{x}{n} [u^{n-1}] e^{xu} e^{-anu} = \frac{x(x - an)^{n-1}}{n!}.$$

Thus $e^{xU(t)} = \sum A_n(x) t^n / n!$. The generating-function characterization of binomial type now proves (27.11). $\square$

Chapter 28

Multivariate Lagrange–Good inversion

One-variable Lagrange inversion converts an implicit equation into a coefficient formula without recursively solving for all earlier coefficients. For a coupled system the missing ingredient is a Jacobian determinant. The following proof is entirely formal and therefore applies over any characteristic-zero coefficient ring.

28.1 Formal multivariate residues

For $\mathbf{z} = (z_1, \dots, z_d)$, put

$$\operatorname{Res}_{\mathbf{z}} F(\mathbf{z}) \, d\mathbf{z} = [z_1^{-1} \cdots z_d^{-1}] F(\mathbf{z}).$$

The multivariate substitution law has the same form as the analytic change of variables formula, provided the substitution has an invertible linear part. For the special substitution used below it can also be checked directly by iterated one-variable residue substitutions.

28.2 Good’s determinant formula

Theorem 28.1 (Multivariate Lagrange–Good inversion). *Let $\mathbf{w}(\mathbf{t})$ be the unique formal solution of*

$$\mathbf{w}_i = t_i \phi_i(\mathbf{w}), \quad \phi_i(\mathbf{0}) \neq 0, \quad 1 \leq i \leq d. \quad (28.1)$$

For a multi-index $\mathbf{n} = (n_1, \dots, n_d) \in \mathbb{N}^d$ and a formal series F , one has

$$[\mathbf{t}^{\mathbf{n}}] F(\mathbf{w}(\mathbf{t})) = [\mathbf{x}^{\mathbf{n}}] F(\mathbf{x}) \prod_{i=1}^d \phi_i(\mathbf{x})^{n_i} \det \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \quad (28.2)$$

Proof. Represent the desired coefficient as an iterated formal residue:

$$[\mathbf{t}^{\mathbf{n}}] F(\mathbf{w}(\mathbf{t})) = \operatorname{Res}_{\mathbf{t}} F(\mathbf{w}(\mathbf{t})) \prod_{i=1}^d t_i^{-n_i-1} \, d\mathbf{t}.$$

Make the substitution $t_i = x_i / \phi_i(\mathbf{x})$. Its Jacobian entries are

$$\frac{\partial t_i}{\partial x_j} = \frac{1}{\phi_i} \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right).$$

Therefore

$$\begin{aligned} F(\mathbf{x}) \prod_i \left(\frac{x_i}{\phi_i(\mathbf{x})} \right)^{-n_i-1} \det \left(\frac{\partial \mathbf{t}}{\partial \mathbf{x}} \right) \\ = F(\mathbf{x}) \prod_i x_i^{-n_i-1} \phi_i(\mathbf{x})^{n_i} \det \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right). \end{aligned}$$

The determinant here equals the determinant in (28.2). Indeed, write $X = \text{diag}(x_i)$, $\Phi = \text{diag}(\phi_i)$, and $J = (\partial_j \phi_i)$. The first matrix is $I - X\Phi^{-1}J$, whereas the displayed matrix in the theorem is $I - \Phi^{-1}JX$; the identity $\det(I - AB) = \det(I - BA)$ shows equality. Taking the residue in $\mathbf{x}$ is precisely coefficient extraction on the right side of (28.2). $\square$

Remark 28.2 (Reduction to one variable). For $d = 1$ the determinant is $1 - x\phi'(x)/\phi(x)$. Formal integration by parts turns Good’s formula into

$$[t^n]H(w(t)) = \frac{1}{n}[x^{n-1}]H'(x)\phi(x)^n,$$

which is Lagrange–Bürmann. The determinant is therefore the genuine multivariate replacement for the one-variable integration-by-parts simplification.

28.3 A coupled exponential system

Consider

$$w_1 = t_1 e^{w_2}, \quad w_2 = t_2 e^{w_1}. \quad (28.3)$$

The determinant in Good’s formula is

$$\det \begin{pmatrix} 1 & -x_1 \\ -x_2 & 1 \end{pmatrix} = 1 - x_1 x_2.$$

For $m \geq 1$ and $n \geq 0$,

$$[t_1^m t_2^n]w_1 = [x_1^m x_2^n]x_1 e^{mx_2 + nx_1} (1 - x_1 x_2).$$

When $m \geq 1$ and $n \geq 1$, direct extraction gives

$$[t_1^m t_2^n]w_1 = \frac{n^{m-1} m^{n-1}}{(m-1)! n!}. \quad (28.4)$$

Indeed, the first determinant term contributes $n^{m-1} m^n / ((m-1)! n!)$ and the second contributes $(m-1)/m$ times that quantity. Their difference is (28.4). This example illustrates the role of the determinant: omitting it produces a systematic overcount.

28.4 Analytic obstruction and the characteristic determinant

Suppose now that the ϕ_i are analytic near the origin. The implicit function theorem continues the solution as long as the Jacobian of the map

$$\mathbf{x} \mapsto \left(\frac{x_1}{\phi_1(\mathbf{x})}, \dots, \frac{x_d}{\phi_d(\mathbf{x})} \right)$$

remains nonsingular and no ϕ_i becomes singular. Its determinant differs from the determinant in (28.2) only by the nonzero factor $\prod_i \phi_i^{-1}$. Thus the same determinant that corrects formal coefficient extraction also identifies the characteristic variety on which analytic inversion can break down.

Chapter 29

Classical reversion: Kepler's equation

Kepler's equation is an instructive bridge between local power-series reversion and global Fourier analysis. For eccentricity $|e| < 1$, it is

$$M = E - e \sin E. \quad (29.1)$$

Since $1 - e \cos E > 0$ on the real axis, it has a unique real solution $E = E(M)$, and $E(M + 2\pi) = E(M) + 2\pi$.

29.1 Lagrange's expansion in the eccentricity

Put $u = E - M$. Then

$$u = e \sin(M + u).$$

Treating M as a parameter, Lagrange inversion gives the convergent expansion near $e = 0$

$$E = M + \sum_{r \geq 1} \frac{e^r}{r!} \frac{d^{r-1}}{dM^{r-1}} \sin^r M. \quad (29.2)$$

The first terms are

$$E = M + e \sin M + \frac{e^2}{2} \sin 2M + \frac{e^3}{8} (3 \sin 3M - \sin M) + O(e^4).$$

The formula is a direct instance of reversion with an offset and does not require any special-function theory.

29.2 The Fourier–Bessel expansion

Let J_n denote the Bessel function of the first kind, characterized for integer n by

$$J_n(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(n\theta - z \sin \theta) d\theta. \quad (29.3)$$

Theorem 29.1 (Fourier solution of Kepler's equation). *For $|e| < 1$,*

$$E(M) = M + 2 \sum_{n \geq 1} \frac{J_n(ne)}{n} \sin(nM). \quad (29.4)$$

The series converges absolutely and locally uniformly for real M and e in compact subsets of $(-1, 1)$.

Proof. The periodic function $u(M) = E(M) - M$ is odd. Write $u(M) = \sum_{n \geq 1} b_n \sin(nM)$. Its sine coefficient is

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} u(M) \sin(nM) \, dM.$$

Because $u(\pm\pi) = 0$, integration by parts gives

$$b_n = \frac{1}{n\pi} \int_{-\pi}^{\pi} u'(M) \cos(nM) \, dM.$$

Under the change of variable $M = E - e \sin E$,

$$u'(M) \, dM = dE - dM = e \cos E \, dE.$$

Hence, with $\Theta(E) = n(E - e \sin E)$,

$$b_n = \frac{e}{n\pi} \int_{-\pi}^{\pi} \cos E \cos \Theta(E) \, dE. \quad (29.5)$$

The integral of the derivative of $\sin \Theta(E)$ over a full period is zero:

$$0 = n \int_{-\pi}^{\pi} (1 - e \cos E) \cos \Theta(E) \, dE.$$

Therefore

$$e \int_{-\pi}^{\pi} \cos E \cos \Theta(E) \, dE = \int_{-\pi}^{\pi} \cos \Theta(E) \, dE = 2\pi J_n(ne),$$

where the last equality is (29.3). Substitution into (29.5) yields $b_n = 2J_n(ne)/n$.

For $|e| < 1$, the derivative $1 - e \cos E$ has no real zero. Compactness and the analytic implicit-function theorem continue u to a complex strip whose width is positive uniformly when e ranges over a compact subset of $(-1, 1)$. Fourier coefficients of a function analytic in a strip decay exponentially, which proves absolute and locally uniform convergence. $\square$

The Lagrange series (29.2) organizes the answer by powers of e ; the Bessel series (29.4) organizes it by Fourier modes in M . Expanding $J_n(ne)$ in powers of e and rearranging recovers the Lagrange series term by term.

Chapter 30

Catalan refinements and associahedral faces

The supplied reference material emphasizes three related objects: Catalan numbers, triangulations, and associahedra. The preceding chapters counted vertices. Here Lagrange inversion counts every face and simultaneously derives the Narayana refinement by peaks of a Dyck path.

30.1 Narayana numbers from first-return decomposition

Let $C(z, u)$ count Dyck paths, where z marks semilength and u marks peaks. A nonempty Dyck path has a unique first-return decomposition $UADB$. If A is empty, the initial UD is a peak; otherwise it is not. Therefore

$$C(z, u) = 1 + zC(z, u)(C(z, u) + u - 1). \quad (30.1)$$

Put $Y = C - 1$. Then $Y = z(1 + Y)(u + Y)$. Lagrange inversion gives

$$\begin{aligned} [z^n u^k]Y &= \frac{1}{n} [y^{n-1} u^k] (1 + y)^n (u + y)^n \\ &= \frac{1}{n} \binom{n}{k} \binom{n}{k-1}. \end{aligned}$$

Thus the number of Dyck paths of semilength n with k peaks is the Narayana number

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}, \quad 1 \leq k \leq n. \quad (30.2)$$

Summing over k and using Vandermonde's identity gives

$$\sum_{k=1}^n N(n, k) = \frac{1}{n} \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n} = C_n.$$

The Narayana row is also the h -vector of the type- A associahedron; the Dyck path statistic supplies a combinatorial model for those entries.

30.2 The Kirkman–Cayley dissection formula

A face of codimension r in the associahedron associated with an N -gon is a set of r pairwise noncrossing diagonals. Such a set dissects the polygon into $r + 1$ cells.

Theorem 30.1 (Polygon dissections). *The number of ways to dissect a convex N -gon by r noncrossing diagonals is*

$$D(N, r) = \frac{1}{r+1} \binom{N+r-1}{r} \binom{N-3}{r}, \quad 0 \leq r \leq N-3. \quad (30.3)$$

Consequently the number of $(n-k)$ -dimensional faces of K_{n+1} is

$$\frac{1}{k} \binom{n+k}{k-1} \binom{n-1}{k-1}, \quad 1 \leq k \leq n. \quad (30.4)$$

Proof. Use the planar dual of a dissection, rooted at a boundary edge. Boundary edges other than the root become leaves, and cells become internal vertices of outdegree at least two. If z marks leaves and u marks internal vertices, the corresponding Schröder trees satisfy

$$T = z + u \frac{T^2}{1-T}. \quad (30.5)$$

Indeed, an internal root has an ordered sequence of at least two subtrees. Rearrange this as

$$T = z \frac{1-T}{1-(1+u)T}.$$

Lagrange inversion yields, for $m \geq 1$ and $q \geq 1$,

$$\begin{aligned} [z^m u^q]T &= \frac{1}{m} [t^{m-1} u^q] \left(\frac{1-t}{1-(1+u)t} \right)^m \\ &= \frac{1}{m} [t^{m-1} u^q] \left(1 - \frac{ut}{1-t} \right)^{-m} \\ &= \frac{1}{m} \binom{m+q-1}{q} \binom{m-2}{q-1}. \end{aligned}$$

An N -gon contributes $m = N-1$ nonroot boundary edges, and a dissection by r diagonals has $q = r+1$ cells. Substitution gives

$$\frac{1}{N-1} \binom{N+r-1}{r+1} \binom{N-3}{r} = \frac{1}{r+1} \binom{N+r-1}{r} \binom{N-3}{r},$$

which is (30.3). Finally K_{n+1} corresponds to an $(n+2)$ -gon and codimension $k-1$, so $N = n+2$ and $r = k-1$ give (30.4). $\square$

For K_5 the formula gives 1, 9, 21, 14 faces in dimensions 3, 2, 1, 0, respectively, matching the polyhedral data in the supplied associahedron reference.

Part VIII

Bernoulli, Euler, Genocchi, and Summation Calculus

Chapter 31

Bernoulli polynomials and finite power sums

Bernoulli numbers appeared earlier through alternating Eulerian sums and Stirling transforms. The polynomial family places those identities in an Appell sequence and makes finite summation an operator inverse to forward difference.

31.1 Definition and elementary identities

Define the Bernoulli polynomials $\beta_n(x)$ by

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} \beta_n(x) \frac{t^n}{n!}. \quad (31.1)$$

Thus $\beta_n = \beta_n(0)$ are the Bernoulli numbers in the convention $\beta_1 = -1/2$.

Theorem 31.1 (Bernoulli polynomial calculus). *For $n \geq 0$,*

$$\beta_n(x) = \sum_{k=0}^n \binom{n}{k} \beta_k x^{n-k}, \quad (31.2)$$

$$\beta'_n(x) = n\beta_{n-1}(x), \quad (31.3)$$

$$\beta_n(x+y) = \sum_{k=0}^n \binom{n}{k} \beta_k(x) y^{n-k}, \quad (31.4)$$

$$\beta_n(x+1) - \beta_n(x) = nx^{n-1} \quad (n \geq 1), \quad (31.5)$$

$$\beta_n(1-x) = (-1)^n \beta_n(x), \quad (31.6)$$

$$\beta_n(x) = m^{n-1} \sum_{r=0}^{m-1} \beta_n\left(\frac{x+r}{m}\right) \quad (m \geq 1). \quad (31.7)$$

Proof. Factor (31.1) as $e^{xt} \sum \beta_k t^k / k!$ and compare coefficients to obtain (31.2). Differentiation with respect to x proves (31.3); multiplying by e^{yt} proves the addition law.

For the difference law, subtract the generating functions at $x+1$ and x :

$$\frac{te^{(x+1)t} - te^{xt}}{e^t - 1} = te^{xt}.$$

Coefficient extraction gives (31.5). Replacing t by $-t$ in the EGF and simplifying gives

$$\frac{-te^{-(1-x)t}}{e^{-t} - 1} = \frac{te^{xt}}{e^t - 1},$$

which proves reflection.

For multiplication, compute the EGF of the right side:

$$\begin{aligned} \sum_{n \geq 0} m^{n-1} \sum_{r=0}^{m-1} \beta_n \left(\frac{x+r}{m} \right) \frac{t^n}{n!} &= \frac{1}{m} \sum_{r=0}^{m-1} \frac{mt e^{(x+r)t}}{e^{mt} - 1} \\ &= \frac{te^{xt}}{e^{mt} - 1} \sum_{r=0}^{m-1} e^{rt} \\ &= \frac{te^{xt}}{e^t - 1}. \end{aligned}$$

This proves (31.7). □

31.2 Faulhaber's formula

The difference identity telescopes immediately.

Corollary 31.2 (Finite power sums). *For $p \geq 0$, $N \geq 0$, and arbitrary x ,*

$$\sum_{j=0}^{N-1} (x+j)^p = \frac{\beta_{p+1}(x+N) - \beta_{p+1}(x)}{p+1}. \quad (31.8)$$

In particular,

$$\sum_{j=1}^N j^p = \frac{\beta_{p+1}(N+1) - \beta_{p+1}}{p+1}. \quad (31.9)$$

Proof. Apply (31.5) with $n = p+1$ and argument $x+j$, then sum from $j = 0$ to $N-1$. All intermediate Bernoulli-polynomial values cancel. □

This proof explains the role of Bernoulli polynomials: they are the canonical polynomial antiderivatives for the forward-difference operator.

31.3 Stirling and finite-difference representations

Theorem 31.3 (Bernoulli polynomials from finite differences). *For every $n \geq 0$,*

$$\beta_n(x) = \sum_{k=0}^n \frac{(-1)^k}{k+1} \Delta^k x^n, \quad \Delta^k x^n = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} (x+j)^n. \quad (31.10)$$

At $x = 0$ this becomes the Bernoulli–Stirling formula

$$\beta_n = \sum_{k=0}^n (-1)^k \frac{k!}{k+1} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (31.11)$$

Proof. In the ring of formal series,

$$\begin{aligned} \frac{te^{xt}}{e^t - 1} &= e^{xt} \int_0^1 \frac{du}{1 + u(e^t - 1)} \\ &= \sum_{k \geq 0} \frac{(-1)^k}{k+1} e^{xt} (e^t - 1)^k. \end{aligned}$$

The coefficient of $t^n/n!$ in $e^{xt}(e^t - 1)^k$ is $\Delta^k x^n$, which proves (31.10). At $x = 0$ this coefficient is $k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$, giving (31.11). □

31.4 Fourier series and even zeta values

Theorem 31.4 (Fourier expansion). *For $n \geq 2$ and $0 \leq x \leq 1$,*

$$\beta_n(x) = -n! \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{2\pi i k x}}{(2\pi i k)^n}. \quad (31.12)$$

Consequently, for $m \geq 1$,

$$\beta_{2m}(x) = (-1)^{m+1} \frac{2(2m)!}{(2\pi)^{2m}} \sum_{k \geq 1} \frac{\cos(2\pi k x)}{k^{2m}}, \quad (31.13)$$

$$\beta_{2m+1}(x) = (-1)^{m+1} \frac{2(2m+1)!}{(2\pi)^{2m+1}} \sum_{k \geq 1} \frac{\sin(2\pi k x)}{k^{2m+1}}. \quad (31.14)$$

In particular,

$$\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m} \beta_{2m}}{2(2m)!}. \quad (31.15)$$

Proof. Let $c_k^{(n)} = \int_0^1 \beta_n(x) e^{-2\pi i k x} dx$. For $k \neq 0$ and $n \geq 2$, the endpoint values agree by (31.5), so integration by parts and the Appell identity give

$$c_k^{(n)} = \frac{n}{2\pi i k} c_k^{(n-1)}.$$

At the last step,

$$c_k^{(1)} = \int_0^1 \left(x - \frac{1}{2}\right) e^{-2\pi i k x} dx = -\frac{1}{2\pi i k}.$$

Thus $c_k^{(n)} = -n!/(2\pi i k)^n$. The mean coefficient is zero for $n \geq 1$ because $\int_0^1 \beta_n(x) dx = 0$, a consequence of the Appell identity and $\beta_{n+1}(1) = \beta_{n+1}(0)$. This proves the complex Fourier series. Pairing k with $-k$ gives the sine and cosine forms. Setting $x = 0$ in the even formula gives (31.15). $\square$

Chapter 32

Euler polynomials and Genocchi numbers

Define the Euler polynomials by

$$\frac{2e^{xt}}{e^t + 1} = \sum_{n \geq 0} E_n(x) \frac{t^n}{n!}. \quad (32.1)$$

They are an Appell sequence, but forward difference is replaced by a symmetric average.

Theorem 32.1 (Euler polynomial calculus). *For $n \geq 0$,*

$$E'_n(x) = nE_{n-1}(x), \quad (32.2)$$

$$E_n(x+y) = \sum_{k=0}^n \binom{n}{k} E_k(x) y^{n-k}, \quad (32.3)$$

$$E_n(x+1) + E_n(x) = 2x^n, \quad (32.4)$$

$$E_n(1-x) = (-1)^n E_n(x), \quad (32.5)$$

$$E_n(x) = \frac{2}{n+1} \left(\beta_{n+1}(x) - 2^{n+1} \beta_{n+1}\left(\frac{x}{2}\right) \right). \quad (32.6)$$

Proof. Differentiation and multiplication by e^{yt} prove the first two identities. Adding the EGFs at $x+1$ and x gives $2e^{xt}$, proving (32.4); replacing t by $-t$ proves reflection. Finally,

$$\frac{2}{e^t + 1} = \frac{2}{e^t - 1} - \frac{4}{e^{2t} - 1}.$$

Multiply by e^{xt} , express both terms through Bernoulli-polynomial EGFs, and compare the coefficient of $t^n/n!$. The shift by one index caused by the factor $1/t$ gives exactly (32.6). $\square$

Corollary 32.2 (Alternating power sums). *For $m \geq 1$,*

$$\sum_{j=0}^{m-1} (-1)^j (x+j)^n = \frac{E_n(x) - (-1)^m E_n(x+m)}{2}. \quad (32.7)$$

Proof. Use $2(x+j)^n = E_n(x+j+1) + E_n(x+j)$, multiply by $(-1)^j$, and sum. The interior terms cancel in pairs. $\square$

The centered values

$$\mathcal{E}_n = 2^n E_n\left(\frac{1}{2}\right)$$

are the classical Euler numbers in the hyperbolic-secant convention, because

$$\operatorname{sech} t = \sum_{n \geq 0} \mathcal{E}_n \frac{t^n}{n!}.$$

Reflection about $1/2$ makes every odd $\mathcal{E}_n$ vanish.

32.1 Genocchi numbers

Define G_n by

$$\frac{2t}{e^t + 1} = \sum_{n \geq 0} G_n \frac{t^n}{n!}. \quad (32.8)$$

Theorem 32.3 (Bernoulli and Euler representations). *For all $n \geq 0$,*

$$G_n = 2(1 - 2^n)\beta_n = nE_{n-1}(0) \quad (n \geq 1), \quad (32.9)$$

with $G_0 = 0$.

Proof. Use

$$\frac{2t}{e^t + 1} = \frac{2t}{e^t - 1} - \frac{4t}{e^{2t} - 1}.$$

The first term contributes $2\beta_n$ and the second contributes $2^{n+1}\beta_n$, proving the first equality. On the other hand, $t \cdot 2/(e^t + 1)$ is t times the Euler-polynomial EGF at $x = 0$; shifting indices gives $G_n = nE_{n-1}(0)$. $\square$

Since $\beta_n = 0$ for odd $n > 1$, the same is true of G_n . The first values are

$$G_0 = 0, \quad G_1 = 1, \quad G_2 = -1, \quad G_4 = 1, \quad G_6 = -3, \quad G_8 = 17.$$

32.2 Modified Bernoulli coefficients

A small sequence in the supplied MathWorld material is generated by

$$\frac{1}{2} \log \left(\frac{e^{x/2} - e^{-x/2}}{x/2} \right) = \frac{1}{2} \log 2 + \sum_{r \geq 1} \mathfrak{b}_{2r} x^{2r}. \quad (32.10)$$

Proposition 32.4. *The odd coefficients vanish, and*

$$\mathfrak{b}_{2r} = \frac{\beta_{2r}}{4r(2r)!}. \quad (32.11)$$

Equivalently, for even $n \geq 2$, $\mathfrak{b}_n = \beta_n/(2n n!)$.

Proof. Differentiate the left side of (32.10):

$$\frac{1}{4} \coth \frac{x}{2} - \frac{1}{2x}.$$

The Bernoulli generating function implies

$$\coth \frac{x}{2} = \frac{2}{x} + \sum_{r \geq 1} \frac{2\beta_{2r}}{(2r)!} x^{2r-1}.$$

After cancellation of $1/(2x)$, integrate term by term and use the limiting constant $\frac{1}{2} \log 2$. $\square$

Chapter 33

Bernoulli polynomials of the second kind and complementary Bell numbers

The phrase “Bernoulli numbers of the second kind” refers to a different Sheffer sequence, also called the Cauchy numbers of the first kind. Their appearance in the reference folder is natural here because their coefficients are signed first-kind Stirling numbers, whereas complementary Bell numbers are an alternating second-kind Stirling transform.

33.1 The second-kind Bernoulli–Cauchy sequence

Define $c_n(x)$ by

$$\frac{t(1+t)^x}{\log(1+t)} = \sum_{n \geq 0} c_n(x) \frac{t^n}{n!}. \quad (33.1)$$

This is a Sheffer sequence with prefactor $t/\log(1+t)$ and basic sequence x^n .

Theorem 33.1 (Integral, derivative, and Stirling formulas). *For $n \geq 0$,*

$$c_n(x) = \int_0^1 x + u^n \, du, \quad (33.2)$$

$$c'_n(x) = nx^{n-1}, \quad (33.3)$$

$$c_n(x+1) - c_n(x) = nc_{n-1}(x), \quad (33.4)$$

$$c_n(x+y) = \sum_{k=0}^n \binom{n}{k} c_k(x) y^{n-k}, \quad (33.5)$$

$$c_n(0) = \sum_{k=0}^n \frac{s(n, k)}{k+1}, \quad (33.6)$$

$$c_n(x) = c_n(0) + \sum_{k=1}^n \frac{n}{k} s(n-1, k-1) x^k. \quad (33.7)$$

Proof. Since

$$\frac{t}{\log(1+t)} = \int_0^1 (1+t)^u \, du,$$

multiplication by $(1+t)^x$ and coefficient extraction prove the integral formula. Differentiating the EGF with respect to x gives $t(1+t)^x$, whose coefficient is nx^{n-1} ; this proves the derivative formula. Replacing x by $x+1$ and subtracting multiplies the EGF by t , giving the difference formula. The Sheffer addition law gives (33.5).

At $x = 0$, expand $u^n = \sum_k s(n, k)u^k$ in the integral to obtain (33.6). Finally integrate $c'_n(x) = n \sum_j s(n-1, j)x^j$ from 0 to x , and put $k = j + 1$. $\square$

The reflection formula is centered at $n/2 - 1$.

Proposition 33.2 (Reflection).

$$c_n\left(\frac{n}{2} - 1 - x\right) = (-1)^n c_n\left(\frac{n}{2} - 1 + x\right). \quad (33.8)$$

Proof. The falling factorial satisfies $y^n = (-1)^n n - 1 - y^n$. Apply this in the integral representation and change u to $1 - u$ to get $c_n(x) = (-1)^n c_n(n - 2 - x)$. Recenter by writing $x = n/2 - 1 + v$. $\square$

The first polynomials are

$$\begin{aligned} c_0(x) &= 1, \\ c_1(x) &= x + \frac{1}{2}, \\ c_2(x) &= x^2 - \frac{1}{6}, \\ c_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{1}{4}, \\ c_4(x) &= x^4 - 4x^3 + 4x^2 - \frac{19}{30}. \end{aligned}$$

33.2 Complementary Bell numbers

Define

$$U_n = \sum_{k=0}^n (-1)^k \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (33.9)$$

Thus U_n is the difference between the numbers of set partitions with an even and an odd number of blocks, with the sign convention in (33.9). Equivalently, $U_n = T_n(-1)$ for the Touchard polynomial $T_n(x)$.

Theorem 33.3 (Complementary Bell calculus). *The numbers U_n satisfy*

$$\sum_{n \geq 0} U_n \frac{t^n}{n!} = \exp(1 - e^t), \quad (33.10)$$

$$U_{n+1} = - \sum_{k=0}^n \binom{n}{k} U_k, \quad (33.11)$$

$$U_n = e \sum_{m \geq 0} \frac{(-1)^m m^n}{m!}. \quad (33.12)$$

Proof. Insert $x = -1$ into the Touchard EGF $\sum T_n(x)t^n/n! = \exp(x(e^t - 1))$ to obtain (33.10). Differentiation gives $F'(t) = -e^t F(t)$; the labelled Cauchy product yields the recurrence. Finally, expand

$$e^{1-e^t} = e \sum_{m \geq 0} \frac{(-1)^m e^{mt}}{m!}$$

and compare coefficients. Absolute convergence justifies the interchange for every complex t . $\square$

The initial values are

$$1, -1, 0, 1, 1, -2, -9, -9, 50, 267, 413, \dots$$

The formula is the exact analogue of Dobiński's representation of ordinary Bell numbers, with the Poisson weight analytically continued from $+1$ to -1 .

Chapter 34

Euler–Maclaurin summation with a controlled remainder

The Bernoulli-polynomial difference law is the algebraic core of the Euler–Maclaurin formula. The analytic formula repeatedly transfers derivatives from a function to periodic Bernoulli polynomials, converting a sum into an integral plus endpoint corrections.

34.1 Periodic Bernoulli functions

For $r \geq 1$, let

$$\tilde{\beta}_r(x) = \beta_r(\{x\}),$$

where $\{x\}$ is the fractional part. For $r \geq 2$ this is continuous at integers because $\beta_r(1) = \beta_r(0)$; for $r = 1$ it is the sawtooth function $x - [x] - 1/2$ away from integers. On every open unit interval,

$$\frac{d}{dx} \tilde{\beta}_r(x) = r \tilde{\beta}_{r-1}(x).$$

Theorem 34.1 (Euler–Maclaurin formula). *Let $a < b$ be integers, let $p \geq 1$, and let $f \in C^{2p}([a, b])$. Then*

$$\sum_{k=a}^b f(k) = \int_a^b f(x) \, dx + \frac{f(a) + f(b)}{2} + \sum_{r=1}^p \frac{\beta_{2r}}{(2r)!} (f^{(2r-1)}(b) - f^{(2r-1)}(a)) + R_p, \quad (34.1)$$

$$R_p = -\frac{1}{(2p)!} \int_a^b \tilde{\beta}_{2p}(x) f^{(2p)}(x) \, dx. \quad (34.2)$$

Moreover,

$$|R_p| \leq \frac{2\zeta(2p)}{(2\pi)^{2p}} \int_a^b |f^{(2p)}(x)| \, dx. \quad (34.3)$$

Proof. On one interval $[k, k+1]$, integration by parts gives

$$\int_k^{k+1} f(x) \, dx = \frac{f(k) + f(k+1)}{2} - \int_k^{k+1} \left(x - k - \frac{1}{2}\right) f'(x) \, dx.$$

Sum from $k = a$ to $b-1$ and rearrange:

$$\sum_{k=a}^b f(k) = \int_a^b f(x) \, dx + \frac{f(a) + f(b)}{2} + \int_a^b \tilde{\beta}_1(x) f'(x) \, dx. \quad (34.4)$$

On each unit interval, integrate the last integral by parts using $\tilde{\beta}_2/2!$ as an antiderivative. The boundary contributions at interior integers cancel because $\tilde{\beta}_2$ is continuous and has the common endpoint value β_2 . This gives the first correction and a remainder involving $\tilde{\beta}_2 f''$ with a minus sign.

Continue. The boundary terms involving odd Bernoulli polynomials beyond β_1 vanish because $\beta_{2r+1}(0) = \beta_{2r+1}(1) = 0$ for $r \geq 1$. Every second integration therefore contributes $\beta_{2r}(f^{(2r-1)}(b) - f^{(2r-1)}(a))/(2r)!$. Stopping after the $2p$ th derivative gives (34.1) and (34.2).

The Fourier series (31.13) implies

$$|\tilde{\beta}_{2p}(x)| \leq \frac{2(2p)!}{(2\pi)^{2p}} \zeta(2p).$$

Insert this in the remainder integral to obtain (34.3). $\square$

Remark 34.2. If f is a polynomial of degree at most $2p - 1$, the remainder vanishes. Euler–Maclaurin then reduces to an exact polynomial identity and recovers Faulhaber’s formula. For general analytic f , the series of correction terms is usually asymptotic rather than convergent; the integral remainder specifies exactly what truncation means.

34.2 Harmonic numbers

Applying Euler–Maclaurin to $f(x) = 1/x$ and passing to an infinite endpoint gives, for fixed $p \geq 1$,

$$H_n = \log n + \gamma + \frac{1}{2n} - \sum_{r=1}^{p-1} \frac{\beta_{2r}}{2r n^{2r}} + O_p(n^{-2p}). \quad (34.5)$$

Here $\gamma = \lim_{N \rightarrow \infty} (H_N - \log N)$. To see the coefficients, note that $f^{(2r-1)}(x) = -(2r-1)!x^{-2r}$. The remainder estimate follows directly from (34.3) on the tail interval.

34.3 Stirling’s expansion

Apply Euler–Maclaurin to $f(x) = \log x$. The derivative $f^{(m)}(x) = (-1)^{m-1}(m-1)!x^{-m}$ gives

$$\log n! = \left(n + \frac{1}{2}\right) \log n - n + C + \sum_{r=1}^{p-1} \frac{\beta_{2r}}{2r(2r-1)n^{2r-1}} + O_p(n^{-(2p-1)}). \quad (34.6)$$

The constant is independent of p . Wallis’ product determines it. Indeed, subtracting the corresponding expansions for $(2n)!$, $n!$, and the central binomial coefficient, and comparing with

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}},$$

which follows from Wallis’ inequalities, gives $C = \frac{1}{2} \log(2\pi)$. Hence

$$\log n! \sim \left(n + \frac{1}{2}\right) \log n - n + \frac{1}{2} \log(2\pi) + \sum_{r \geq 1} \frac{\beta_{2r}}{2r(2r-1)n^{2r-1}}. \quad (34.7)$$

Exponentiation begins with

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} + O(n^{-4})\right). \quad (34.8)$$

The coefficients in the exponential form are complete Bell polynomials in the Bernoulli correction terms, providing another direct connection with the coefficient calculus of this monograph.

Part IX

Gamma Products, Barnes Functions, and Darboux Asymptotics

Chapter 35

Pochhammer symbols and polygamma derivative calculus

The rising factorial is simultaneously a polynomial basis, a gamma quotient, and the coefficient sequence of the generalized binomial theorem. Its parameter derivatives are naturally expressed by complete Bell polynomials in finite polygamma differences.

35.1 Equivalent definitions

For $n \in \mathbb{N}$, the Pochhammer symbol is

$$(z)_n = z^{\overline{n}} = z(z+1) \cdots (z+n-1), \quad (z)_0 = 1.$$

Whenever neither side encounters an uncanceled pole,

$$(z)_n = \frac{\Gamma(z+n)}{\Gamma(z)}. \quad (35.1)$$

This follows by iterating $\Gamma(z+1) = z\Gamma(z)$; since both sides are polynomials after cancellation, the identity extends to all z .

Theorem 35.1 (Binomial and Stirling expansions). *For formal t ,*

$$(1-t)^{-z} = \sum_{n \geq 0} (z)_n \frac{t^n}{n!}, \quad (35.2)$$

$$(z)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} z^k, \quad (35.3)$$

$$(-z)_n = (-1)^n z^{\overline{n}}. \quad (35.4)$$

Proof. The series on the right of (35.2) equals 1 at $t = 0$. Its derivative is

$$\sum_{n \geq 0} (z)_{n+1} \frac{t^n}{n!} = \sum_{n \geq 0} (z+n)(z)_n \frac{t^n}{n!} = zF(t) + tF'(t).$$

Thus $(1-t)F'(t) = zF(t)$, whose unique formal solution with $F(0) = 1$ is $(1-t)^{-z}$. The Stirling expansion is the defining change of basis between powers and rising factorials, already proved combinatorially by permutation cycles. The last identity follows by reversing the signs of all n factors. $\square$

A useful forward-difference identity is

$$(z+1)_n - (z)_n = n(z+1)_{n-1}. \quad (35.5)$$

It follows by factoring $(z+1)_{n-1}$ from both terms.

35.2 Polygamma differences

Let

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}, \quad \psi^{(r)}(z) = \frac{d^r}{dz^r} \psi(z).$$

Logarithmic differentiation of the gamma recurrence gives

$$\psi^{(r)}(z+1) - \psi^{(r)}(z) = (-1)^r \frac{r!}{z^{r+1}}. \quad (35.6)$$

Iteration yields the finite difference

$$\Delta_r(z, n) := \psi^{(r)}(z+n) - \psi^{(r)}(z) = (-1)^r r! \sum_{j=0}^{n-1} \frac{1}{(z+j)^{r+1}}. \quad (35.7)$$

Although the terms on the right may have poles at nonpositive integers, the final Pochhammer derivative formulas below extend polynomially after cancellation.

Theorem 35.2 (Parameter derivatives of a Pochhammer symbol). *For $m, n \geq 0$,*

$$\frac{d^m}{dz^m} (z)_n = (z)_n \mathcal{B}_m(\Delta_0(z, n), \Delta_1(z, n), \dots, \Delta_{m-1}(z, n)). \quad (35.8)$$

Proof. Write $(z)_n = e^{L(z)}$ with $L(z) = \log \Gamma(z+n) - \log \Gamma(z)$. Then $L^{(r+1)}(z) = \Delta_r(z, n)$. The exponential specialization of Faà di Bruno states

$$\frac{d^m}{dz^m} e^{L(z)} = e^{L(z)} \mathcal{B}_m(L', L'', \dots, L^{(m)}),$$

which gives the formula. □

For example,

$$\begin{aligned} \frac{d}{dz} (z)_n &= (z)_n \Delta_0, \\ \frac{d^2}{dz^2} (z)_n &= (z)_n (\Delta_0^2 + \Delta_1), \\ \frac{d^3}{dz^3} (z)_n &= (z)_n (\Delta_0^3 + 3\Delta_0\Delta_1 + \Delta_2). \end{aligned}$$

At $z = 1$, let $H_n^{(r)} = \sum_{j=1}^n j^{-r}$. Then

$$\left. \frac{d^m}{dz^m} (z)_n \right|_{z=1} = n! \mathcal{B}_m(H_n, -1!H_n^{(2)}, 2!H_n^{(3)}, \dots, (-1)^{m-1}(m-1)!H_n^{(m)}). \quad (35.9)$$

The first three values are

$$n!H_n, \quad n!(H_n^2 - H_n^{(2)}), \quad n!(H_n^3 - 3H_nH_n^{(2)} + 2H_n^{(3)}).$$

These are the same harmonic combinations that describe near-diagonal first-kind Stirling numbers, because both arise from elementary symmetric functions in reciprocal linear factors.

Chapter 36

The Barnes G -function

The gamma function interpolates factorials; the Barnes G -function interpolates products of factorials. It is therefore the natural special function for triangular products that arise after summing logarithms or integrating $\log \Gamma$.

36.1 Canonical product and functional equation

Define

$$G(z+1) = (2\pi)^{z/2} \exp\left(-\frac{z + (1+\gamma)z^2}{2}\right) \times \prod_{k=1}^{\infty} \left\{ \left(1 + \frac{z}{k}\right)^k \exp\left(\frac{z^2}{2k} - z\right) \right\}. \quad (36.1)$$

For z in a compact set, the logarithm of the k th factor after its explicit exponential correction is

$$k \log\left(1 + \frac{z}{k}\right) - z + \frac{z^2}{2k} = O(k^{-2}),$$

uniformly. Hence the product converges normally and defines an entire function of z .

Theorem 36.1 (Barnes functional equation). *The product (36.1) satisfies*

$$G(z+1) = \Gamma(z)G(z), \quad G(1) = 1. \quad (36.2)$$

Consequently, for integers $n \geq 1$,

$$G(n+1) = 1!2!\cdots(n-1)!. \quad (36.3)$$

Proof. The value $G(1) = 1$ follows by putting $z = 0$ in the product. Form the quotient of (36.1) at $z+1$ and z . After cancellation of the quadratic exponential normalizations and shifting one product index, the remaining product is the reciprocal of

$$e^{\gamma z} \prod_{k \geq 1} \left(1 + \frac{z}{k}\right) e^{-z/k},$$

which is $\Gamma(z+1)$ by the Weierstrass product for the gamma function. Accounting for the elementary factor z gives $G(z+2)/G(z+1) = \Gamma(z+1)$, equivalent to (36.2). Iterating from $G(1) = 1$ proves the integer formula. $\square$

36.2 Alexeiewsky's integral identity

Theorem 36.2 (Integral of $\log \Gamma$). *For z in a simply connected domain avoiding the gamma poles and with consistent logarithm branches,*

$$\log G(z+1) = \frac{z}{2} \log(2\pi) - \frac{z(z+1)}{2} + z \log \Gamma(z+1) - \int_0^z \log \Gamma(1+t) dt. \quad (36.4)$$

Proof. Differentiate the logarithm of the normally convergent product (36.1). Termwise differentiation gives

$$\begin{aligned} \frac{d}{dz} \log G(z+1) &= \frac{1}{2} \log(2\pi) - \frac{1}{2} - (1+\gamma)z + \sum_{k \geq 1} z \left(\frac{1}{k} - \frac{1}{k+z} \right) \\ &= \frac{1}{2} \log(2\pi) - \frac{1}{2} - z + z\psi(z+1). \end{aligned} \quad (36.5)$$

where the standard series $\psi(z+1) = -\gamma + \sum_{k \geq 1} (1/k - 1/(k+z))$ was used.

Differentiate the right side of (36.4). The derivative of $z \log \Gamma(z+1)$ is $\log \Gamma(z+1) + z\psi(z+1)$, and the derivative of the integral cancels the logarithm. The result is exactly (36.5). Both sides vanish at $z = 0$, so they are equal. $\square$

36.3 Asymptotic expansion

Theorem 36.3 (Barnes asymptotic expansion). *Uniformly in sectors $|\arg z| \leq \pi - \delta$, for every fixed $p \geq 1$,*

$$\begin{aligned} \log G(z+1) &= \frac{z^2}{2} \log z - \frac{3z^2}{4} + \frac{z}{2} \log(2\pi) - \frac{1}{12} \log z + \zeta'(-1) \\ &\quad + \sum_{r=1}^{p-1} \frac{\beta_{2r+2}}{2r(2r+2)z^{2r}} + O_{p,\delta}(z^{-2p}). \end{aligned} \quad (36.6)$$

Proof. Insert the Euler–Maclaurin expansion of $\log \Gamma(z+1)$ into (36.4) and integrate term by term. The leading terms integrate as

$$\int t \log t dt = \frac{t^2}{2} \log t - \frac{t^2}{4}, \quad \int \log t dt = t \log t - t.$$

After cancellation with the explicit quadratic and gamma terms in (36.4), the displayed polynomial and logarithmic terms remain. A term $\beta_{2r}/(2r(2r-1)t^{2r-1})$ in $\log \Gamma$ contributes, after the cancellations, the term indexed by $r-1$ in the sum above. The integrated Euler–Maclaurin remainder is $O(z^{-2p})$ uniformly in the stated sectors.

The integration leaves one constant. Evaluate along positive integers and use $G(n+1) = \prod_{k=1}^{n-1} k!$; Euler–Maclaurin applied to the resulting double sum identifies the constant as $\zeta'(-1)$. Equivalently, defining the Glaisher–Kinkelin constant A by

$$A = \lim_{n \rightarrow \infty} \frac{\prod_{k=1}^n k^k}{n^{n^2/2 + n/2 + 1/12} e^{-n^2/4}}, \quad (36.7)$$

one obtains $\zeta'(-1) = 1/12 - \log A$. $\square$

36.4 Hyperfactorials

Let $H_n^* = \prod_{k=1}^n k^k$. Counting the exponent of each integer in a triangular product gives

$$H_n^* = \frac{(n!)^n}{G(n+1)}. \quad (36.8)$$

Indeed, $(n!)^n$ contains k to exponent n , whereas $G(n+1) = 1!2! \cdots (n-1)!$ contains it to exponent $n-k$. Combining Stirling's formula with (36.6) yields

$$H_n^* \sim A n^{n^2/2+n/2+1/12} e^{-n^2/4}. \quad (36.9)$$

Thus the constant in the Barnes asymptotic is exactly the normalization constant governing the hyperfactorial.

Chapter 37

Darboux's principle and coefficient asymptotics

The coefficient of a power series is a Fourier coefficient on a circle. If a function is smooth on that circle, repeated integration by parts makes its coefficients small. If smoothness fails at an isolated algebraic singularity, the local singular terms dominate. This is Darboux's principle in a form suited to the inversion problems of this monograph.

37.1 Exact algebraic-singularity coefficients

For $\lambda \notin \{0, 1, 2, \dots\}$,

$$[z^n] \left(1 - \frac{z}{\rho}\right)^\lambda = \rho^{-n} (-1)^n \binom{\lambda}{n} = \rho^{-n} \frac{\Gamma(n - \lambda)}{\Gamma(-\lambda)\Gamma(n + 1)}. \quad (37.1)$$

The gamma-ratio expansion, obtained from Euler–Maclaurin applied to $\log \Gamma(n - \lambda) - \log \Gamma(n + 1)$, gives

$$[z^n] \left(1 - \frac{z}{\rho}\right)^\lambda = \frac{\rho^{-n} n^{-\lambda-1}}{\Gamma(-\lambda)} \left(1 + \frac{\lambda(\lambda + 1)}{2n} + \frac{\lambda(\lambda + 1)(\lambda + 2)(3\lambda + 1)}{24n^2} + O(n^{-3})\right). \quad (37.2)$$

For a pole $(1 - z/\rho)^{-\alpha}$, set $\lambda = -\alpha$.

37.2 A finite-smoothness Darboux theorem

Theorem 37.1 (Darboux subtraction principle). *Suppose F is analytic in $|z| < \rho$ and can be written near the boundary as*

$$F(z) = \sum_{j=1}^J c_j \left(1 - \frac{z}{\rho}\right)^{\lambda_j} + R(z), \quad (37.3)$$

where branches are fixed near $z = \rho$. Assume that the boundary function $r(\theta) = R(\rho e^{i\theta})$ extends to a 2π -periodic function whose p th derivative is integrable and whose lower derivatives match at the endpoints. Then

$$[z^n] F(z) = \sum_{j=1}^J c_j \rho^{-n} \frac{\Gamma(n - \lambda_j)}{\Gamma(-\lambda_j)\Gamma(n + 1)} + O(\rho^{-n} n^{-p}). \quad (37.4)$$

Proof. The singular terms have the exact coefficients (37.1). For the remainder, Cauchy's coefficient formula on the circle gives

$$[z^n]R(z) = \frac{\rho^{-n}}{2\pi} \int_{-\pi}^{\pi} r(\theta) e^{-in\theta} d\theta.$$

Integrate by parts p times. The endpoint terms vanish because the derivatives match periodically, and therefore

$$[z^n]R(z) = \frac{\rho^{-n}}{2\pi(in)^p} \int_{-\pi}^{\pi} r^{(p)}(\theta) e^{-in\theta} d\theta.$$

The integral is bounded by its L^1 norm, proving the stated error. $\square$

The theorem says precisely what “the nearest singularity controls the coefficients” means: subtract enough terms of the local singular expansion so that the boundary remainder has enough derivatives, and the rest is a smaller Fourier coefficient.

37.3 Catalan numbers revisited

The Catalan generating function

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z} = \frac{2}{1 + \sqrt{1 - 4z}}$$

has its first singularity at $\rho = 1/4$. Its local expansion is

$$C(z) = 2 - 2(1 - 4z)^{1/2} + 2(1 - 4z) - 2(1 - 4z)^{3/2} + \dots$$

Integer powers contribute only finitely many coefficients, while the half-integer powers control the tail. Darboux's theorem gives the leading term

$$C_n \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}.$$

Combining the exact formula $C_n = (n+1)^{-1} \binom{2n}{n}$ with Stirling's expansion refines this to

$$C_n = \frac{4^n}{\sqrt{\pi} n^{3/2}} \left(1 - \frac{9}{8n} + \frac{145}{128n^2} + O(n^{-3}) \right). \quad (37.5)$$

37.4 Square-root singularities of inverse functions

Theorem 37.2 (Generic inverse singularity). *Let f be analytic near τ , and suppose*

$$f(\tau) = \rho, \quad f'(\tau) = 0, \quad f''(\tau) \neq 0.$$

Then every local inverse branch meeting τ has a Puiseux expansion

$$g(z) = \tau - c \left(1 - \frac{z}{\rho} \right)^{1/2} + O \left(1 - \frac{z}{\rho} \right), \quad c^2 = -\frac{2\rho}{f''(\tau)}, \quad (37.6)$$

with the sign of c determined by the branch. If ρ is the unique dominant singularity and the remaining boundary is sufficiently smooth, then

$$[z^n]g(z) \sim \frac{c}{2\sqrt{\pi}} \rho^{-n} n^{-3/2}. \quad (37.7)$$

Proof. Taylor expansion at τ gives

$$z - \rho = \frac{f''(\tau)}{2}(w - \tau)^2 + O((w - \tau)^3).$$

Solving first for $(w - \tau)^2$ and then recursively for $w - \tau$ in powers of $(1 - z/\rho)^{1/2}$ proves the Puiseux expansion and the value of c^2 . The coefficient of the leading singular term satisfies

$$-c[z^n](1 - z/\rho)^{1/2} \sim \frac{c}{2\sqrt{\pi}}\rho^{-n}n^{-3/2},$$

because $\Gamma(-1/2) = -2\sqrt{\pi}$. Darboux subtraction makes the remaining terms lower order. $\square$

Example 37.3 (Rooted labelled trees and Lambert W). The tree function $T(z)$ satisfies $T = ze^T$, so $z = f(T) = Te^{-T}$. Its critical point is $\tau = 1$, with $\rho = e^{-1}$ and $f''(1) = -e^{-1} = -\rho$. Hence $c = \sqrt{2}$ on the branch with $T(z) < 1$ for $0 < z < e^{-1}$. Since $[z^n]T(z) = n^{n-1}/n!$,

$$\frac{n^{n-1}}{n!} \sim \frac{e^n}{\sqrt{2\pi} n^{3/2}},$$

which is also immediate from Stirling's formula. The singularity analysis explains why the universal exponent $n^{-3/2}$ appears in tree enumeration.

Chapter 38

A unified coefficient algorithm layer

The formulas in this monograph are constructive. This chapter collects implementation-neutral recurrences that compute the main transforms without symbolic expansion of large powers. They also serve as independent certificates for tables and examples.

38.1 Arithmetic with ordinary series

Let $A(z) = \sum_{n \geq 0} a_n z^n$ with a_0 invertible.

Reciprocal. For $C = 1/A = \sum c_n z^n$,

$$c_0 = a_0^{-1}, \quad c_n = -a_0^{-1} \sum_{j=1}^n a_j c_{n-j}. \quad (38.1)$$

Quotient. For $C = B/A$, with $B = \sum b_n z^n$,

$$c_n = a_0^{-1} \left(b_n - \sum_{j=1}^n a_j c_{n-j} \right). \quad (38.2)$$

Logarithm and exponential. Assume $a_0 = 1$. If $L = \log A = \sum_{n \geq 1} \ell_n z^n$, then $A' = AL'$ gives

$$\ell_n = a_n - \frac{1}{n} \sum_{j=1}^{n-1} j \ell_j a_{n-j}. \quad (38.3)$$

Conversely, if $A = e^L$,

$$a_0 = 1, \quad a_n = \frac{1}{n} \sum_{j=1}^n j \ell_j a_{n-j}. \quad (38.4)$$

Arbitrary powers. For $C = A^\alpha$, logarithmic differentiation $AC' = \alpha A' C$ yields

$$c_0 = 1, \quad c_n = \frac{1}{n} \sum_{j=1}^n ((\alpha + 1)j - n) a_j c_{n-j}. \quad (38.5)$$

Each recurrence follows by equating the coefficient of z^n in the displayed defining identity, so it is valid over any characteristic-zero coefficient ring.

38.2 Reversion by triangular recursion

Let $F(z) = a_1z + a_2z^2 + \cdots$ with a_1 invertible, and let $G(z) = b_1z + b_2z^2 + \cdots$ satisfy $F(G(z)) = z$. Then $b_1 = a_1^{-1}$ and, for $n \geq 2$,

$$b_n = -a_1^{-1} \sum_{k=2}^n a_k [z^n] G(z)^k. \quad (38.6)$$

The coefficient on the right depends only on $b_1, \dots, b_{n-1}$, so this is a triangular algorithm. Ordinary partial Bell polynomials compute the inner coefficient efficiently:

$$[z^n] G(z)^k = \hat{B}_{n,k}(b_1, b_2, \dots).$$

Lagrange inversion gives a nonrecursive formula; the triangular recurrence is often preferable for exact computation because every step can be checked by a single composition coefficient.

38.3 Good inversion by truncated multivariate arithmetic

For a system $w_i = t_i \phi_i(\mathbf{w})$, coefficients up to a prescribed multi-degree can be obtained in either of two independently checkable ways:

1. fixed-point iteration in the complete local ring, truncating after each substitution; and
2. direct extraction from Good's determinant formula (28.2).

The first method is structurally simple; the second computes a selected coefficient without constructing the whole solution. Agreement between them is a strong certificate because the two calculations use different algebraic paths.

Part X

Archive Concordance and Reference Record

Chapter 39

Concordance with the source archive

The supplied archive contains five overlapping LaTeX drafts, a Wikipedia reference directory, and a MathWorld reference directory. The drafts repeat a large common core but differ in breadth, notation, proof detail, examples, and source-audit material. The present monograph uses the broadest draft as a structural spine, merges the unique portions of the other four, and then adds the genuinely new reference material. “Normalized” below means algebraically equivalent after the notation and boundary conventions of the opening chapters are imposed; “strengthened” means that a proof, convergence hypothesis, determinant, or remainder estimate has been supplied.

Source	Distinct contribution	Treatment here
<code>combinatorial_efficient_calculus.tex</code>	The widest treatment of Stirling, Bell, Eulerian, Bell-polynomial, Faà di Bruno, Lagrange, Lambert W , and asymptotic material; extensive congruence, probability, geometry, and provenance sections.	Used as the structural spine. Repeated material from the other drafts was deduplicated against it. Its source ledger and notation audit were rewritten for the actual five-draft archive.
<code>Combinatorial_Formulae_and_Inversion_Theorems.tex</code>	A strong foundations chapter; detailed set-partition, permutation, and probability interpretations; compact maps of formula families; alternative proofs of inverse relations and source corrections.	Unique bijections, probability interpretations, normalization checks, and compact dependency explanations were retained and reconciled with the spine’s more extensive formula tables.
<code>Formulae_Compendium.tex</code>	Multinomial Leibniz rules, coordinate-free Fréchet Faà di Bruno, inverse derivative recursions, nested composition, Good’s multivariate inversion, partition-lattice Möbius inversion, Sheffer sequences, exponential Riordan arrays, Abel polynomials, and coefficient algorithms.	Rewritten and strengthened in Chapters 25–28 and 38. Good’s determinant receives a complete formal residue proof and a genuinely coupled worked example.

Continued on the next page

Source	Distinct contribution	Treatment here
Formulae_Monograph.tex (package draft)	A proof-oriented reorganization, low-order verification tables, explicit source certificates, and complementary asymptotic and implementation notes.	Verification procedures and proof variants were consolidated rather than repeated. Schematic asymptotic placeholders were replaced by defined saddle, Euler–Maclaurin, or Darboux remainders.
Formulae_Monograph.tex (source-and-PDF draft)	The most complete treatment of inverse derivatives, arbitrary-point reversion, Kepler’s equation, and a reproducible coefficient-computation layer.	The inverse-operator formula and the full Fourier–Bessel derivation of Kepler’s solution appear in Chapters 25 and 29; the computation layer is merged into Chapter 38.
Wikipedia/	Reference articles on Stirling, Bell, Eulerian, Catalan, Faà di Bruno, Lagrange inversion and reversion, and associahedra, including polygon triangulations and face data.	Used as a coverage audit. Existing core material was checked and deduplicated; the Narayana refinement and the Kirkman–Cayley formula for every associahedral face are proved in Chapter 30.
MathWorld/	Reference material on Barnes G , Bernoulli numbers and both kinds of Bernoulli polynomials, Euler and Genocchi polynomials, modified Bernoulli coefficients, complementary Bell numbers, Dobiński’s formula, Euler–Maclaurin summation, Pochhammer symbols, polygamma functions, and Darboux’s formula.	Integrated in Parts VII and VIII. Each formula used in the development is derived from a generating function, gamma product, formal residue, Fourier coefficient, or periodic-Bernoulli integration argument.
Authoritative online sources and primary papers	NIST DLMF sections on Bernoulli/Euler polynomials, Euler–Maclaurin, gamma and Barnes functions; primary literature on multivariate Lagrange inversion, Riordan arrays, and analytic combinatorics.	Used to verify conventions, domains, determinant orientation, and asymptotic coefficients. Bibliographic entries below identify the specific sources.

39.1 Formula-status ledger

Every mathematical display retained from the five drafts or promoted from the reference directories falls into one of the following four classes.

1. *Exact identities* are reproduced, usually in a notation that prevents collisions, and proved by a bijection, coefficient extraction, finite differences, matrix inversion, or a generating-function

calculation.

2. *Equivalent normal forms* are consolidated. For example, the multiplicity sum, Bell-polynomial identity, and set-partition form of Faà di Bruno’s formula are one theorem viewed in three coordinate systems.
3. *Corrected statements* retain the source’s intended content but repair an index shift, a missing range or analyticity hypothesis, or a suppressed Jacobian. The type-B Eulerian and delta-function reversion formulas are the main examples.
4. *Schematic or computational statements* are replaced by something checkable. Undefined asymptotic placeholders become recursively defined saddle coefficients; finite tables and modular searches come with exact recurrences or certificate procedures.

The resulting distinction is important: a theorem is not silently weakened into numerical evidence, and a numerical observation is not silently promoted to a theorem.

39.2 Implementation-neutral algorithms

The transient software lists in encyclopedic source material are best replaced by algorithms whose correctness follows from the proved recurrences. To compute all entries through row N , initialize the boundary values and fill either Stirling triangle by its two-term recurrence; this uses $O(N^2)$ arithmetic operations and $O(N)$ memory if only the preceding row is retained. Bell numbers are then row sums of the second-kind triangle, or may be produced by the binomial recurrence. Eulerian numbers admit the same triangular complexity. Partial Bell polynomials can be filled by the pointing recurrence, with exact polynomial or symbolic-series arithmetic as appropriate. For inverse series, Lagrange–Bürmann gives each target coefficient independently from a truncated power of the defining series; Newton iteration is asymptotically faster for very high orders, while the Lagrange form is usually more transparent for symbolic coefficients. Every one of these algorithms is valid over any coefficient ring in which the displayed divisions make sense.

39.3 Dependency map

The logical flow of the monograph may be summarized as

factorial bases and finite differences $\longrightarrow$ Stirling and Eulerian transforms
 $\longrightarrow$ partition lattices, Bell calculus, and Sheffer–Riordan arrays
 $\longrightarrow$ composition, differentiation, and implicit inversion
 $\longrightarrow$ Bernoulli summation, gamma products, and Darboux asymptotics.

Formal residues turn implicit substitution into Lagrange–Good inversion; periodic Bernoulli polynomials turn discrete sums into integrals; and singular expansions turn local analytic geometry into coefficient asymptotics. Identities that look unrelated in the separate drafts thus repeatedly reduce to four operations: change of basis, labelled decomposition, coefficient extraction, and controlled boundary analysis.

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